

Pycno Run 5

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Set Up

Load Libraries

Values Set

```
IPC_name = "Moa"  
Target_name = "Pycno"
```

Load Data Pycno Plate Run 5

```
sample_set_up <- read_excel(here("raw_data", "qPCR_raw_data", "2025-12-23_161523_Pycno_5_v3.xls"), sheet="Raw")  
amplification_data <- read_excel(here("raw_data", "qPCR_raw_data", "2025-12-23_161523_Pycno_5_v3.xls"), sheet="Amplification")  
results_data <- read_excel(here("raw_data", "qPCR_raw_data", "2025-12-23_161523_Pycno_5_v3.xls"), sheet="Results")
```

Merge Tables

```
amplification_data %>%  
  left_join(sample_set_up) -> combined_qPCR_data
```

Now we are ready to start analyzing the data.

First thing we want to do is check to see how our internal positive control is working.

Moa Map

```
columns = seq(from =1, to=24, by=1)  
columns_t= as.tibble(columns) %>% dplyr::rename("Column"=value)  
rows = capitalize(letters[1:16])  
rows_t= as.tibble(rows) %>% dplyr::rename("Row"=value)
```

```

cross_join(columns_t, rows_t) -> full_plate

combined_qPCR_data %>%
  filter(., `Target Name`==IPC_name) %>%
  mutate(., Row = str_sub(`Well Position`,start = 1L, end = 1L),
          Column= as.numeric(str_sub(`Well Position`,start = 2L, end = -1L ))) -> moa_samples

full_plate %>%
  left_join(moa_samples) -> moa_plate

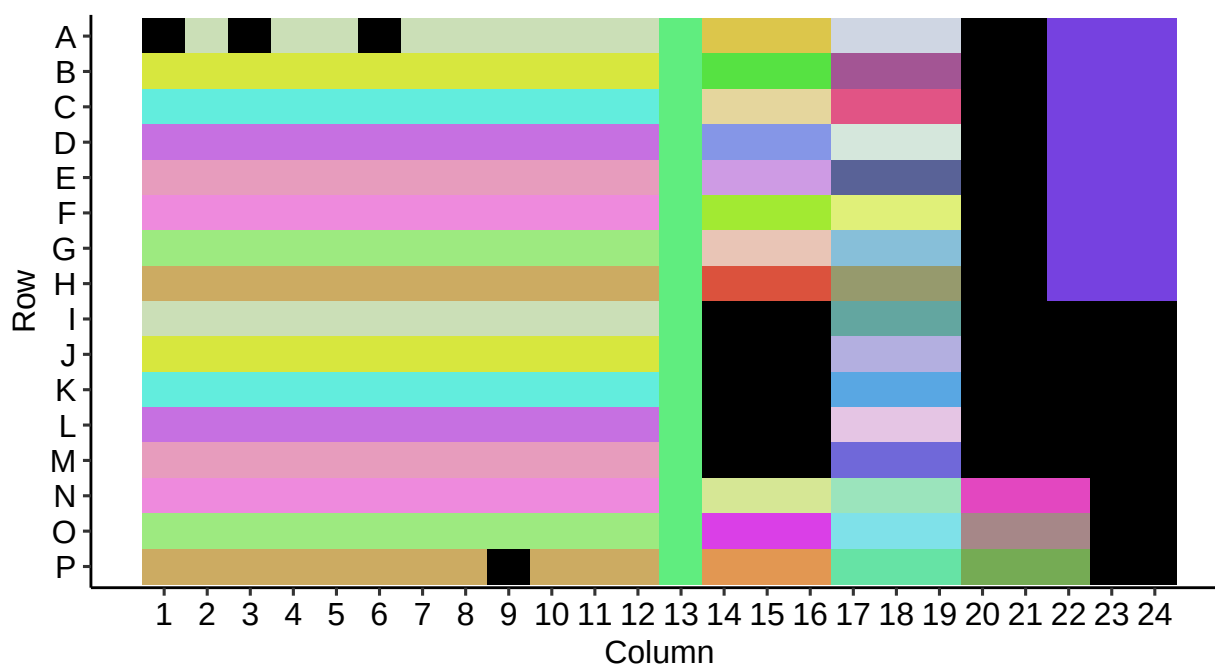
palette <- distinctColorPalette(length(moa_plate$`Sample Name` %>% unique()))

moa_plate %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x=Column, y=fct_rev(Row), fill=`Sample Name` )) +geom_tile() +scale_x_continuous(breaks=col

```

Moa-IPC Standards & Samples

		E-0.1		E2697.3T		E2700.3T		E2705.NC		E3122.NC
		E-0.5		E2698.2T		E2701.1T		E2706.NC		E4_V4
		E0_V4		E2698.1T		E2701.2T		E2707.1T		EXT_Blank
		E0.4_V4		E2698.3T		E2701.3T		E2707.2T		NTC
Sample Name		E0.7_V4		E2699.1T		E2702.NC		E2707.3T		NA
		E1_V4		E2699.2T		E2703.1T		E2708.1T		
		E2_V4		E2699.3T		E2703.2T		E2708.2T		
		E2697.1T		E2700.1T		E2703.3T		E2708.3T		
		E2697.2T		E2700.2T		E2704.NC		E3_V4		



Pycno Map

```
combined_qPCR_data %>%
  filter(., `Target Name`==Target_name) %>%
  mutate(., Row = str_sub(`Well Position`,start = 1L, end = 1L),
          Column= as.numeric(str_sub(`Well Position`,start = 2L, end = -1L ))) -> pycno_samples

full_plate %>%
  left_join(pycno_samples) -> pycno_plate
```

```

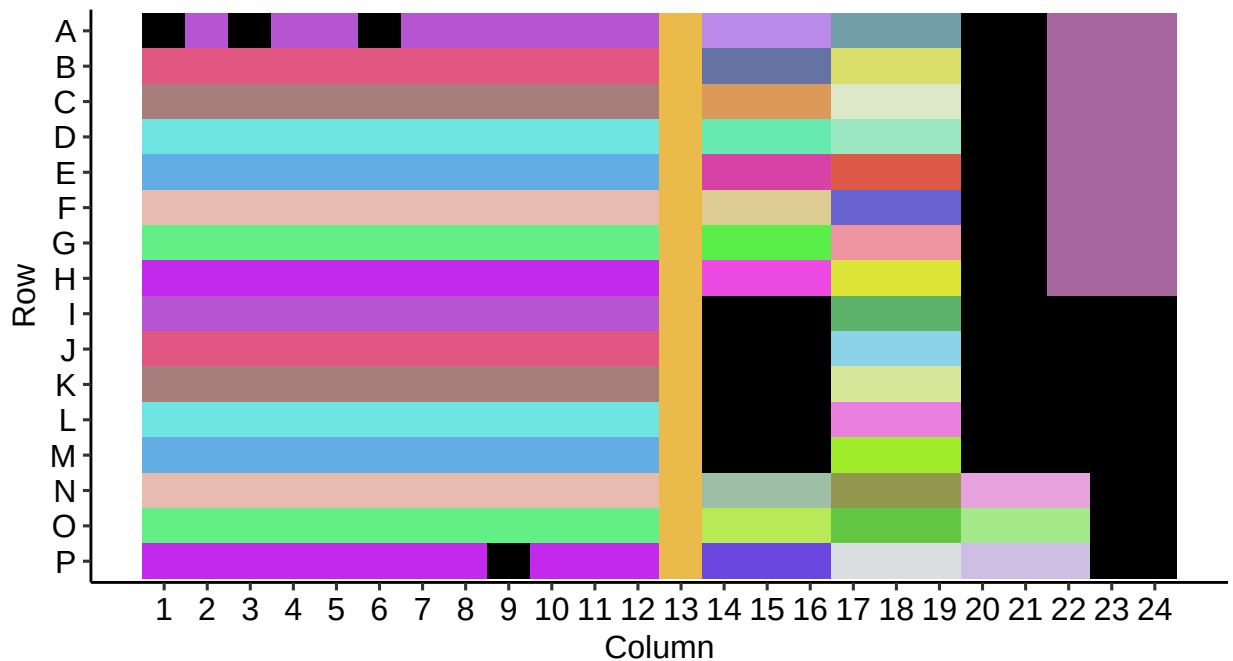
palette <- distinctColorPalette(length(pycno_plate$`Sample Name` %>% unique()))

pycno_plate %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x=Column, y=fct_rev(Row), fill=`Sample Name` )) +geom_tile() +scale_x_continuous(breaks=col)

```

Pycno Standards & Samples

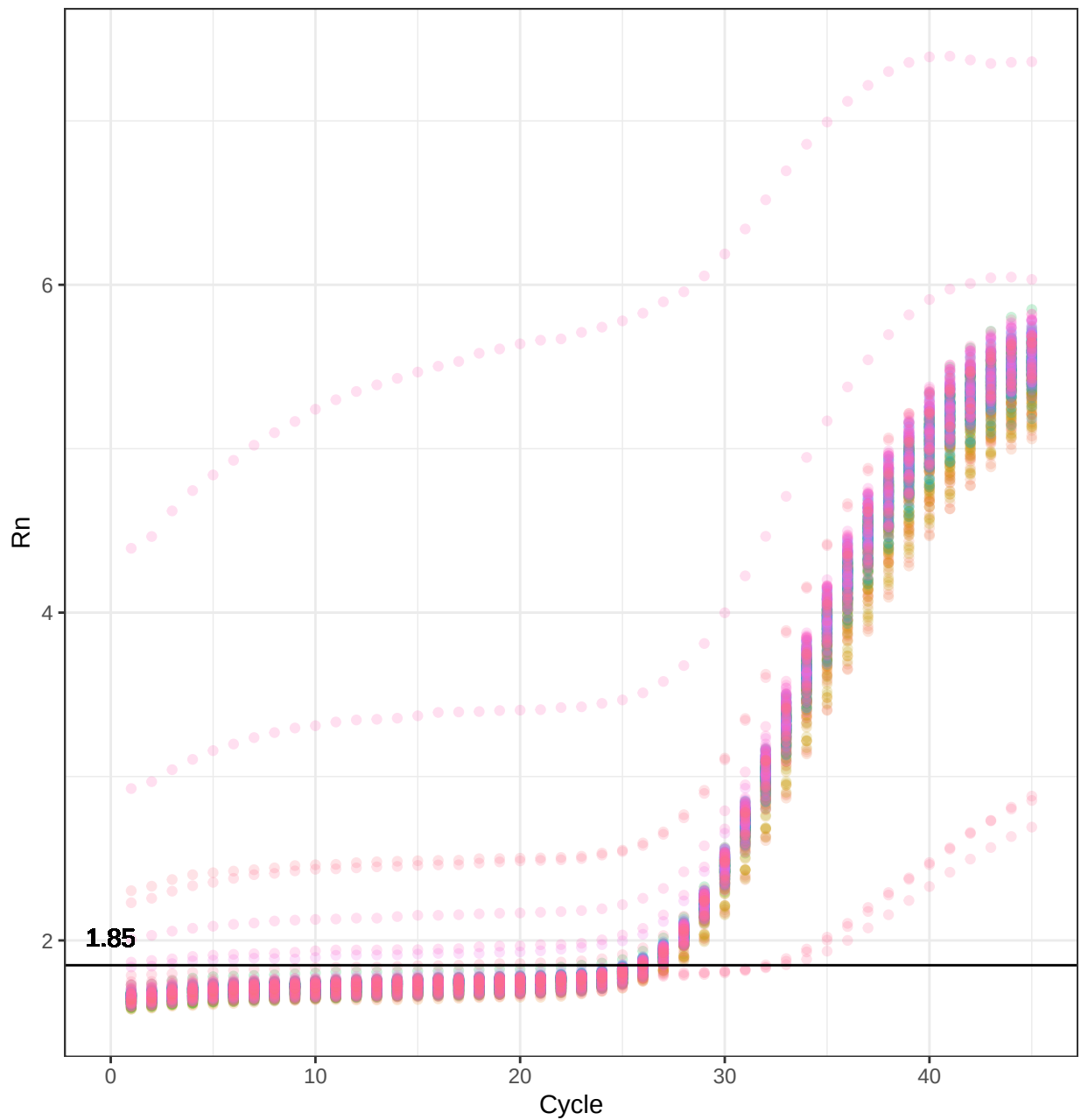
	 E-0.1	 E2697.3T	 E2700.3T	 E2705.NC	 E3122.NC
	 E-0.5	 E2698.2T	 E2701.1T	 E2706.NC	 E4_V4
	 E0_V4	 E2698.1T	 E2701.2T	 E2707.1T	 EXT_Blank
	 E0.4_V4	 E2698.3T	 E2701.3T	 E2707.2T	 NTC
Sample Name	 E0.7_V4	 E2699.1T	 E2702.NC	 E2707.3T	 NA
	 E1_V4	 E2699.2T	 E2703.1T	 E2708.1T	
	 E2_V4	 E2699.3T	 E2703.2T	 E2708.2T	
	 E2697.1T	 E2700.1T	 E2703.3T	 E2708.3T	
	 E2697.2T	 E2700.2T	 E2704.NC	 E3_V4	



Moa Rn

```
combined_qPCR_data %>%  
filter(., str_detect(`Target Name`,IPC_name)) -> moa_standard  
  
moa_standard$Rn %>% max() -> max_Rn_moa  
h_moa= max_Rn_moa*0.25  
  
moa_standard %>%  
  # filter(., `Well Position`=="A24") %>% A24 did not perform well  
ggplot(., aes(y=Rn, x=Cycle, color=`Well Position`)) +  
  geom_point(alpha=0.2) + geom_hline(yintercept =h_moa) +  
  geom_text(aes(0,h_moa,label = round(h_moa,2), vjust = -1),color="black") +theme_bw() +ggtitle("Moa Rn")
```

Moa Rn



There is quite a bit of spread in our IPC. Will remove any samples outside normal distribution. #####
Remove Samples with Anomolous Outlier IPC values

```
moa_standard %>%
  filter(., Cycle ==40) -> moa_standard_c40

moa_standard_c40 %>%
  dplyr::summarise(mean_Rn = ci(Rn, confidence = 0.99)[1],
    lower_ci_mpg = ci(Rn, confidence = 0.99)[2],
    upper_ci_mpg = ci(Rn, confidence = 0.99)[3],
    sd_Rn = ci (Rn, confidence = 0.99)[4]) -> CI0.95_moa
```

```

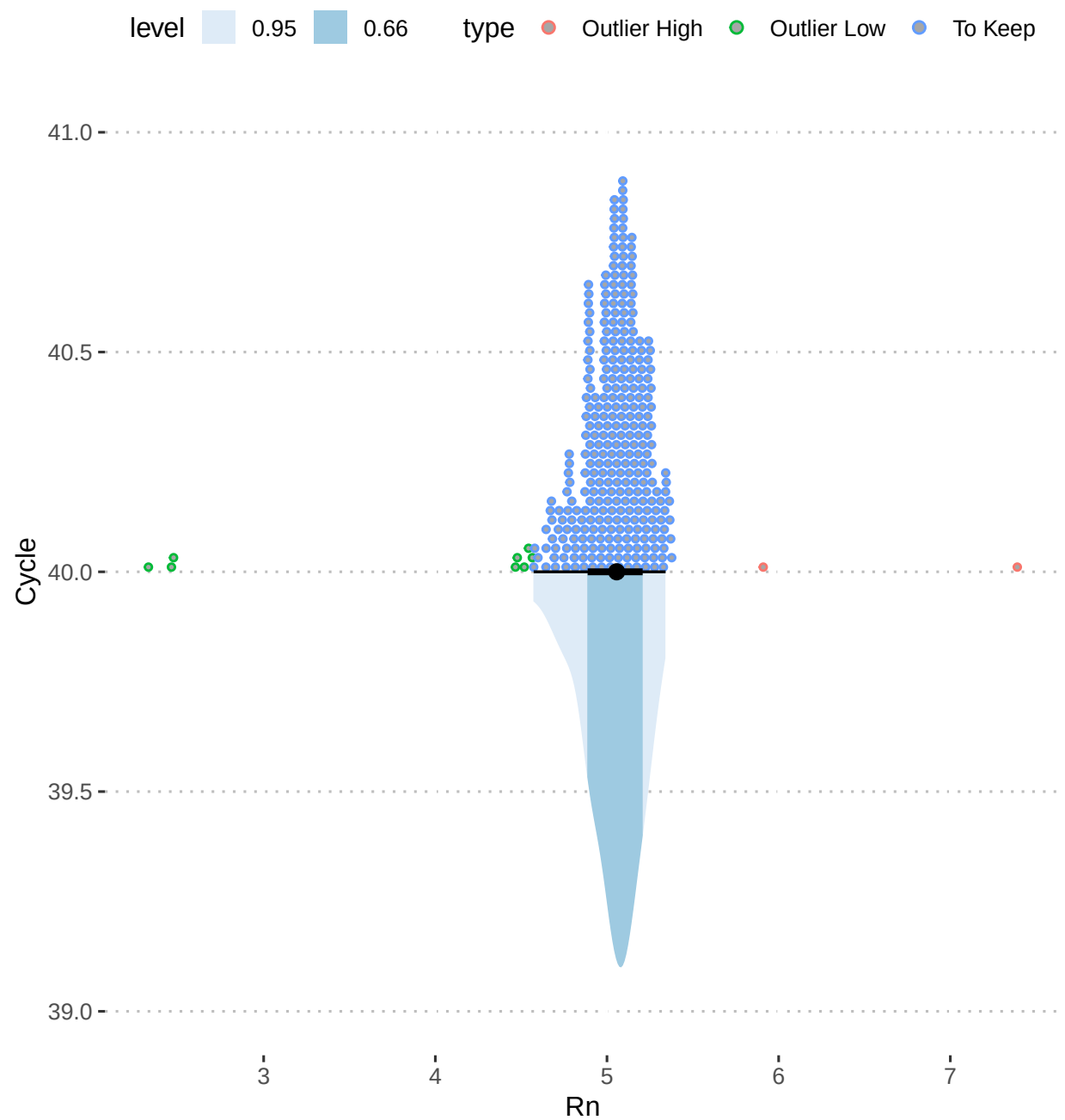
Sum = groupwiseMean(Rn ~ Cycle, data=moa_standard_c40, conf = 0.99)

moa_standard_c40 %>%
  summarise(
    Q1 = quantile(Rn, 0.25, na.rm = TRUE),
    Q3 = quantile(Rn, 0.75, na.rm = TRUE),
    IQR = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR,
    upper_bound = Q3 + 1.5 * IQR
  ) -> moa_iqr

moa_standard_c40 %>%
  mutate(., type=case_when(Rn > moa_iqr$upper_bound ~"Outlier High",
                           Rn < moa_iqr$lower_bound ~"Outlier Low",
                           TRUE~"To Keep")) -> moa_standard_c40

moa_standard_c40 %>%
  ggplot(., aes(x= Rn, y=Cycle)) + geom_dots(layout='weave', side = "top", aes(color=type)) + stat_slab

```



```
moa_standard_c40 %>%
  filter(., type!="To Keep") -> moa_outliers_to_drop
```

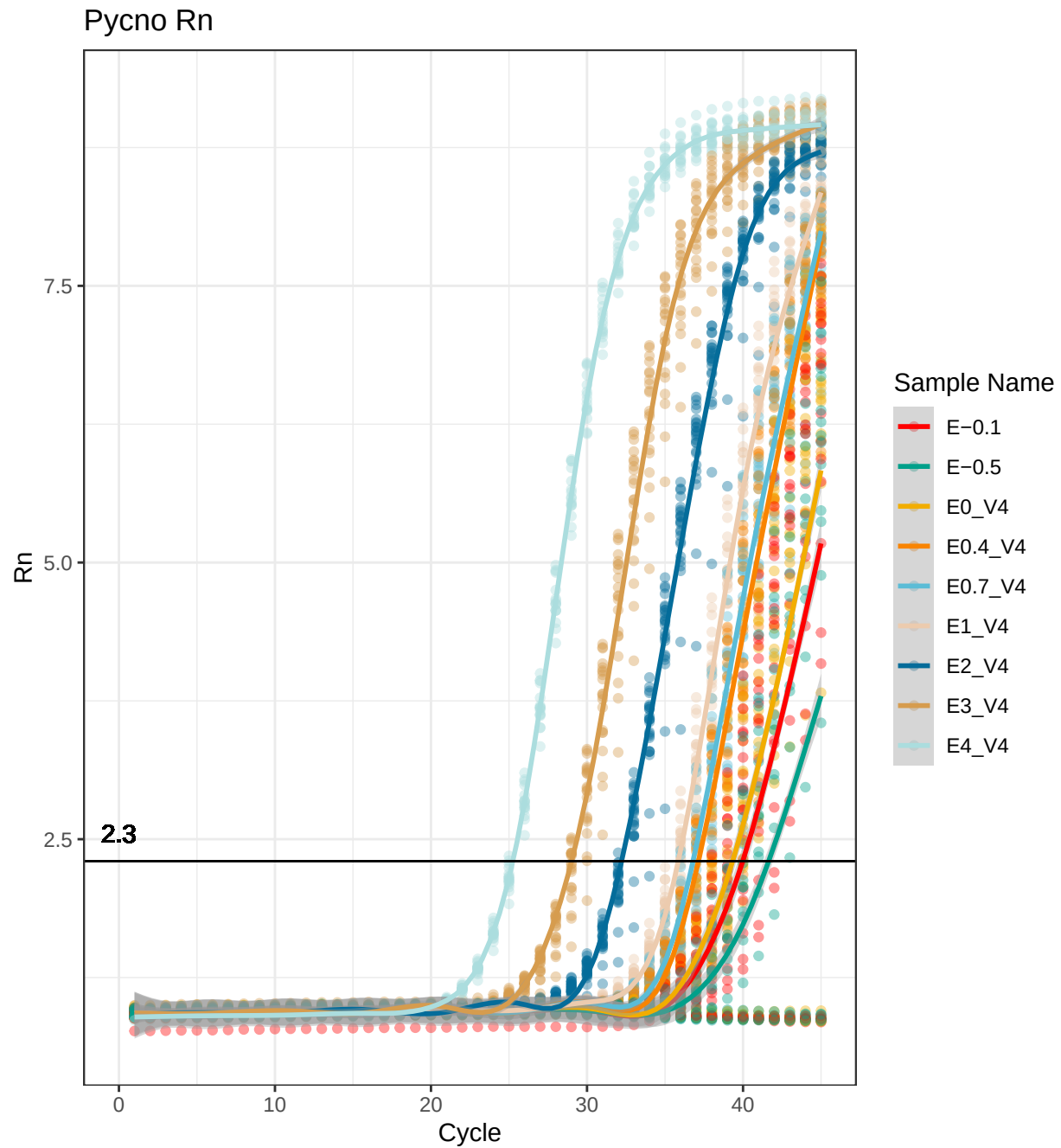
```
combined_qPCR_data %>%
  filter(., `Well Position` != moa_outliers_to_drop$`Well Position`) -> combined_qPCR_data_cleaned

results_data %>%
  filter(., `Well Position` != moa_outliers_to_drop$`Well Position`) -> results_data_cleaned
```


Filter Data

Pycno Rn

```
combined_qPCR_data_cleaned %>%  
filter(., str_detect(`Target Name`,Target_name)) %>%  
  filter(., Task=="STANDARD")-> pycno_standard  
  
pycno_standard$Rn %>% max() -> max_Rn  
h= max_Rn*0.25  
  
pycno_standard %>%  
ggplot(., aes(y=Rn, x=Cycle, color=`Sample Name`)) +geom_point(alpha=0.4) +geom_smooth() + geom_hline(y=  
  geom_text(aes(0,h,label = round(h,2), vjust = -1),color="black") +theme_bw() + ggtitle("Pycno Rn") +s
```



Looks great. We expect higher concentration standards to surpass the threshold (here set to 25% of the maximum Rn fluorescence value) at fewer cycles and we see that in the data.

Pycno Ct versus Quantity

```
pycno_standard %>%
  mutate(., Threshold = if_else(Rn > h, "Above","Below")) -> pycno_standard

pycno_standard %>%
```

```

group_by(`Well Position`, `Sample Name`) %>%
filter(., rank(Threshold, ties.method="first")==1) %>%
filter(., Threshold == "Above") -> pycno_standard_above

pycno_standard %>%
ungroup() %>%
filter(., !`Well Position` %in% pycno_standard_above$`Well Position`) %>%
group_by(`Well Position`, `Sample Name`) %>%
filter(., rank(Threshold, ties.method="last")==1) -> pycno_standard_below

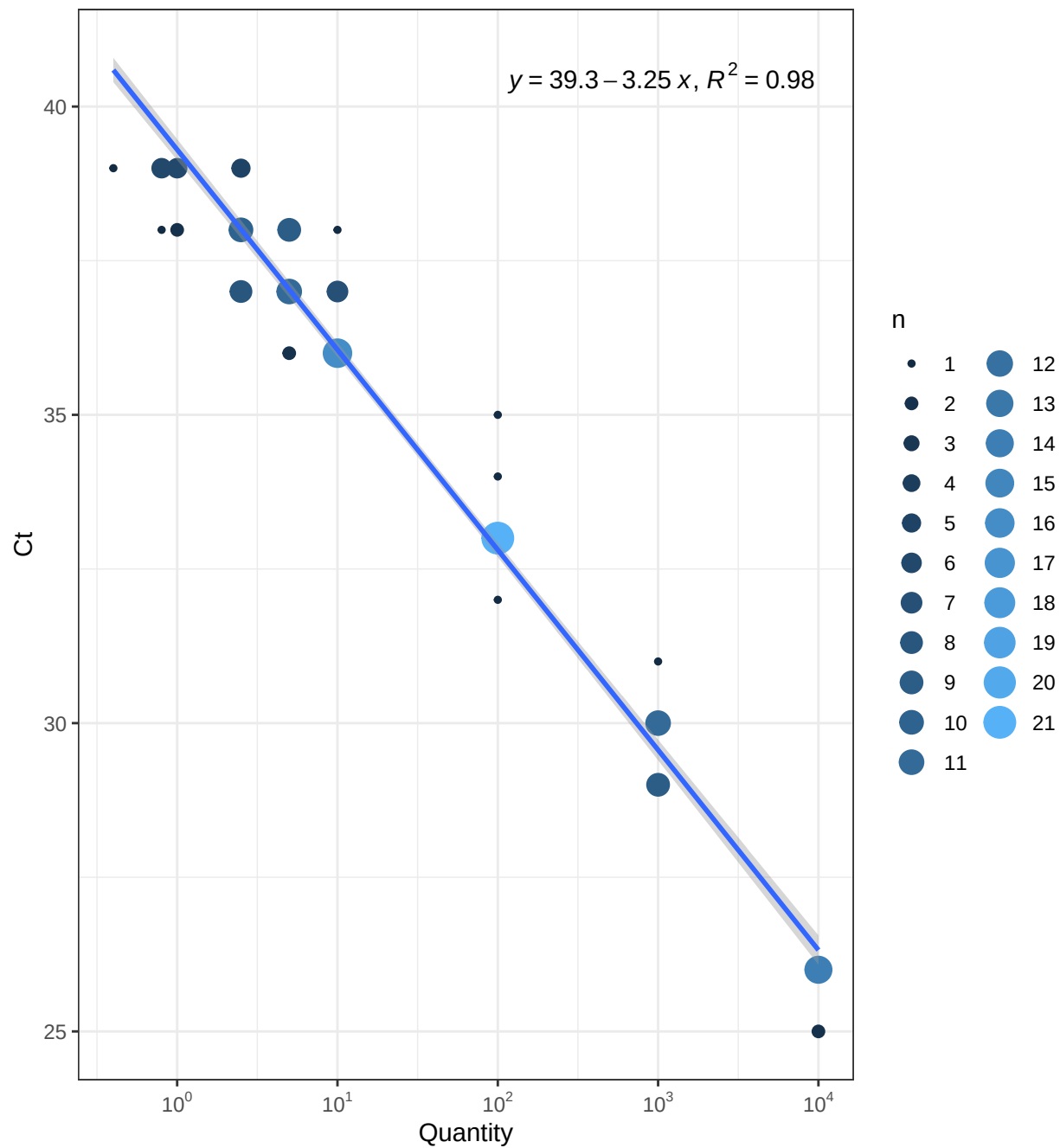
bind_rows(pycno_standard_above, pycno_standard_below) -> pycno_standard_threshold

set_breaks = function(limits) {
  seq(limits[1], limits[2], by = 1)
}

pycno_standard_threshold %>%
filter(., Cycle < 40 ) %>%
ggplot(., aes(x=`Quantity`, y = Cycle)) +geom_count(aes(color = ..n.., size = ..n..)) +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
  label.y = "top") +theme_bw() +ylab("Ct") -> run5_ct_versus_quant

run5_ct_versus_quant

```



```
# View the structure of the generated plot object
# The computed data is stored within the plot object
grob_data_run5 <- ggplot_build(run5_ct_versus_quant)$data[[3]]

# The result is a list; unlist it to get a character vector
coef_char_run5 <- unlist(grob_data_run5$coefs)

standard_curve_coefs_run5 <- enframe(coef_char_run5, name = "Metric", value = "Value") %>%
  mutate( Metric = recode(Metric,
    "(Intercept)" = "Intercept",
```

```

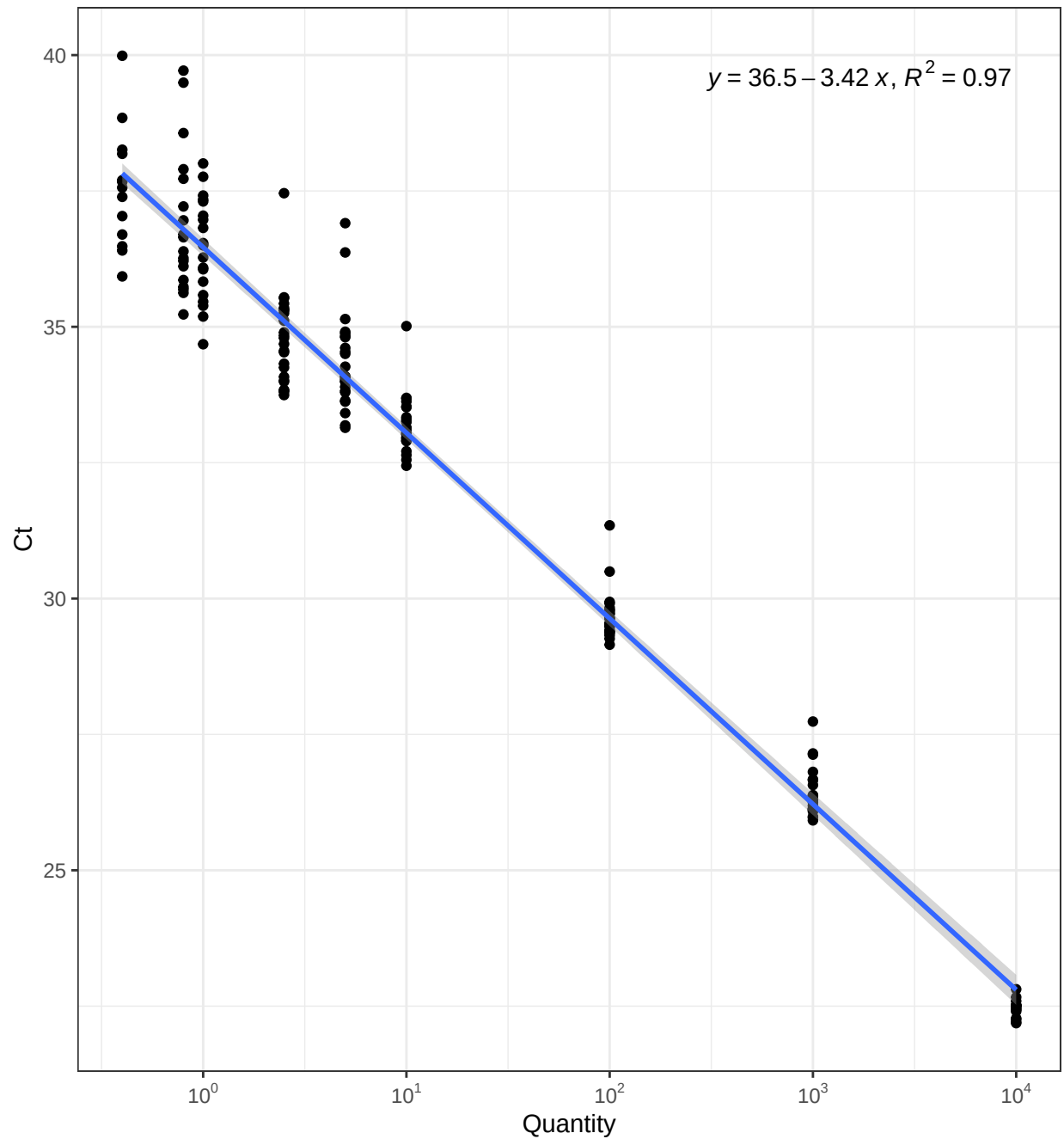
      "x" = "Slope"))

# Extract the R^2 value from the 'r.squared' column
r_squared_value_run5 <- grob_data_run5$r.squared

results_data_cleaned %>%
  filter(., `Target Name`==Target_name) %>%
  filter(., Task=="STANDARD") %>%
  mutate(., CT=as.numeric(CT)) -> pycno_results_standard

pycno_results_standard %>%
  ggplot(., aes(x=`Quantity`, y = CT)) +geom_point() +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
    label.y = "top") +theme_bw() +ylab("Ct")

```



FHL Analysis Transplanted Individual Detection

Merge Sample Metadata

```
sample_metadata<- read_excel(here("raw_data","qPCR_raw_data","ElectronicArchive_CruiseSampleMetadata_20200601.xlsx"))
results_data_cleaned %>%
  filter(.,`Target Name`=="Pycno") %>%
```

```

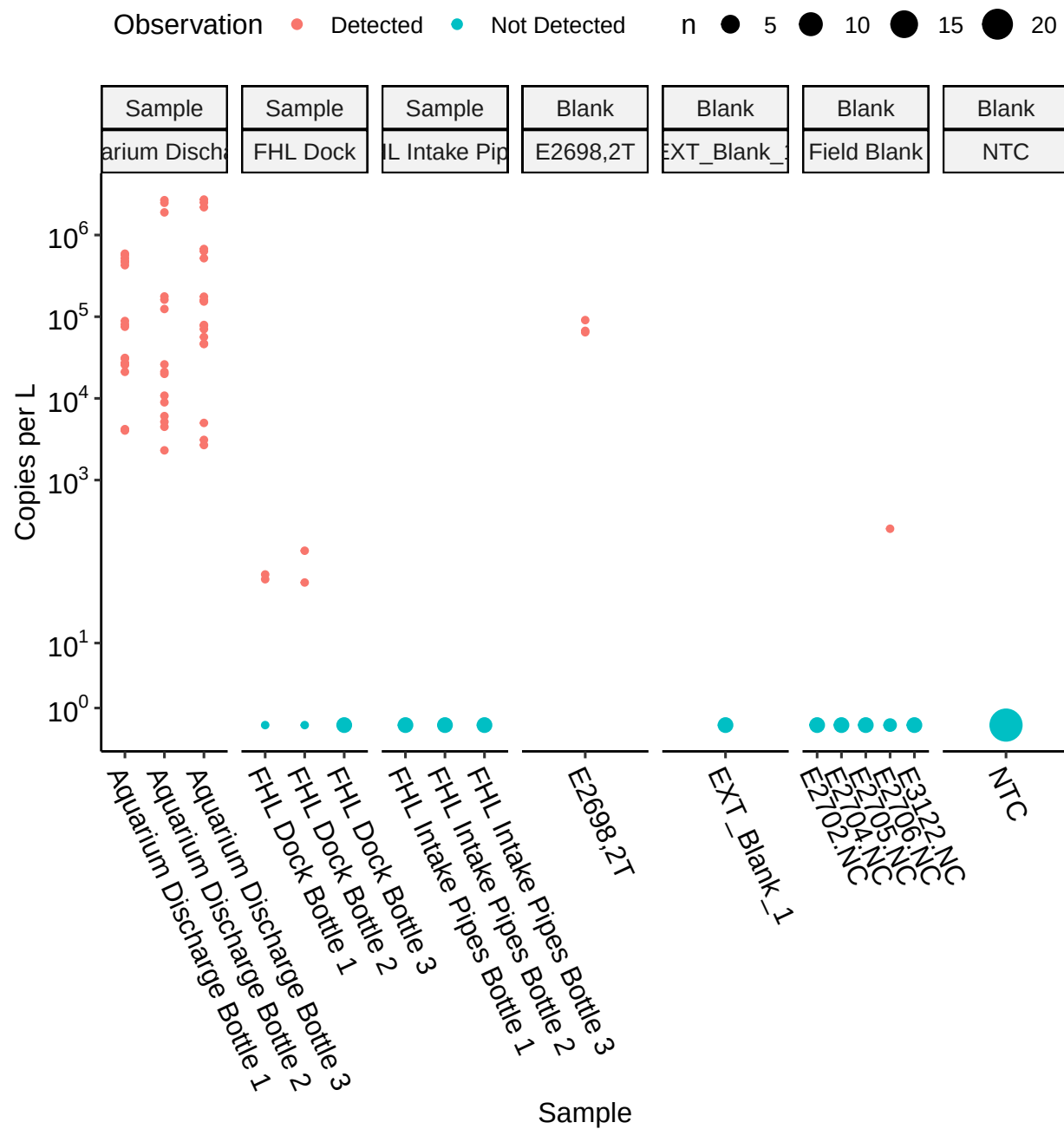
filter(., !str_detect(`Task`, "STANDARD")) %>%
mutate(., Quantity = if_else(is.na(Quantity), 0, Quantity)) %>%
mutate(., Concentration = (Quantity*10*(100/2)/ 1)) %>% # Quantity is the concentration generated by
mutate(., Observation = if_else(Quantity ==0, "Not Detected", "Detected")) %>%
mutate(., `Sample Name` = if_else(str_detect(`Sample Name`, "Negative Control"), "NTC", `Sample Name`)) %>%
ungroup() %>%
left_join(sample_metadata, by=c("Sample Name"="Sample_Name")) %>%
mutate(., `Sample Name` = if_else(`Sample Name`=="Ext_blank_UWFH25", "Extraction Blank", `Sample Name`)) %>%
mutate(., Negative_control = if_else(is.na(Negative_control), TRUE, Negative_control)) %>%
group_by(`Sample Name`) %>%
mutate(Technical_Replicate = row_number(`Sample Name`)) %>%
mutate(., Station = if_else(is.na(Station) & Negative_control==TRUE | Task=="NTC", `Sample Name`, Station)) %>%
mutate(., Name_plotting = str_c(Station, " Bottle ", Biological_Replicate, " PCR ", Technical_Replicate)) %>%
mutate(., Name_plotting = if_else(is.na(Name_plotting), str_c(`Sample Name`, " PCR ", Technical_Replicate))) %>%
mutate(., Name_plotting_b = str_c(Station, " Bottle ", Biological_Replicate)) %>%
mutate(., Name_plotting_b = if_else(is.na(Name_plotting_b), `Sample Name`, Name_plotting_b)) %>%
mutate(., Type = if_else(is.na(Station), "Extraction Blank", Station)) %>%
mutate(., Type = if_else(str_detect(Type, ".NC"), "Field Blank", Type)) %>%
mutate(., Sample_Type = if_else(Negative_control==FALSE, "Sample", "Blank")) -> pycno_sample_data_4_plotting

```

```

pycno_sample_data_4_plotting %>%
mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
ggplot(., aes(x=Name_plotting_b, y=Concentration, color=Observation)) + geom_count() +
scale_y_continuous(
trans = scales::pseudo_log_trans(base = 10),
breaks = c(1,10,1000,10000,100000,1000000),
labels = scales::trans_format("log10", scales::math_format(10^.x))
) +theme_pubr()+theme(axis.text.x = element_text(angle = -65, vjust = 1, hjust=0)) +facet_wrap(

```



```
# pycno_sample_data_4_plotting_run1 %>%
#   filter(., str_detect(Name_plotting_b, "Aquarium")) %>%
#   group_by(`Target Name`) %>%
#   dplyr::summarise(mean(Concentration))
```

Figure 6a2

```
pycno_sample_data_4_plotting$Type2 <- pycno_sample_data_4_plotting$Type
pycno_sample_data_4_plotting$Type2 <- as.factor(pycno_sample_data_4_plotting$Type2)
```



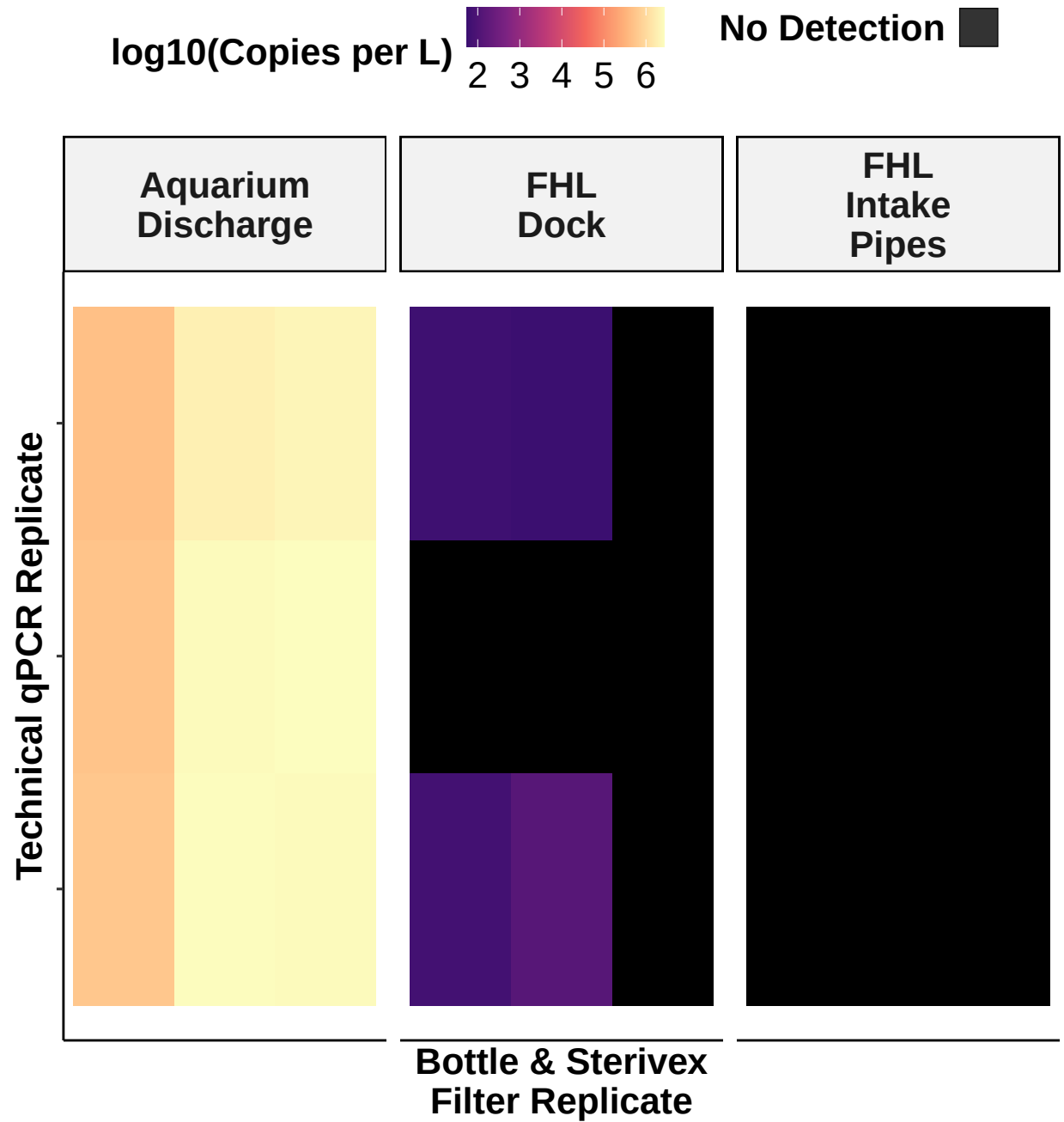
```

levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "Aquarium Disch"] <- "FHL Intake Pipe"
levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "FHL Dock"] <- "FHL Intake Pipe"
levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "FHL Intake Pipe"] <- "FHL Dock"

pycno_sample_data_4_plotting %>%
  filter(., str_detect(Field_collection_method,"Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
  mutate(across(Sample_Type, ~factor(., levels=c("Sample","Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., Concentration=if_else(Concentration==0,NA,Concentration) ) %>%
  mutate(., Plate = "5")-> plate_5_data_analysis

plate_5_data_analysis %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=log10(Concentration), color="")) + geom_tile() + theme_minimal() +
  guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +
  labs(fill = "log10(Copies per L)") -> a2
a2

```



```
ggsave(filename = here("Figures","FHL_mesocosm_field_trial_run5.png"),
        width=10,
        height = 4,
        dpi = 400,
        units = c("in"))
```

Figure s6a2

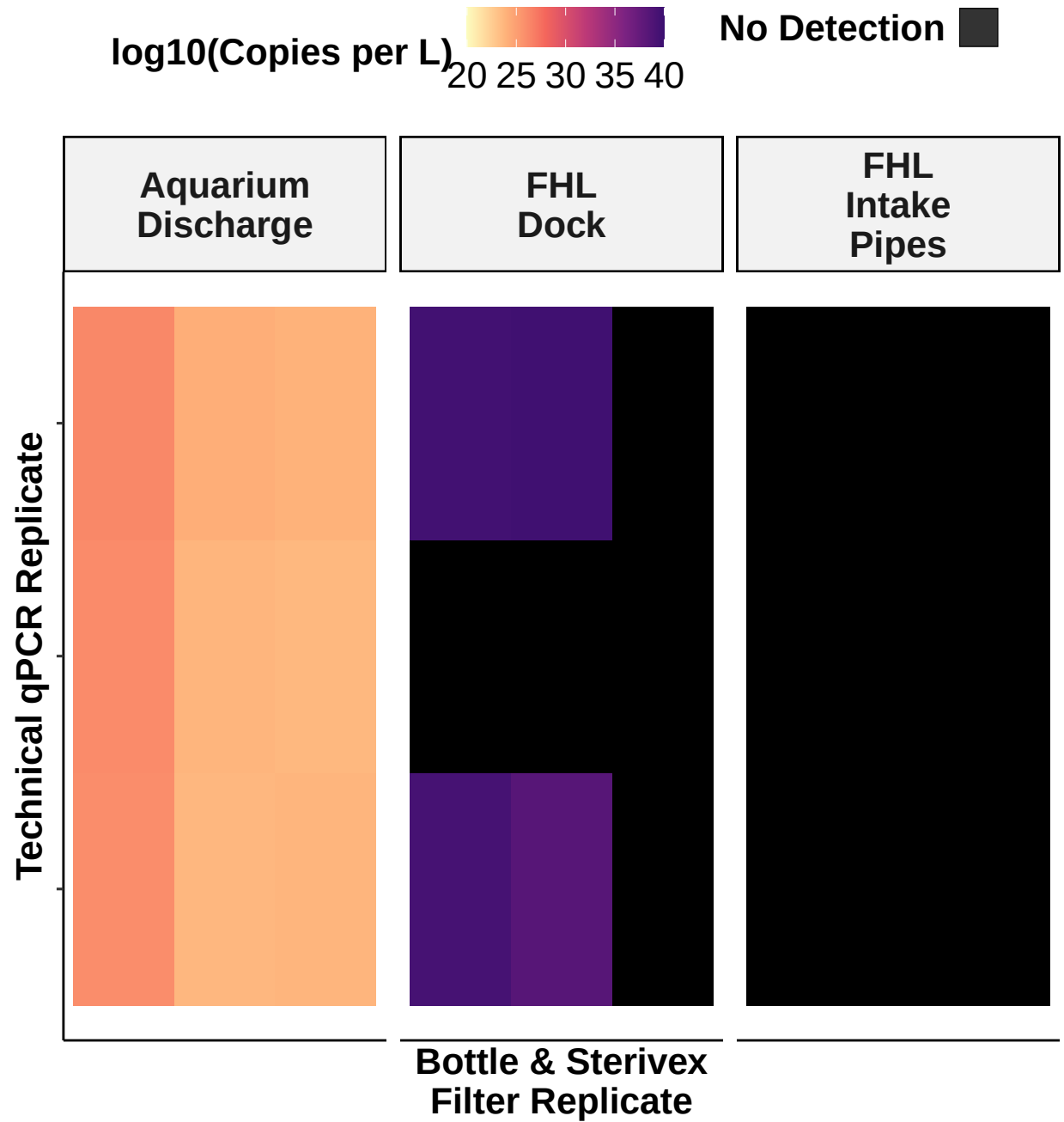
pycno_sample_data_4_plotting

```
## # A tibble: 114 x 62
## # Groups:   Sample Name [31]
##   Well 'Well Position' Omit 'Sample Name' 'Target Name' Task Reporter
##   <dbl> <chr>          <lgl> <chr>          <chr>          <chr> <chr>
## 1 14 A14                FALSE E2697.1T      Pycno          UNKNOWN FAM
## 2 15 A15                FALSE E2697.1T      Pycno          UNKNOWN FAM
## 3 16 A16                FALSE E2697.1T      Pycno          UNKNOWN FAM
## 4 17 A17                FALSE E2698.1T      Pycno          UNKNOWN FAM
## 5 18 A18                FALSE E2698.1T      Pycno          UNKNOWN FAM
## 6 19 A19                FALSE E2698.1T      Pycno          UNKNOWN FAM
## 7 22 A22                FALSE NTC          Pycno          NTC FAM
## 8 23 A23                FALSE NTC          Pycno          NTC FAM
## 9 24 A24                FALSE NTC          Pycno          NTC FAM
## 10 38 B14               FALSE E2697.2T      Pycno          UNKNOWN FAM
## # i 104 more rows
## # i 55 more variables: Quencher <chr>, CT <chr>, 'Ct Mean' <dbl>,
## # 'Ct SD' <dbl>, Quantity <dbl>, 'Quantity Mean' <dbl>, 'Quantity SD' <dbl>,
## # 'Y-Intercept' <dbl>, 'R(superscript 2)' <dbl>, Slope <dbl>,
## # Efficiency <dbl>, 'Automatic Ct Threshold' <lgl>, 'Ct Threshold' <dbl>,
## # 'Automatic Baseline' <lgl>, 'Baseline Start' <dbl>, 'Baseline End' <dbl>,
## # 'Amp Status' <chr>, Comments <lgl>, 'Cq Conf' <dbl>, HIGHSD <chr>, ...
```

```
pycno_sample_data_4_plotting$Type2 <- pycno_sample_data_4_plotting$Type
pycno_sample_data_4_plotting$Type2 <- as.factor(pycno_sample_data_4_plotting$Type2)
levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "Aquarium Discl]
levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "FHL Dock"] <-
levels(pycno_sample_data_4_plotting$Type2)[levels(pycno_sample_data_4_plotting$Type2) == "FHL Intake Pi]

pycno_sample_data_4_plotting %>%
  filter(., str_detect(Field_collection_method,"Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
  mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., CT=as.numeric(CT)) %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=CT, color="")) + geom_tile() +theme_pubr()+theme(a
  guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +
  labs(fill = "log10(Copies per L)") -> sa2

sa2
```



```
ggsave(filename = here("Figures", "FHL_mesocosm_field_trial_Ct_run5.png"),
        width=10,
        height = 4,
        dpi = 400,
        units = c("in"))
```

Here we see ~2 million copies per L of Pycnopodia target DNA in the Friday Harbor Aquarium discharge water. The discharge is from a flow through system with 39 individual stars of at least 40 cm in diameter.

We were able to detect the 10 transplanted stars from at least one technical replicate from each of the three surface eDNA samples taken ~5m above the individuals. We only detected a single individual located at

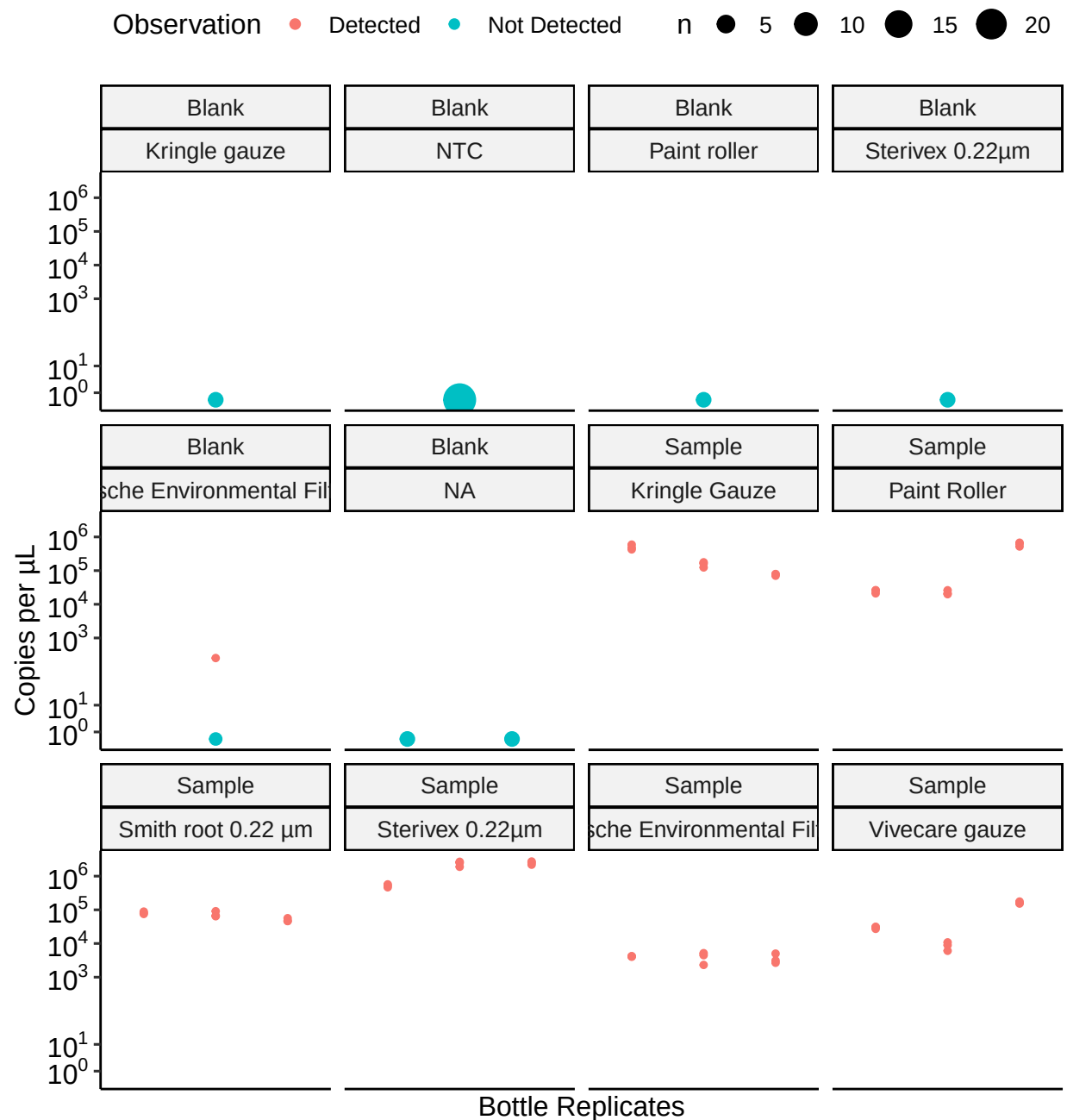
10m depth from one of three surface samples taken at the Friday Harbor Lab intake pipes.

These results demonstrate that the eDNA qPCR assay can detect Pycnopodia from both mesocosms and the field.

FHL Analysis Passive Versus Active Filtration

```
results_data_cleaned %>%
  filter(., `Target Name`=="Pycno") %>%
  filter(., !str_detect(`Task`, "STANDARD")) %>%
  mutate(., `Sample Name`=str_replace(`Sample Name`, ",", ".")) %>%
  left_join(sample_metadata, by=c("Sample Name"="Sample_Name")) %>%
  group_by(`Sample Name`) %>%
  mutate(., Negative_control = if_else(is.na(Negative_control), TRUE, Negative_control)) %>%
  mutate(Technical_Replicate = row_number(`Sample Name`)) %>%
  mutate(., Station = if_else(is.na(Station) & Negative_control=="TRUE" | Task=="NTC", `Sample Name`, S
  mutate(., Name_plotting = str_c(Station, " ", Field_collection_method, " ", Biological_Replicate, " ", Tec
  mutate(., Name_plotting = if_else(is.na(Name_plotting), str_c(`Sample Name`, " ", Technical_Replicate),
  mutate(., Name_plotting_b = str_c(Station, " ", Field_collection_method, " ", Biological_Replicate)) %>%
  mutate(., Name_plotting_b = if_else(is.na(Name_plotting_b), `Sample Name`, Name_plotting_b)) %>%
  mutate(., Quantity=if_else(is.na(Quantity), 0, Quantity)) %>%
  mutate(., Concentration_L = (Quantity*10*(100/2)/ 1)) %>% # Quantity is the concentration generated b
  mutate(., Observation = if_else(Quantity ==0, "Not Detected", "Detected")) %>%
  mutate(., Type = if_else(is.na(Station), "Extraction Blank", Station )) %>%
  mutate(., Type = if_else(str_detect(Type, ".NC"), "Field Blank", Type )) %>% mutate(., Sample_Type = i
  mutate(., Field_collection_method = if_else(Task=="NTC", "NTC", Field_collection_method)) %>%
  mutate(., Field_collection_method = if_else(Type=="Extraction Blank", "Extraction Blank", Field_collection

pycno_sample_data_4_plotting %>%
  filter(., !str_detect(Name_plotting_b, "FHL")) %>%
  ggplot(., aes(x=Name_plotting_b, y=Concentration_L, color=Observation)) + geom_count() +
  scale_y_continuous(
    trans = scales::pseudo_log_trans(base = 10),
    breaks = c(1,10,1000,10000,100000,1000000,10000000),
    labels = scales::trans_format("log10", scales::math_format(10^.x))
  ) +theme_pubr()+theme(axis.text.x = element_text(angle = -65, vjust = 1, hjust=-0.05)) +facet_wrap
```



We had minor contamination in the passive filter. It's 2 orders of magnitude less than any of the samples so not a major concern. Currently not adjusting the values here, but could subtract this value from all observed positive results.

```

pycno_sample_data_4_plotting %>%
  filter(., !str_detect(Name_plotting_b, "FHL")) %>%
  filter(., Sample_Type!="Blank") %>%
  mutate(., Method= case_when(str_detect(Field_collection_method,"Sterivex")~"Active Filtration",str_de
    TRUE~"Passive Filtration"
  )) ->pycno_sample_data_4_plotting_fm

```

Kruskal Wallace test

```
# Perform one-way ANOVA
kw_test_results <- kruskal.test(Concentration_L ~ Field_collection_method, pycno_sample_data_4_plotting_fm$Concentration_L, pycno_sample_data_4_plotting_fm$Field_collection_method)

# Perform post-hoc Tukey HSD test
pairwise.wilcox.test(pycno_sample_data_4_plotting_fm$Concentration_L, pycno_sample_data_4_plotting_fm$Field_collection_method,
  p.adjust.method = "bonferroni") -> pairwise_results

full_p_matrix <- fullPTable(pairwise_results$p.value)
multcompLetters(full_p_matrix) -> cld

cld_tibble_optionA <- as_tibble(cld$Letters, rownames = "group") %>%
  dplyr::select(Field_collection_method=group, Letters=value)

pairwise_results
```

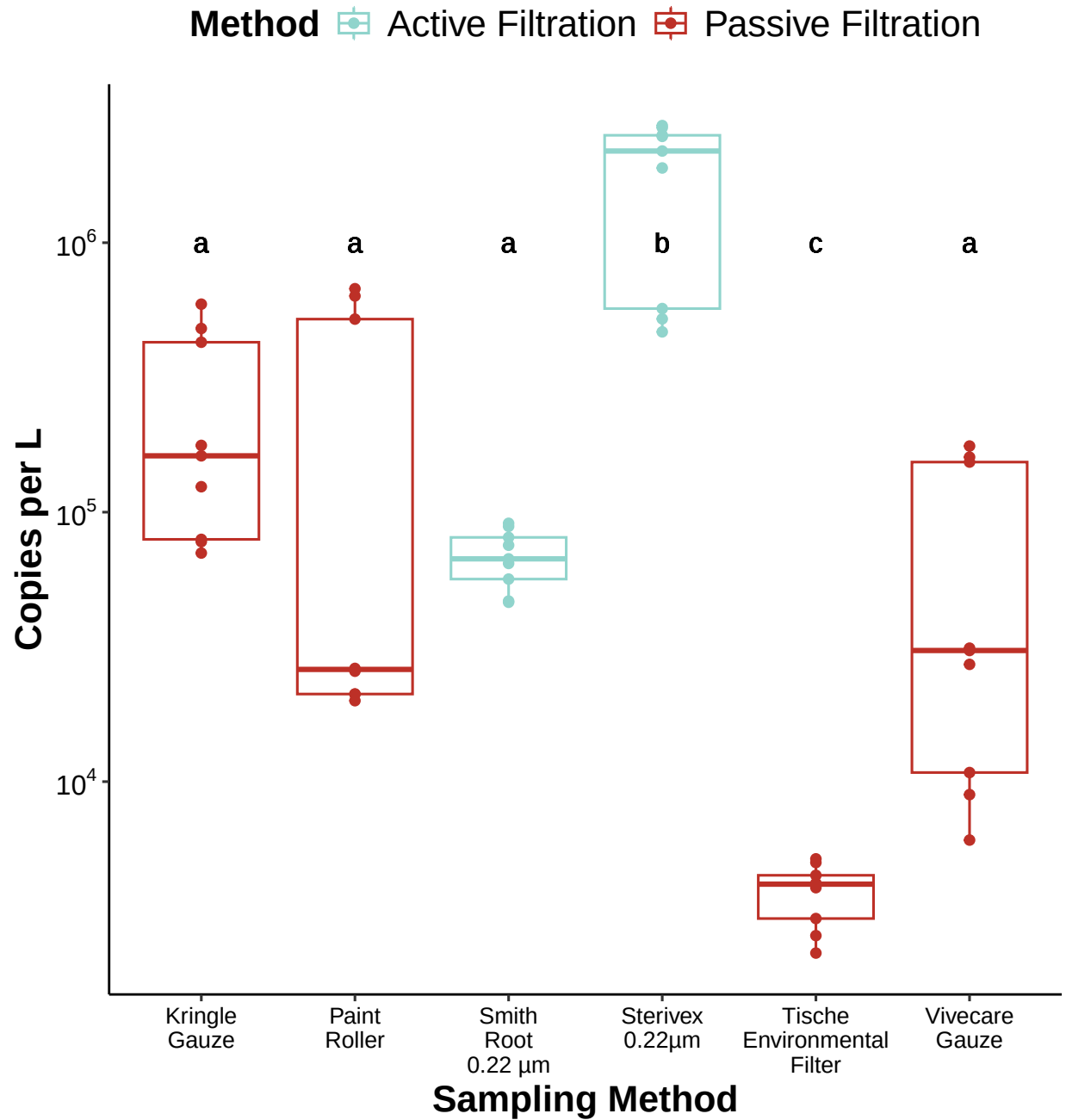
```
##
## Pairwise comparisons using Wilcoxon rank sum exact test
##
## data: pycno_sample_data_4_plotting_fm$Concentration_L and pycno_sample_data_4_plotting_fm$Field_collection_method
##
##          Kringle Gauze Paint Roller Smith root 0.22 µm
## Paint Roller          1.00000      -          -
## Smith root 0.22 µm    0.08453      1.00000      -
## Sterivex 0.22µm       0.00740      0.02777      0.00062
## Tische Environmental Filter 0.00062      0.00062      0.00062
## Vivecare gauze       0.21288      1.00000      1.00000
##          Sterivex 0.22µm Tische Environmental Filter
## Paint Roller          -          -
## Smith root 0.22 µm    -          -
## Sterivex 0.22µm       -          -
## Tische Environmental Filter 0.00062      -
## Vivecare gauze       0.00062      0.00062
##
## P value adjustment method: bonferroni
```

```
pycno_sample_data_4_plotting_fm$Field_collection_method2 <- pycno_sample_data_4_plotting_fm$Field_collection_method
pycno_sample_data_4_plotting_fm$Field_collection_method2 <- as.factor(pycno_sample_data_4_plotting_fm$Field_collection_method2)
levels(pycno_sample_data_4_plotting_fm$Field_collection_method2)[levels(pycno_sample_data_4_plotting_fm$Field_collection_method2) == "Paint Roller"] <- "K"
levels(pycno_sample_data_4_plotting_fm$Field_collection_method2)[levels(pycno_sample_data_4_plotting_fm$Field_collection_method2) == "Smith root 0.22 µm"] <- "S"
levels(pycno_sample_data_4_plotting_fm$Field_collection_method2)[levels(pycno_sample_data_4_plotting_fm$Field_collection_method2) == "Sterivex 0.22µm"] <- "St"
levels(pycno_sample_data_4_plotting_fm$Field_collection_method2)[levels(pycno_sample_data_4_plotting_fm$Field_collection_method2) == "Tische Environmental Filter"] <- "T"
levels(pycno_sample_data_4_plotting_fm$Field_collection_method2)[levels(pycno_sample_data_4_plotting_fm$Field_collection_method2) == "Vivecare gauze"] <- "V"

pycno_sample_data_4_plotting_fm %>%
  left_join(cld_tibble_optionA) %>%
  ggplot(., aes(x=Field_collection_method2, y=Concentration_L)) +
  geom_boxplot(aes(color=Method)) +
  geom_point(aes(color=Method)) +
```

```
scale_y_continuous(
  trans = scales::pseudo_log_trans(base = 10),
  breaks = c(1,10,1000,10000,100000,1000000,10000000,100000000),
  labels = scales::trans_format("log10", scales::math_format(10^.x))
) +theme_pubr() + ylab("Copies per L") +xlab("Sampling Method") +scale_color_manual(values=wes_pa
```

a1




```
ggsave(filename = here("Figures", "Figure_S4_FHL_mesocosm_passive_active_comparison.png"),
        width=10,
        height = 6,
        dpi = 400,
        units = c("in"))
```

```
pycno_sample_data_4_plotting %>%
  filter(., !str_detect(Name_plotting_b, "FHL")) %>%
  filter(., Sample_Type!="Blank") %>%
  mutate(., Method= case_when(str_detect(Field_collection_method, "Sterivex")~"Active Filtration", str_detect(
    TRUE~"Passive Filtration"
  )) %>%
  group_by(Field_collection_method, Method) %>%
  dplyr::summarize(Max=max(Concentration_L), Min=min(Concentration_L), Mean=mean(Concentration_L), median=
    IQR = IQR(Concentration_L)) %>% left_join(cld_tibble_optionA) %>% kable(digits=2)
```

Field_collection_method	Method	Max	Min	Mean	median	IQR	Letters
Kringle Gauze	Passive Filtration	591429.81	70507.28	243273.57	161825.16	347914.58	a
Paint Roller	Passive Filtration	674890.14	19967.89	218885.19	26087.71	499322.37	a
Smith root 0.22 µm	Active Filtration	90907.27	46306.81	68512.41	67092.76	24153.36	a
Sterivex 0.22µm	Active Filtration	2720547.61	466701.11	1779847.76	2188961.67	1934360.78	b
Tische Environmental Filter	Passive Filtration	5168.52	2310.91	3906.84	4163.78	1387.46	c
Vivacare gauze	Passive Filtration	175940.34	6068.63	67154.02	30661.80	142606.52	a

Here we see that Sterivex 0.22 µm filters captured nearly an order of magnitude more Pycnopodia DNA than either Smith Root active filters or any of the passive filters deployed here. This is notable given that Sterivex have a smaller surface area (10cm²) than the Smith Root 47mm filter (17.3 cm²) and suggests that differences in the housing or sample integrity are driving these apparent differences. These results highlight that passive filters can be used for field applications to detect Pycnopodia, albeit at a significant disadvantage in that even the best performing materials captured less than a 10th of the DNA captured by active sterivex filtration methods.

Load Data Pycno qPCR Plate 1

```
sample_set_up_run1 <- read_excel(here("raw_data", "qPCR_raw_data", "2025-05-23_132340_Pycno_test_1.xls"),
amplification_data_run1 <- read_excel(here("raw_data", "qPCR_raw_data", "2025-05-23_132340_Pycno_test_1.xls"),
results_data_run1<- read_excel(here("raw_data", "qPCR_raw_data", "2025-05-23_132340_Pycno_test_1.xls"), sl
```

Merge Tables

```

amplification_data_run1 %>%
  left_join(sample_set_up_run1) %>%
  mutate(., `Sample Name` = if_else(is.na(`Sample Name`),str_c(`Target Name`,`_`,Task,"_", Quantity), `Sample Name`))
  mutate(., `Sample Name` = if_else(is.na(`Sample Name`),"Negative Control", `Sample Name` ))-> combined_qPCR_data_run1

```

Now we are ready to start analyzing the data.

First thing we want to do is check to see how our internal positive control is working.

Moa Map

```

columns = seq(from =1, to=24, by=1)
columns_t= as.tibble(columns) %>% dplyr::rename("Column"=value)
rows = capitalize(letters[1:16])
rows_t= as.tibble(rows) %>% dplyr::rename("Row"=value)

cross_join(columns_t, rows_t) -> full_plate

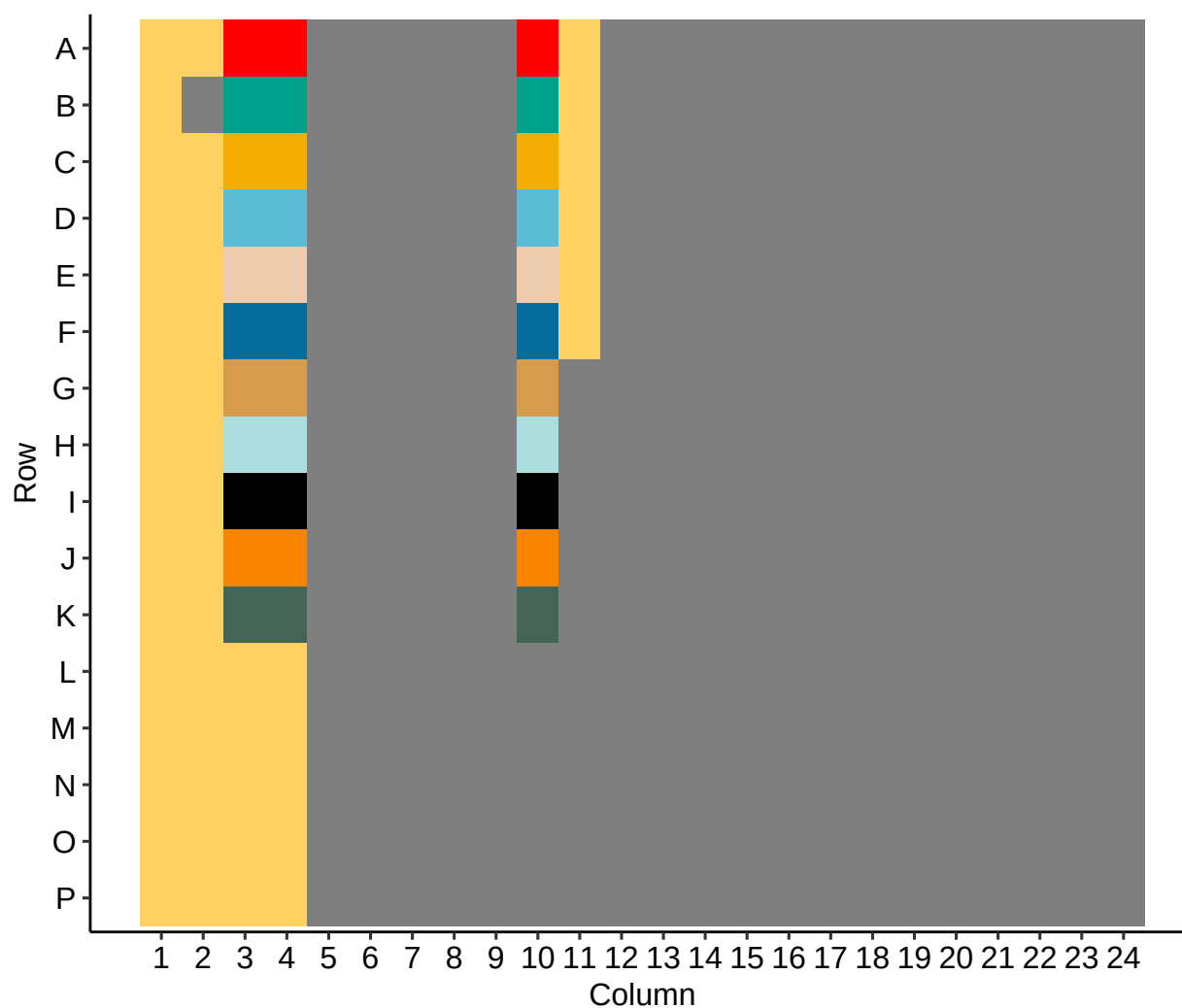
combined_qPCR_data_run1 %>%
  filter(., `Target Name`=="Moa-IPC") %>%
  mutate(., Row = str_sub(`Well Position`,start = 1L, end = 1L),
         Column= as.numeric(str_sub(`Well Position`,start = 2L, end = -1L ))) -> moa_samples_run1

full_plate %>%
  left_join(moa_samples_run1) -> moa_plate_run1

moa_plate_run1 %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x=Column, y=fct_rev(Row), fill=`Sample Name` )) +geom_tile() +scale_x_continuous(breaks=columns)

```

Moa-IPC Standards & Samples



Pycno Map

```
combined_qPCR_data_run1 %>%
  filter(., `Target Name`=="Pycno") %>%
  mutate(., Row = str_sub(`Well Position`,start = 1L, end = 1L),
         Column= as.numeric(str_sub(`Well Position`,start = 2L, end = -1L ))) -> pycno_samples_run1

full_plate %>%
  left_join(pycno_samples_run1) -> pycno_plate_run1
```

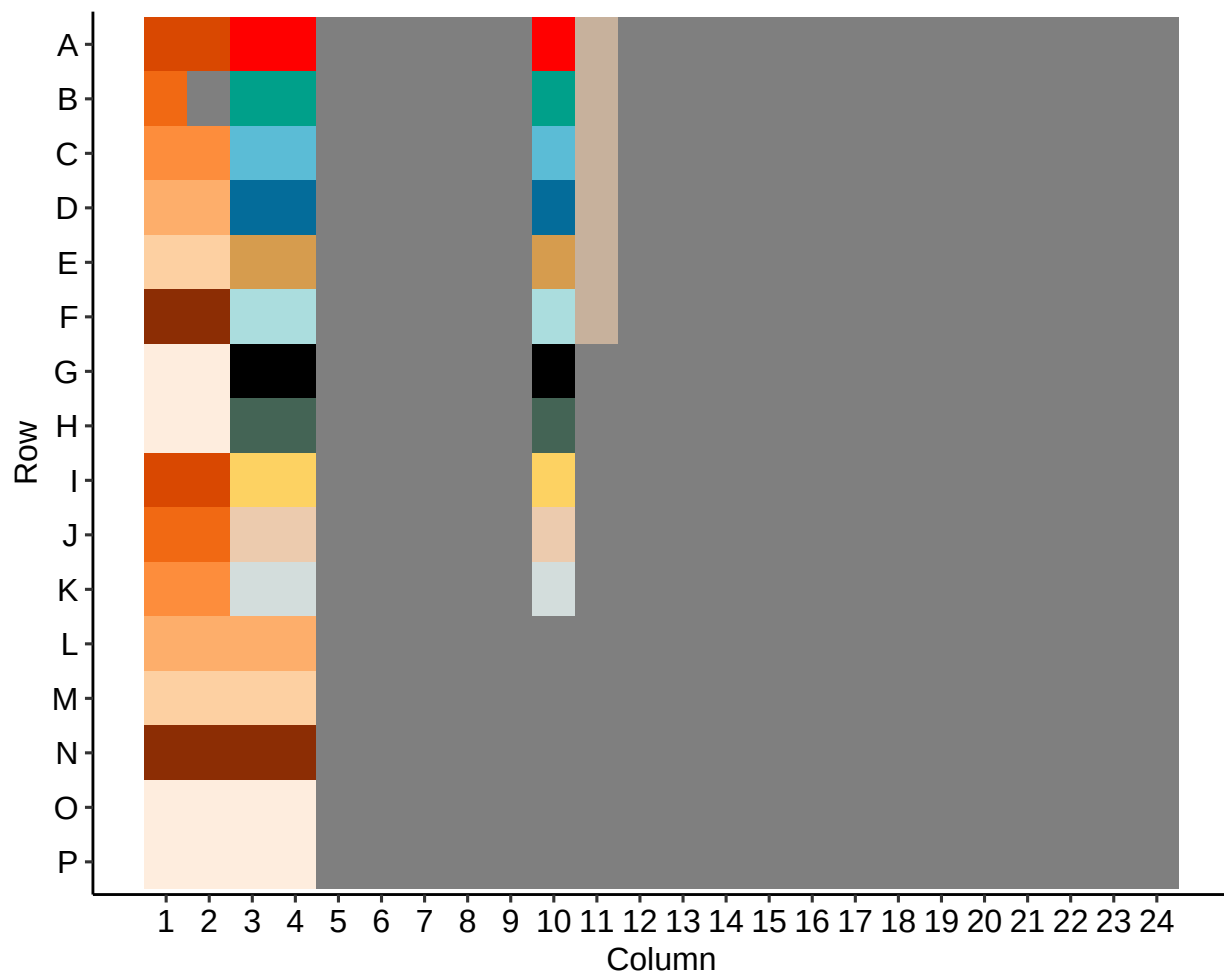
```

pycno_plate_run1 %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x=Column, y=fct_rev(Row), fill=`Sample Name` )) +geom_tile() +scale_x_continuous(breaks=col

```

Pycno Standards & Samples

E2221.1T		E2234.1T.DockPositive		E2235.2T.IntakeNeg		Pycno_STANDARD_1	
E2221.2T		E2234.2T.DockPositive		E2235.3T.IntakeNeg		Pycno_STANDARD_10	
E2221.3T		E2234.3T.DockPositive		Ext_blank_UWFH25		Pycno_STANDARD_100	
E2228.NC.DI		E2235.1T.IntakeNeg		Negative Control		Pycno_STANDARD_1000	

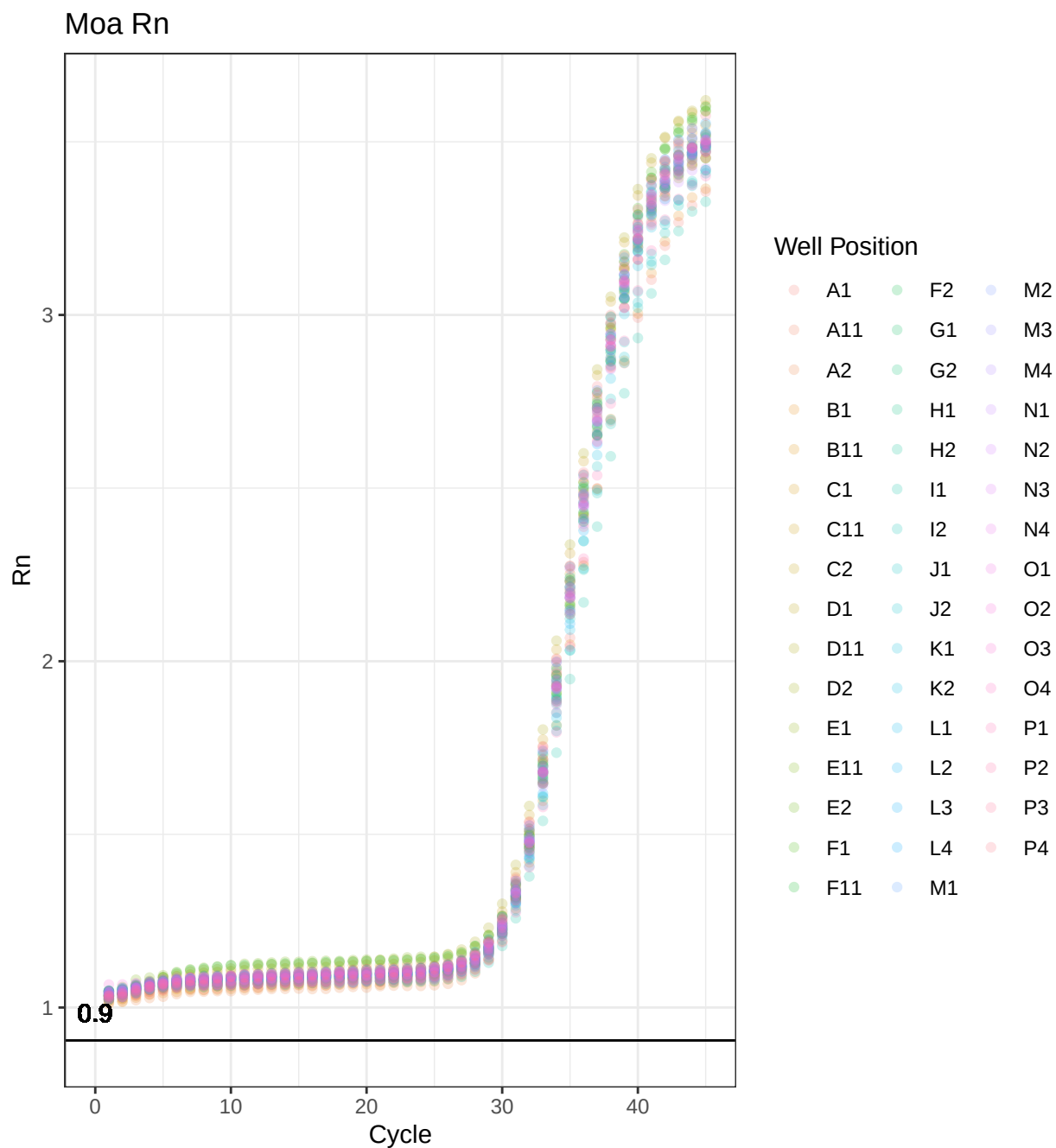


Moa Rn

```
combined_qPCR_data_run1 %>%
filter(., str_detect(`Sample Name`, "Moa-IPC_STANDARD")) -> moa_standard_run1

moa_standard_run1$Rn %>% max() -> max_Rn_moa_run1
h_moa_run1= max_Rn_moa_run1*0.25

moa_standard_run1 %>%
ggplot(., aes(y=Rn, x=Cycle, color=`Well Position`)) +
  geom_point(alpha=0.2) + geom_hline(yintercept =h_moa_run1) +
  geom_text(aes(0,h_moa_run1,label = round(h_moa_run1,2), vjust = -1),color="black") +theme_bw() +ggtitle("Moa Rn")
```



There is quite a bit of spread in our IPC. Will remove any samples outside normal distribution. #####
 Remove Samples with Anomolous Outlier IPC values

```
moa_standard_run1 %>%
  filter(., Cycle ==40) -> moa_standard_c40_run1

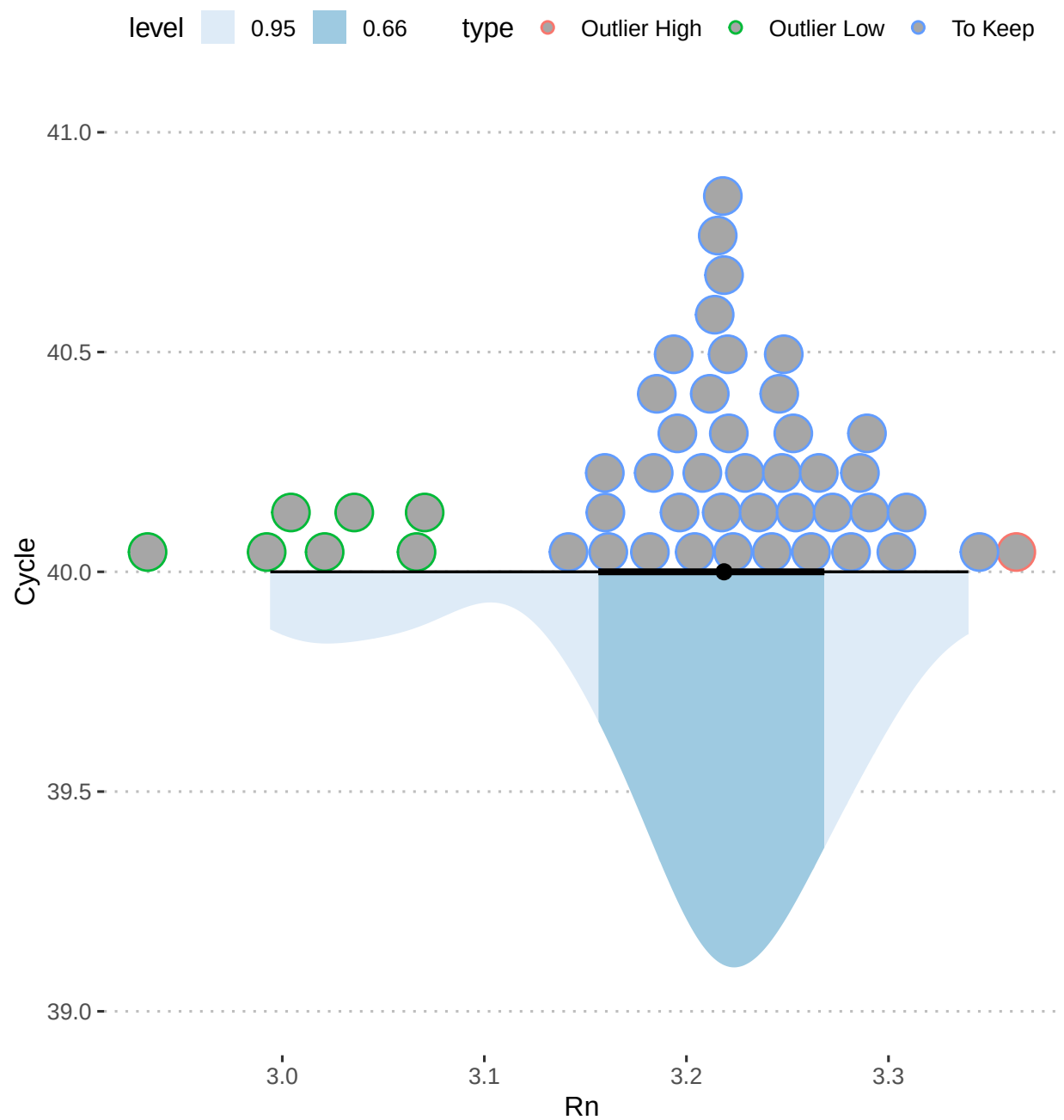
moa_standard_c40_run1 %>%
  dplyr::summarise(mean_Rn = ci(Rn, confidence = 0.99)[1],
                    lower_ci_mpg = ci(Rn, confidence = 0.99)[2],
                    upper_ci_mpg = ci(Rn, confidence = 0.99)[3],
                    sd_Rn = ci (Rn, confidence = 0.99)[4]) -> CIO.95_moa_run1

Sum_run1 = groupwiseMean(Rn ~ Cycle, data=moa_standard_c40_run1, conf = 0.99)

moa_standard_c40_run1 %>%
  summarise(
    Q1 = quantile(Rn, 0.25, na.rm = TRUE),
    Q3 = quantile(Rn, 0.75, na.rm = TRUE),
    IQR = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR,
    upper_bound = Q3 + 1.5 * IQR
  ) -> moa_iqr_run1

moa_standard_c40_run1 %>%
  mutate(., type=case_when(Rn > moa_iqr_run1$upper_bound ~"Outlier High",
                           Rn < moa_iqr_run1$lower_bound ~"Outlier Low",
                           TRUE~"To Keep")) -> moa_standard_c40_run1

moa_standard_c40_run1 %>%
  ggplot(., aes(x= Rn, y=Cycle)) + geom_dots(layout='weave', side = "top", aes(color=type)) + stat_slab
```



```
moa_standard_c40_run1 %>%
  filter(., type!="To Keep") -> moa_outliers_to_drop_run1
```

```
combined_qPCR_data_run1 %>%
  filter(., `Well Position` != moa_outliers_to_drop_run1$`Well Position`) -> combined_qPCR_data_cleaned
```

```
results_data_run1 %>%
  filter(., `Well Position` != moa_outliers_to_drop$`Well Position`) -> results_data_cleaned_run1
```

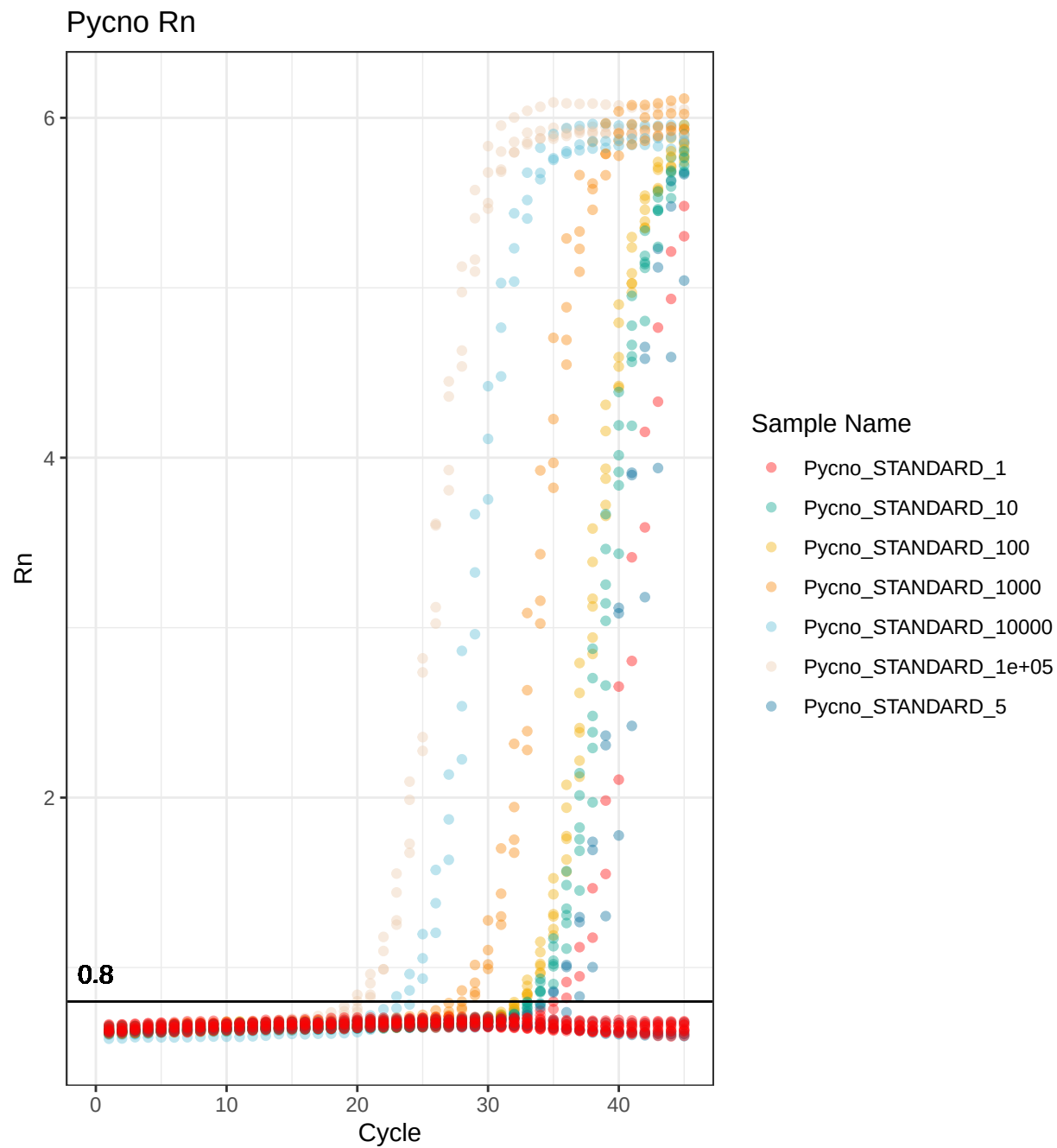
Filter Data

Pycno Rn

```
combined_qPCR_data_run1 %>%
  filter(., str_detect(`Sample Name`, "Pycno_STANDARD")) -> pycno_standard_run1

pycno_standard_run1$Rn %>% max() -> max_Rn_run1
h_run1= 0.8

pycno_standard_run1 %>%
  ggplot(., aes(y=Rn, x=Cycle, color=`Sample Name`)) +geom_point(alpha=0.4) + geom_hline(yintercept =h_run1)
  geom_text(aes(0,h_run1,label = round(h_run1,2), vjust = -1),color="black") +theme_bw() + ggtitle("Pycno")
```

```
pycno_standard_run1 %>%
  mutate(., Threshold = if_else(Rn > h_run1, "Above", "Below")) -> pycno_standard_run1

pycno_standard_run1 %>%
  group_by(`Well Position`, `Sample Name`) %>%
  filter(., rank(Threshold, ties.method="first")==1) %>%
  filter(., Threshold == "Above") -> pycno_standard_above_run1
```

```

pycno_standard_run1 %>%
  ungroup() %>%
  filter(., !`Well Position` %in% pycno_standard_above_run1$`Well Position`) %>%
  group_by(`Well Position`, `Sample Name`) %>%
  filter(., rank(Threshold, ties.method="last")==1) -> pycno_standard_below_run1

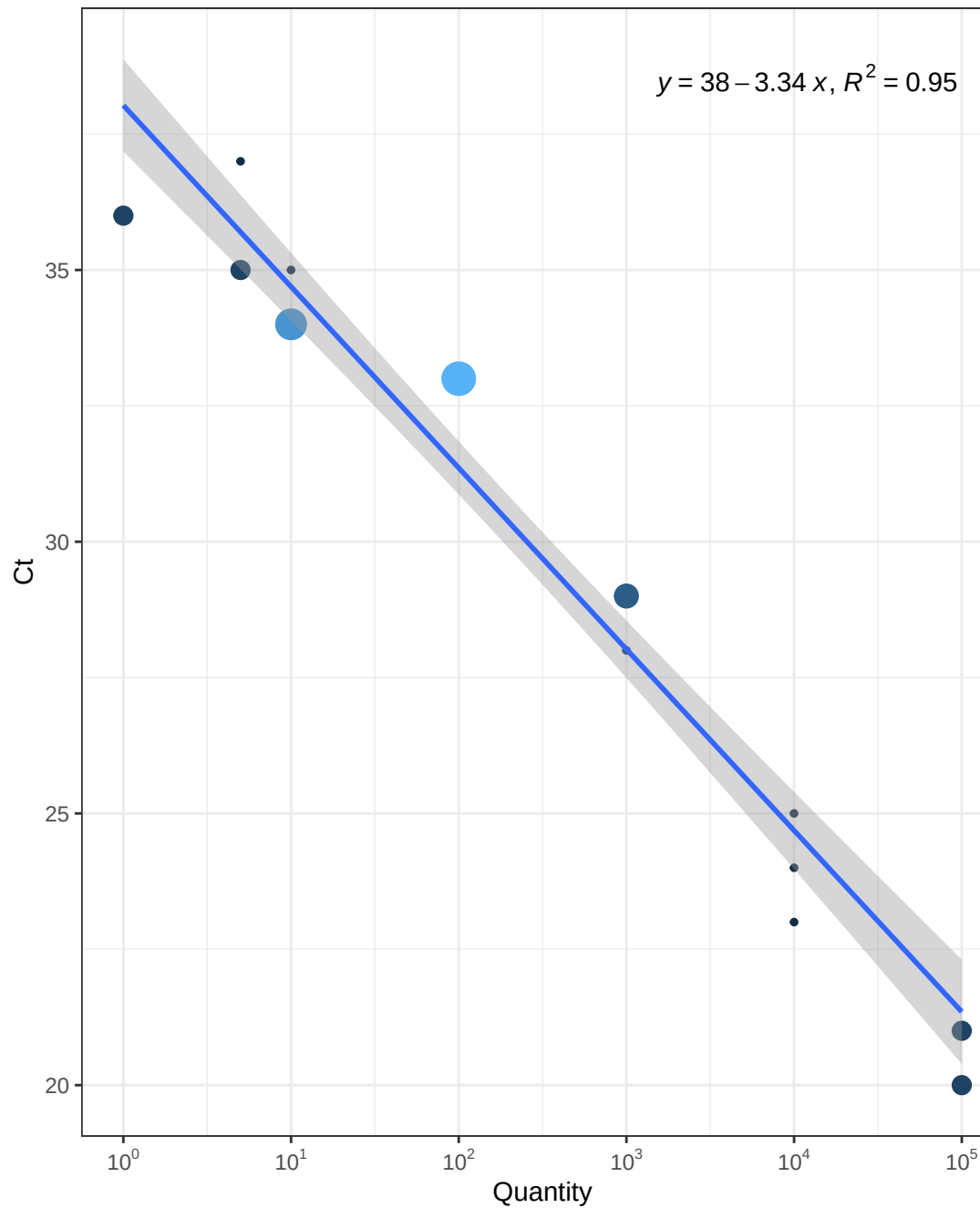
bind_rows(pycno_standard_above_run1, pycno_standard_below_run1) -> pycno_standard_threshold_run1

set_breaks = function(limits) {
  seq(limits[1], limits[2], by = 1)
}

pycno_standard_threshold_run1 %>%
  filter(., Cycle < 40 ) %>%
  ggplot(., aes(x=`Quantity`, y = Cycle)) +geom_count(aes(color = ..n.., size = ..n..)) +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
  label.y = "top") +theme_bw() +ylab("Ct") -> run1_ct_versus_quant

run1_ct_versus_quant

```



Pycno Ct versus Quantity

```
# View the structure of the generated plot object
# The computed data is stored within the plot object
grob_data_run1 <- ggplot_build(run1_ct_versus_quant)$data[[3]]

# The result is a list; unlist it to get a character vector
coef_char_run1 <- unlist(grob_data_run1$coefs)

standard_curve_coefs_run1 <- enframe(coef_char_run1, name = "Metric", value = "Value") %>%
  mutate( Metric = recode(Metric,
    "(Intercept)" = "Intercept",
```

```

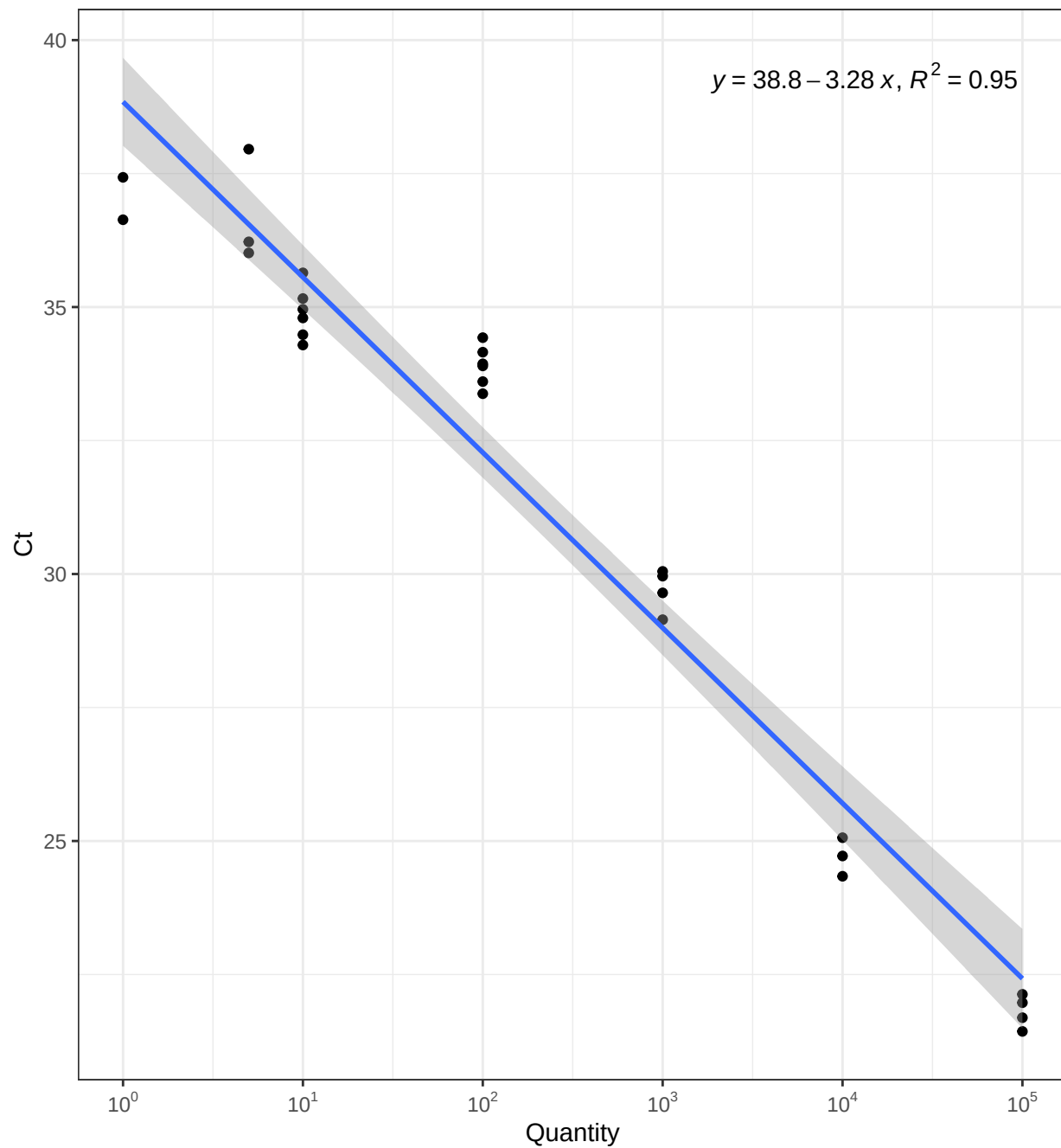
      "x" = "Slope"))

# Extract the R^2 value from the 'r.squared' column
r_squared_value_run1 <- grob_data_run1$r.squared

results_data_cleaned_run1 %>%
  filter(., `Target Name`==Target_name) %>%
  filter(., Task=="STANDARD") %>%
  mutate(., CT=as.numeric(CT)) -> pycno_results_standard_run1

pycno_results_standard_run1 %>%
  ggplot(., aes(x=`Quantity`, y = CT)) +geom_point() +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
    label.y = "top") +theme_bw() +ylab("Ct")

```



```
results_data_cleaned_run1 %>%
  filter(., `Target Name`==Target_name) %>%
  filter(., Task!="UNKNOWN") %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., Lab="NOAA PMEL OME") %>%
  dplyr::select(Target=`Target Name`, Lab, Cq=CT, SQ=Quantity) %>%
  mutate(., TemplateVolume=2,
    ReactionVolume=10) %>%
  mutate(., Cq = replace(Cq, Cq>40, NA)) -> qPCR_Pycno_1_standards

write.csv(qPCR_Pycno_1_standards, file = here("LOD_LOQ/qPCR_Pycno_1_standards.csv"), row.names = F)
```

FHL Analysis Transplanted Individual Detection

Merge Sample Metadata

```
sample_metadata<- read_excel(here("raw_data","qPCR_raw_data","ElectronicArchive_CruiseSampleMetadata_20180101.xlsx"))

results_data_cleaned_run1 %>%
  filter(., `Target Name`=="Pycno") %>%
  filter(., !str_detect(`Task`, "STANDARD")) %>%
  mutate(., Quantity=if_else(is.na(Quantity), 0, Quantity)) %>%
  mutate(., Concentration = (Quantity*10*(100/2)/ 1)) %>% # Quantity is the concentration generated by
  mutate(., Observation = if_else(Quantity ==0, "Not Detected", "Detected")) %>%
  mutate(., `Sample Name`= if_else(str_detect(`Sample Name`, "Negative Control"), "NTC", `Sample Name`)) %>%
  ungroup() %>%
  left_join(sample_metadata, by=c("Sample Name"="Field_Name")) %>%
  mutate(., `Sample Name` = if_else(`Sample Name`=="Ext_blank_UWFH25", "Extraction Blank", `Sample Name`)) %>%
  mutate(., Negative_control = if_else(is.na(Negative_control), TRUE, Negative_control)) %>%
  group_by(`Sample Name`) %>%
  mutate(Technical_Replicate = row_number(`Sample Name`)) %>%
  mutate(., Station = if_else(is.na(Station) & Negative_control==TRUE | Task=="NTC", `Sample Name`, Station)) %>%
  mutate(., Name_plotting = str_c(Station, " Bottle ", Biological_Replicate, " PCR ", Technical_Replicate)) %>%
  mutate(., Name_plotting = if_else(is.na(Name_plotting), str_c(`Sample Name`, " PCR ", Technical_Replicate), Name_plotting)) %>%
  mutate(., Name_plotting_b = str_c(Station, " Bottle ", Biological_Replicate)) %>%
  mutate(., Name_plotting_b = if_else(is.na(Name_plotting_b), `Sample Name`, Name_plotting_b)) %>%
  mutate(., Type = if_else(is.na(Station), "Extraction Blank", Station)) %>%
  mutate(., Type = if_else(str_detect(Type, ".NC"), "Field Blank", Type)) %>%
  mutate(., Sample_Type = if_else(Negative_control==FALSE, "Sample", "Blank")) -> pycno_sample_data_4_plotting_run1
```

Figure sa3

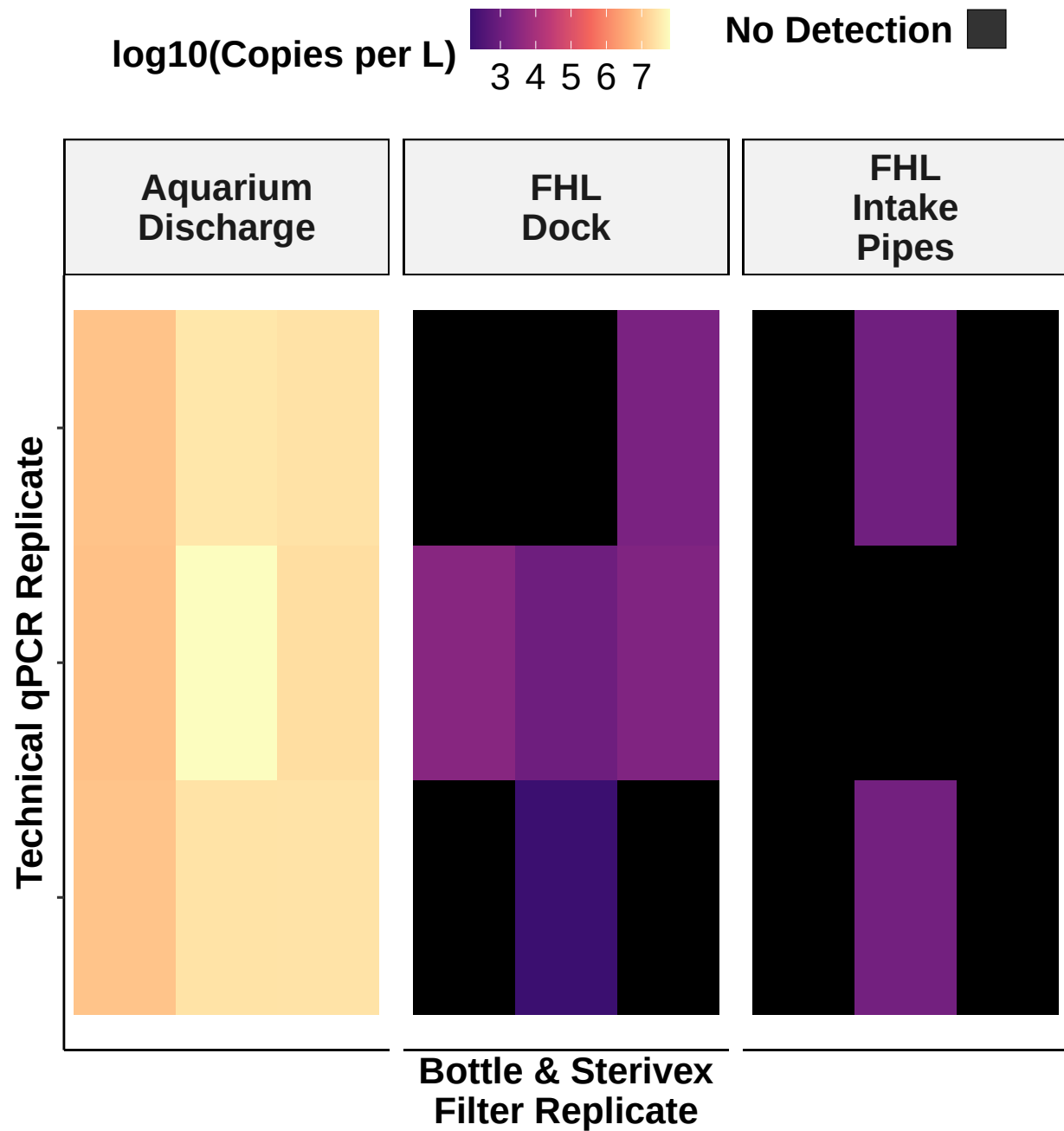
```
pycno_sample_data_4_plotting_run1$Type2 <- pycno_sample_data_4_plotting_run1$Type
pycno_sample_data_4_plotting_run1$Type2 <- as.factor(pycno_sample_data_4_plotting_run1$Type2)
levels(pycno_sample_data_4_plotting_run1$Type2)[levels(pycno_sample_data_4_plotting_run1$Type2) == "Aquatic"] <- "FHL"
levels(pycno_sample_data_4_plotting_run1$Type2)[levels(pycno_sample_data_4_plotting_run1$Type2) == "FHL"] <- "FHL"
levels(pycno_sample_data_4_plotting_run1$Type2)[levels(pycno_sample_data_4_plotting_run1$Type2) == "FHL"] <- "FHL"

pycno_sample_data_4_plotting_run1 %>%
  filter(., str_detect(Field_collection_method, "Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
  mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., Concentration=if_else(Concentration==0, NA, Concentration)) %>%
  mutate(., Plate = "1") -> plate_1_data_analysis

plate_1_data_analysis %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=log10(Concentration), color="")) + geom_tile() + theme_minimal()
```

```
guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +  
labs(fill = "log10(Copies per L)") -> sa3
```

sa3



```
ggsave(filename = here("Figures", "FHL_mesocosm_field_trial_run1.png"),  
width=10,  
height = 4,  
dpi = 400,  
units = c("in"))
```

```

pyno_sample_data_4_plotting_run1 %>%
  filter(., str_detect(Station, "Aquarium")) %>%
  ungroup() %>%
  dplyr::summarise(mean(Concentration), sd(Concentration))

```

```

## # A tibble: 1 x 2
##   'mean(Concentration)' 'sd(Concentration)'
##           <dbl>           <dbl>
## 1      23305480.      16578262.

```

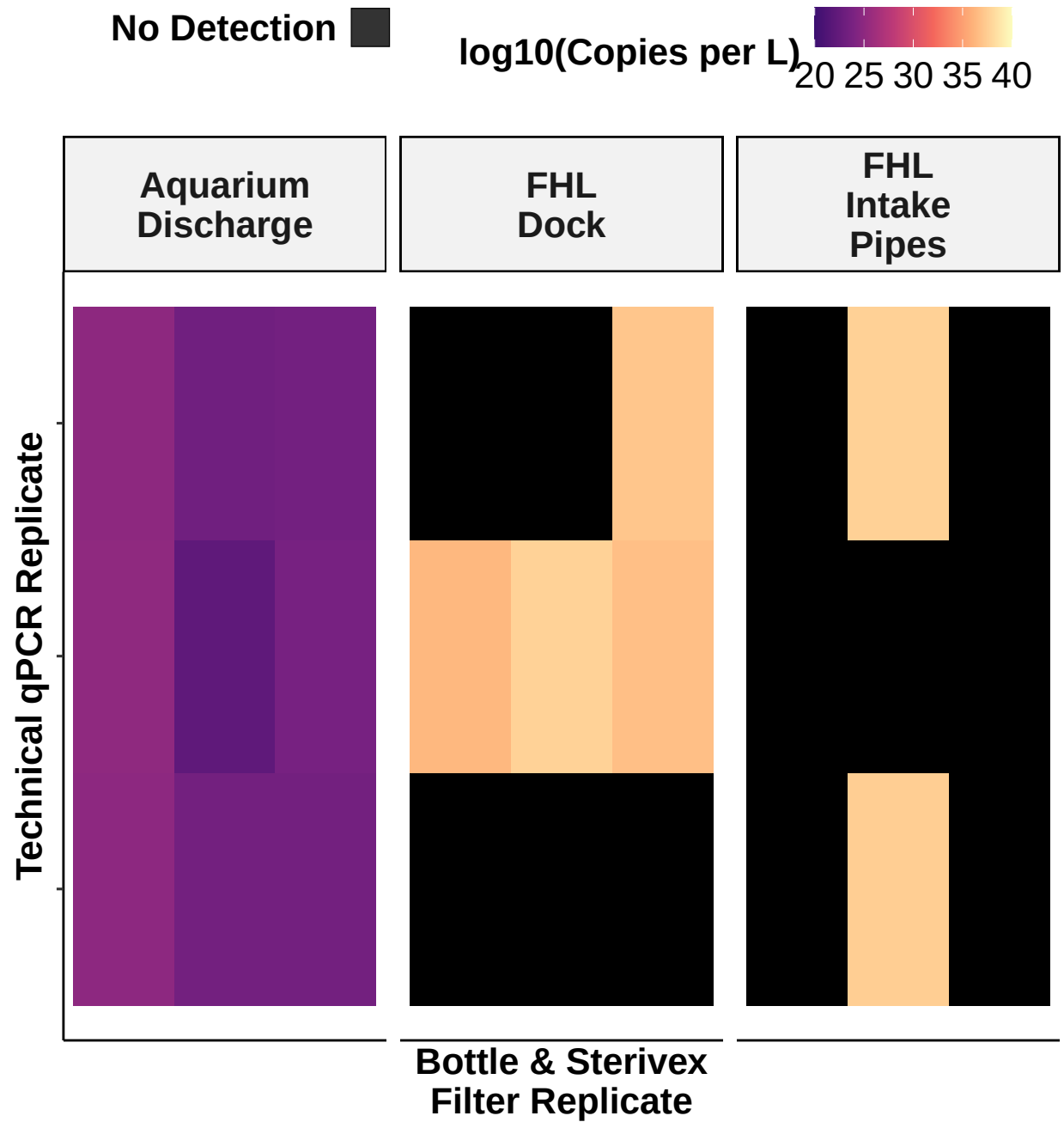
Figure sa4

```

pyno_sample_data_4_plotting_run1 %>%
  filter(., str_detect(Field_collection_method, "Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
  mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT = replace(CT, CT>40, NA)) %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=CT, color="")) + geom_tile() + theme_pubr() + theme(a.
    guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +
  labs(fill = "log10(Copies per L)") -> sa4

```

sa4



```
ggsave(filename = here("Figures", "FHL_mesocosm_field_trial_Ct_run1.png"),
        width=10,
        height = 4,
        dpi = 400,
        units = c("in"))
```

Load Data Pycno qPCR Plate 3

```
sample_set_up_run3 <- read_excel(here("raw_data", "qPCR_raw_data", "Pycno_test_3.xls"), sheet="Sample Set")
amplification_data_run3 <- read_excel(here("raw_data", "qPCR_raw_data", "Pycno_test_3.xls"), sheet="Amplification")
results_data_run3 <- read_excel(here("raw_data", "qPCR_raw_data", "Pycno_test_3.xls"), sheet="Results", sheet_index=1)
```

Merge Tables

```
amplification_data_run3 %>%
  left_join(sample_set_up_run3) %>%
  mutate(., `Sample Name` = if_else(is.na(`Sample Name`), str_c(`Target Name`, "_", Task, "_", Quantity), `Sample Name`))
  mutate(., `Sample Name` = if_else(is.na(`Sample Name`), "Negative Control", `Sample Name`)) -> combined_data_run3
```

Now we are ready to start analyzing the data.

First thing we want to do is check to see how our internal positive control is working.

Moa Map

```
columns = seq(from = 1, to = 24, by = 1)
columns_t = as_tibble(columns) %>% dplyr::rename("Column" = value)
rows = capitalize(letters[1:16])
rows_t = as_tibble(rows) %>% dplyr::rename("Row" = value)

cross_join(columns_t, rows_t) -> full_plate

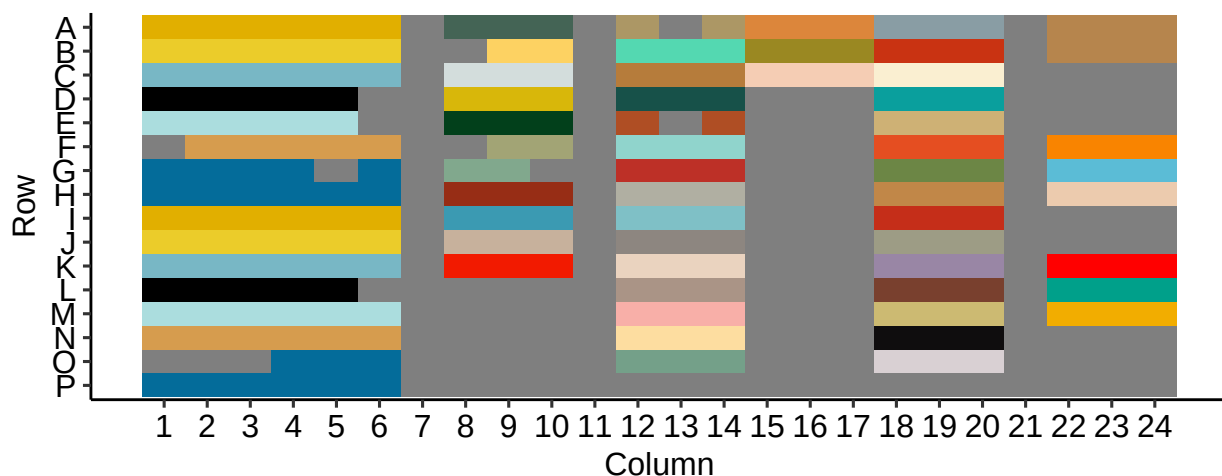
combined_qPCR_data_run3 %>%
  filter(., `Target Name` == "MOA") %>%
  mutate(., Row = str_sub(`Well Position`, start = 1L, end = 1L),
         Column = as.numeric(str_sub(`Well Position`, start = 2L, end = -1L))) -> moa_samples_run3

full_plate %>%
  left_join(moa_samples_run3) -> moa_plate_run3

moa_plate_run3 %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x = Column, y = fct_rev(Row), fill = `Sample Name`)) + geom_tile() + scale_x_continuous(breaks = columns)
```

Moa-IPC Standards & Samples

BHAK-1619 1:10	E2697.3T.FH25	M0D023141A 1:10	M0D056876Z 1:10	
BHAK-1619 1:100	E2702.NC.FH25	M0D023141A 1:100	M0D056876Z 1:100	
BHAK-1619 1:1000	E2707.1T.FH25	M0D023141A 1:1000	M0D056876Z 1:1000	
BHAK1597 1:10	E2707.2T.FH25	M0D052273Y 1:10	M0D061183Q 1:10	
BHAK1597 1:100	E2707.3T.FH25	M0D052273Y 1:100	M0D061183Q 1:100	
BHAK1597 1:1000	E2708.1T.FH25	M0D052273Y 1:1000	M0D061183Q 1:1000	
E0	E2708.2T.FH25	M0D053804V 1:10	M0D061197E 1:10	
E0.7	E2708.3T.FH25	M0D053804V 1:100	M0D061197E 1:100	
E1	E3	M0D053804V 1:1000	M0D061197E 1:1000	
E2	E4	M0D056848X 1:10	M0D064351M 1:10	
E2697.1T.FH25	E5	M0D056848X 1:100	M0D064351M 1:100	
E2697.2T.FH25	Ext_Blank	M0D056848X 1:1000	M0D064351M 1:1000	



Pycno Map

```
combined_qPCR_data_run3 %>%
  filter(., `Target Name`=="Pycno") %>%
  mutate(., Row = str_sub(`Well Position`,start = 1L, end = 1L),
         Column= as.numeric(str_sub(`Well Position`,start = 2L, end = -1L ))) -> pycno_samples_run3

full_plate %>%
  left_join(pycno_samples_run3) -> pycno_plate_run3
```

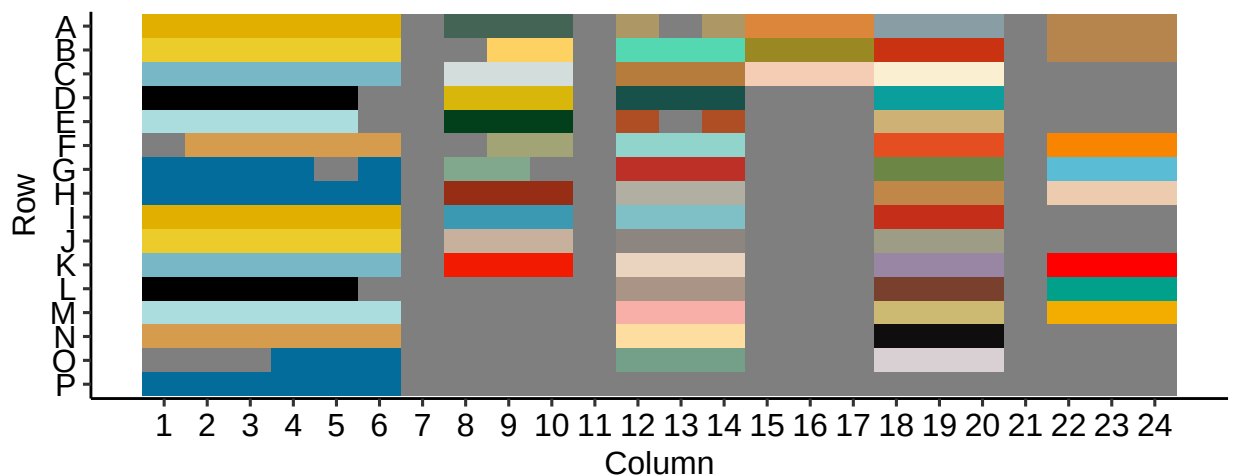
```

pycno_plate_run3 %>%
  mutate(Row = factor(Row, levels = rows)) %>%
  ggplot(aes(x=Column, y=fct_rev(Row), fill=`Sample Name` )) +geom_tile() +scale_x_continuous(breaks=col

```

Pycno Standards & Samples

BHAK-1619 1:10	E2697.3T.FH25	M0D023141A 1:10	M0D056876Z 1:10
BHAK-1619 1:100	E2702.NC.FH25	M0D023141A 1:100	M0D056876Z 1:100
BHAK-1619 1:1000	E2707.1T.FH25	M0D023141A 1:1000	M0D056876Z 1:1000
BHAK1597 1:10	E2707.2T.FH25	M0D052273Y 1:10	M0D061183Q 1:10
BHAK1597 1:100	E2707.3T.FH25	M0D052273Y 1:100	M0D061183Q 1:100
BHAK1597 1:1000	E2708.1T.FH25	M0D052273Y 1:1000	M0D061183Q 1:1000
E0	E2708.2T.FH25	M0D053804V 1:10	M0D061197E 1:10
E0.7	E2708.3T.FH25	M0D053804V 1:100	M0D061197E 1:100
E1	E3	M0D053804V 1:1000	M0D061197E 1:1000
E2	E4	M0D056848X 1:10	M0D064351M 1:10
E2697.1T.FH25	E5	M0D056848X 1:100	M0D064351M 1:100
E2697.2T.FH25	Ext_Blank	M0D056848X 1:1000	M0D064351M 1:1000

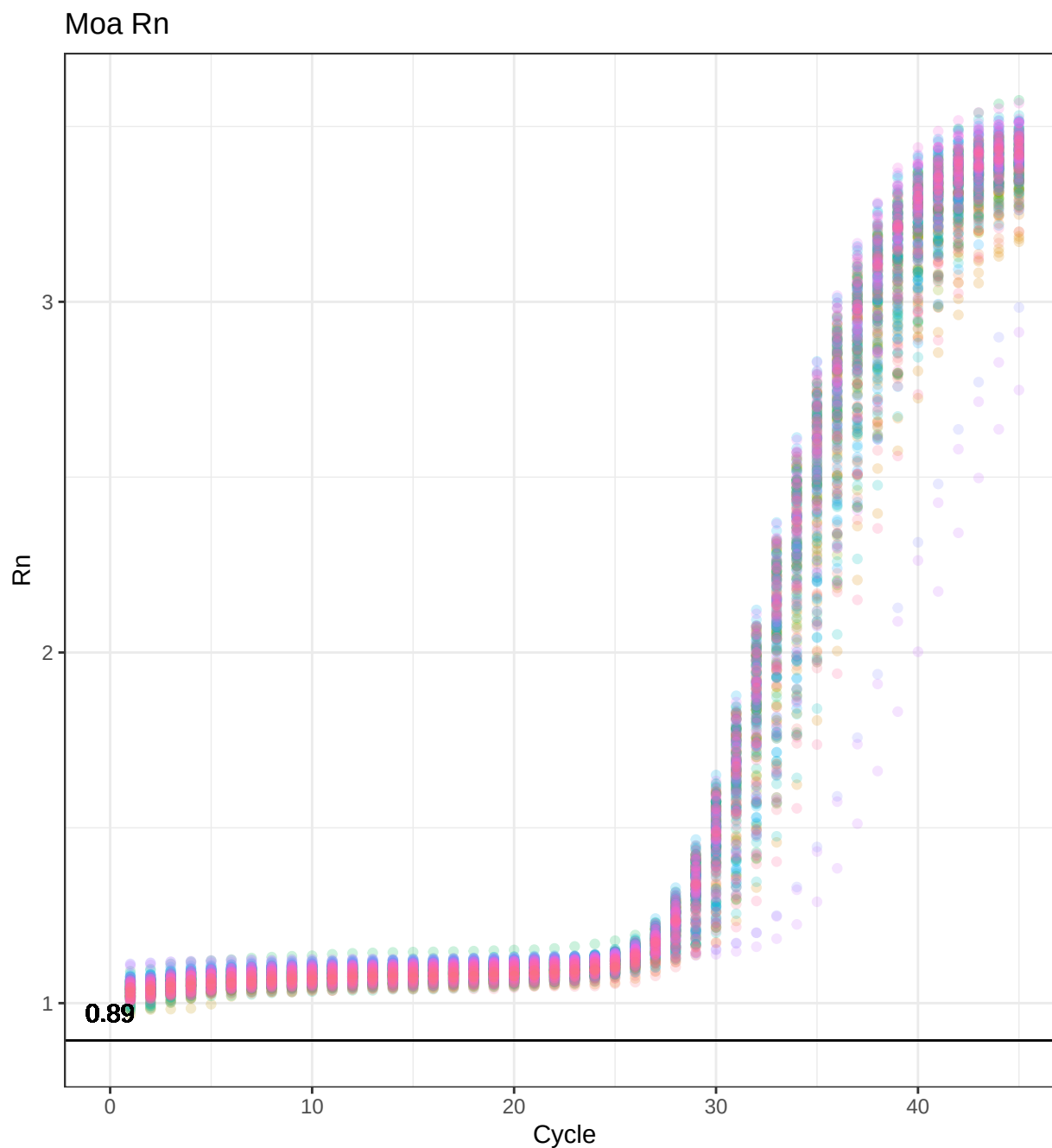


Moa Rn

```
combined_qPCR_data_run3 %>%
  filter(`Target Name`=="MOA" & Task=="STANDARD") -> moa_standard_run3

moa_standard_run3$Rn %>% max() -> max_Rn_moa_run3
h_moa_run3= max_Rn_moa_run3*0.25

moa_standard_run3 %>%
  ggplot(., aes(y=Rn, x=Cycle, color=`Well Position`)) +
  geom_point(alpha=0.2) + geom_hline(yintercept =h_moa_run3) +
  geom_text(aes(0,h_moa_run3,label = round(h_moa_run3,2), vjust = -1),color="black") +theme_bw() +ggtitle("Moa Rn")
```



There is quite a bit of spread in our IPC. Will remove any samples outside normal distribution. #####
Remove Samples with Anomolous Outlier IPC values

```
moa_standard_run3 %>%
  filter(., Cycle ==40) -> moa_standard_c40_run3

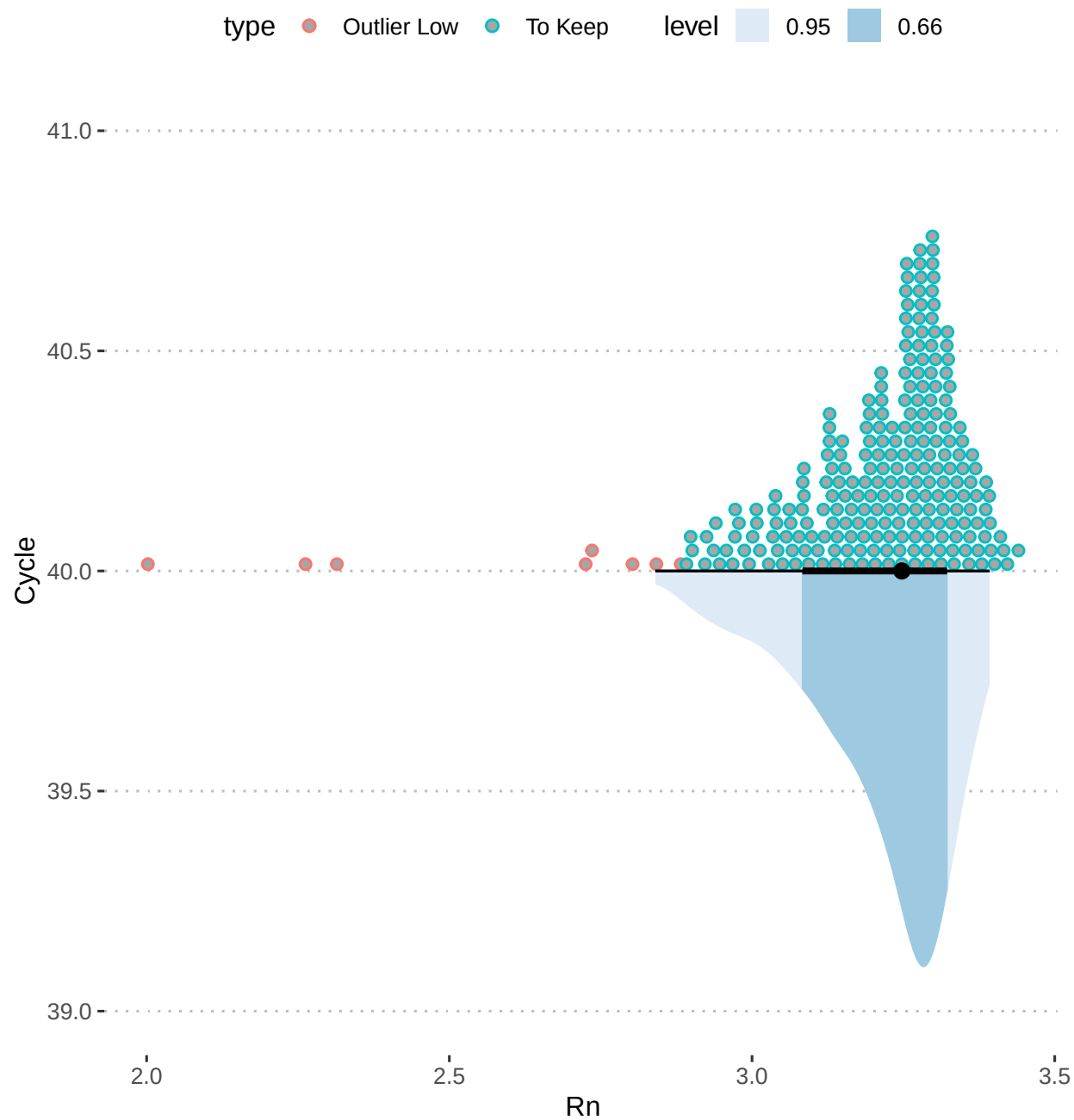
moa_standard_c40_run3 %>%
  dplyr::summarise(mean_Rn = ci(Rn, confidence = 0.99)[1],
    lower_ci_mpg = ci(Rn, confidence = 0.99)[2],
    upper_ci_mpg = ci(Rn, confidence = 0.99)[3],
    sd_Rn = ci (Rn, confidence = 0.99)[4]) -> CIO.95_moa_run3

Sum_run3 = groupwiseMean(Rn ~ Cycle, data=moa_standard_c40_run3, conf = 0.99)

moa_standard_c40_run3 %>%
  summarise(
    Q1 = quantile(Rn, 0.25, na.rm = TRUE),
    Q3 = quantile(Rn, 0.75, na.rm = TRUE),
    IQR = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR,
    upper_bound = Q3 + 1.5 * IQR
  ) -> moa_iqr_run3

moa_standard_c40_run3 %>%
  mutate(., type=case_when(Rn > moa_iqr_run3$upper_bound ~"Outlier High",
    Rn < moa_iqr_run3$lower_bound ~"Outlier Low",
    TRUE~"To Keep")) -> moa_standard_c40_run3

moa_standard_c40_run3 %>%
  ggplot(., aes(x= Rn, y=Cycle)) + geom_dots(layout='weave', side = "top", aes(color=type)) + stat_slab
```



```
moa_standard_c40_run3 %>%
  filter(., type!="To Keep") -> moa_outliers_to_drop_run3
```

```
combined_qPCR_data_run3 %>%
  filter(., `Well Position` != moa_outliers_to_drop_run3$`Well Position`) -> combined_qPCR_data_cleaned
```

```
results_data_run3 %>%
  filter(`Well Position` != moa_outliers_to_drop$`Well Position`) -> results_data_cleaned_run3
```

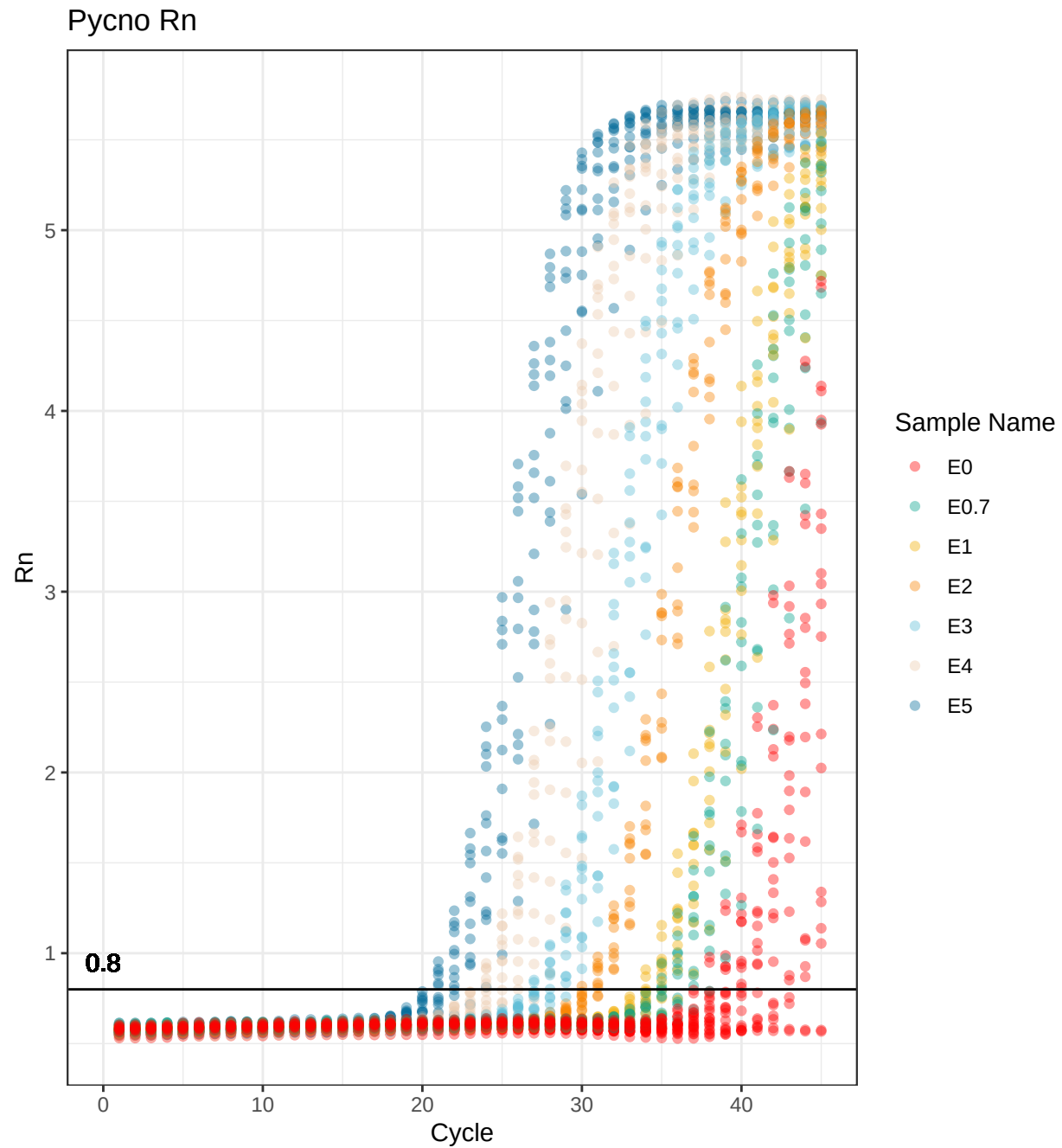
Filter Data

Pycno Rn

```
combined_qPCR_data_run3 %>%
  filter(`Target Name`=="Pycno" & Task=="STANDARD")-> pycno_standard_run3

pycno_standard_run3$Rn %>% max() -> max_Rn_run3
h_run3= 0.8

pycno_standard_run3 %>%
  ggplot(., aes(y=Rn, x=Cycle, color=`Sample Name`)) +geom_point(alpha=0.4) + geom_hline(yintercept =h_run3)
  geom_text(aes(0,h_run3,label = round(h_run3,2), vjust = -1),color="black") +theme_bw() + ggtitle("Pycno")
```

```
pycno_standard_run3 %>%
  mutate(., Threshold = if_else(Rn > h_run3, "Above", "Below")) -> pycno_standard_run3

pycno_standard_run3 %>%
  group_by(`Well Position`, `Sample Name`) %>%
  filter(., rank(Threshold, ties.method="first")==1) %>%
  filter(., Threshold == "Above") -> pycno_standard_above_run3
```

```

pycno_standard_run3 %>%
  ungroup() %>%
  filter(., !`Well Position` %in% pycno_standard_above_run3$`Well Position`) %>%
  group_by(`Well Position`, `Sample Name`) %>%
  filter(., rank(Threshold, ties.method="last")==1) -> pycno_standard_below_run3

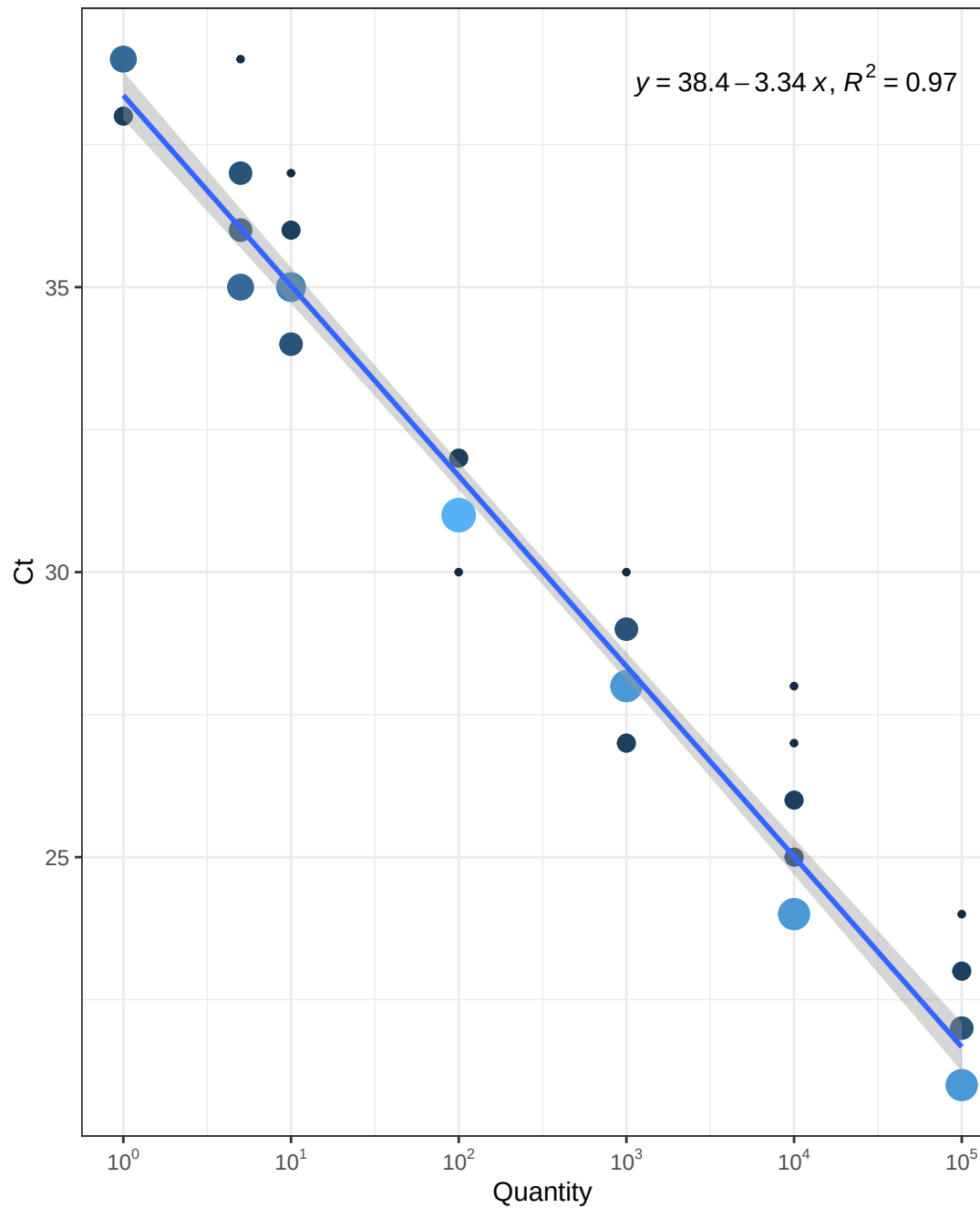
bind_rows(pycno_standard_above_run3, pycno_standard_below_run3) -> pycno_standard_threshold_run3

set_breaks = function(limits) {
  seq(limits[1], limits[2], by = 1)
}

pycno_standard_threshold_run3 %>%
  filter(., Cycle < 40 ) %>%
  ggplot(., aes(x=`Quantity`, y = Cycle)) +geom_count(aes(color = ..n.., size = ..n..)) +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
    label.y = "top") +theme_bw() +ylab("Ct") -> run3_ct_versus_quant

run3_ct_versus_quant

```



Pycno Ct versus Quantity

```
# View the structure of the generated plot object
# The computed data is stored within the plot object
grob_data_run3 <- ggplot_build(run3_ct_versus_quant)$data[[3]]

# The result is a list; unlist it to get a character vector
coef_char_run3 <- unlist(grob_data_run3$coefs)

standard_curve_coefs_run3 <- enframe(coef_char_run3, name = "Metric", value = "Value") %>%
  mutate( Metric = recode(Metric,
    "(Intercept)" = "Intercept",
```

```

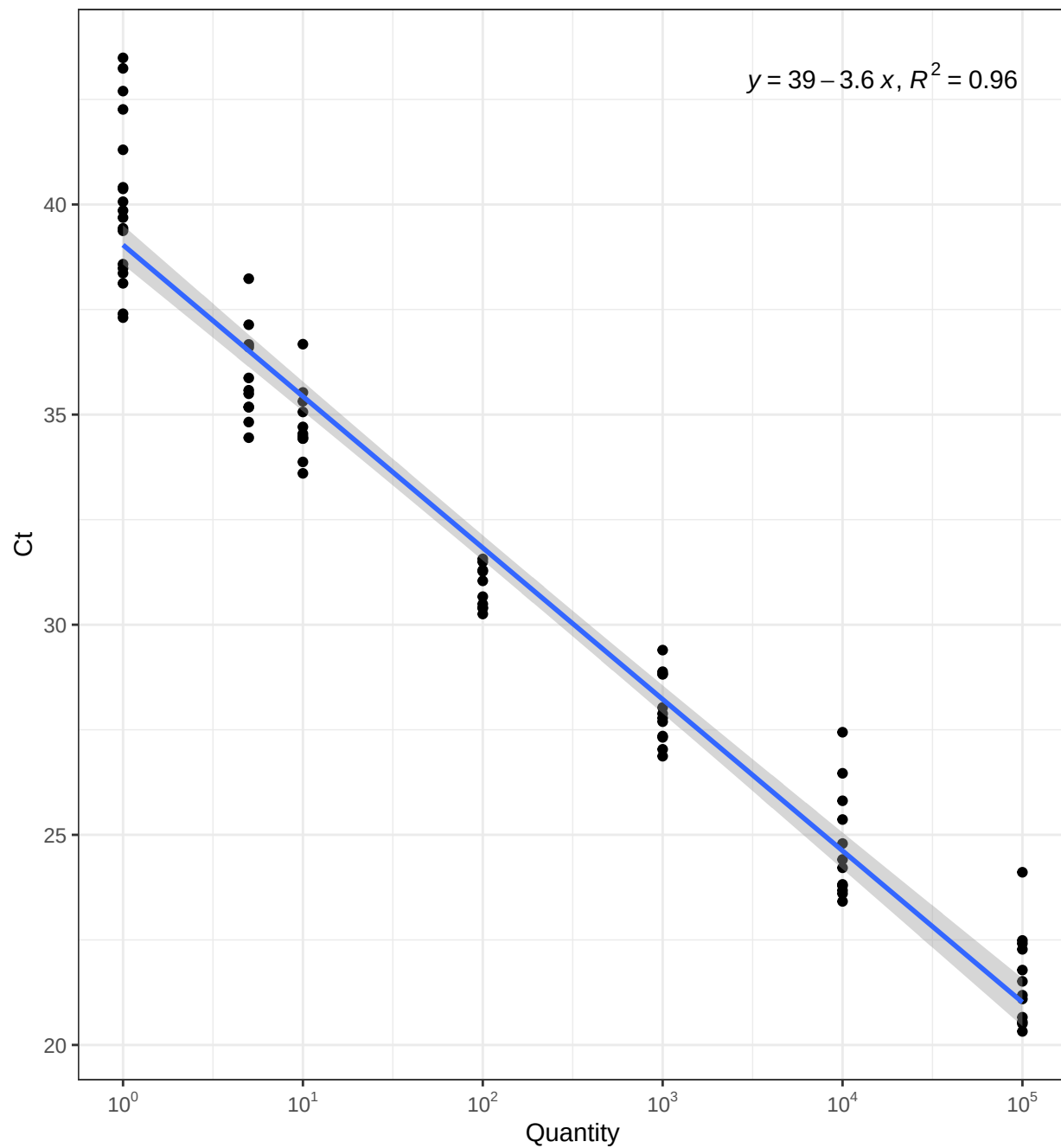
      "x" = "Slope"))

# Extract the R^2 value from the 'r.squared' column
r_squared_value_run3 <- grob_data_run3$r.squared

results_data_cleaned_run3 %>%
  filter(., `Target Name`==Target_name) %>%
  filter(., Task=="STANDARD") %>%
  mutate(., CT=as.numeric(CT)) -> pycno_results_standard_run3

pycno_results_standard_run3 %>%
  ggplot(., aes(x=`Quantity`, y = CT)) +geom_point() +
  scale_x_log10("Quantity",
    breaks = trans_breaks("log10", function(x) 10^x),
    labels = trans_format("log10", math_format(10^.x))) +
  guides(color = 'legend') +
  scale_size_continuous(breaks = set_breaks)+
  scale_color_continuous(breaks = set_breaks)+
  stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")),label.x = "right",
    label.y = "top") +theme_bw() +ylab("Ct")

```



```
results_data_cleaned_run3 %>%
  filter(., `Target Name`==Target_name) %>%
  filter(., Task!="UNKNOWN") %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., Lab="NOAA PMEL OME") %>%
  dplyr::select(Target=`Target Name`, Lab, Cq=CT, SQ=Quantity) %>%
  mutate(., TemplateVolume=2,
    ReactionVolume=10) %>%
  mutate(., Cq = replace(Cq, Cq>40, NA)) -> qPCR_Pycno_3_standards

write.csv(qPCR_Pycno_3_standards, file = here("LOD_LOQ/qPCR_Pycno_3_standards.csv"), row.names = F)
```

```
results_data_cleaned_run3 %>%
  filter(., `Target Name` == "Pycno") %>%
  filter(., str_detect(`Sample Name`, "FH25"))
```

```
## # A tibble: 27 x 30
##   Well 'Well Position' Omit 'Sample Name' 'Target Name' Task Reporter
##   <dbl> <chr>          <lgl> <chr>          <chr>          <chr> <chr>
## 1     8 A8            FALSE E2697.1T.FH25 Pycno          UNKNOWN FAM
## 2     9 A9            FALSE E2697.1T.FH25 Pycno          UNKNOWN FAM
## 3    10 A10           FALSE E2697.1T.FH25 Pycno          UNKNOWN FAM
## 4    33 B9            FALSE E2697.2T.FH25 Pycno          UNKNOWN FAM
## 5    34 B10           FALSE E2697.2T.FH25 Pycno          UNKNOWN FAM
## 6    56 C8            FALSE E2697.3T.FH25 Pycno          UNKNOWN FAM
## 7    57 C9            FALSE E2697.3T.FH25 Pycno          UNKNOWN FAM
## 8    58 C10           FALSE E2697.3T.FH25 Pycno          UNKNOWN FAM
## 9    80 D8            FALSE E2707.1T.FH25 Pycno          UNKNOWN FAM
## 10   81 D9            FALSE E2707.1T.FH25 Pycno          UNKNOWN FAM
## # i 17 more rows
## # i 23 more variables: Quencher <chr>, CT <chr>, 'Ct Mean' <dbl>,
## #   'Ct SD' <dbl>, Quantity <dbl>, 'Quantity Mean' <dbl>, 'Quantity SD' <dbl>,
## #   'Y-Intercept' <dbl>, 'R(superscript 2)' <dbl>, Slope <dbl>,
## #   Efficiency <dbl>, 'Automatic Ct Threshold' <lgl>, 'Ct Threshold' <dbl>,
## #   'Automatic Baseline' <lgl>, 'Baseline Start' <dbl>, 'Baseline End' <dbl>,
## #   'Amp Status' <chr>, Comments <lgl>, 'Cq Conf' <dbl>, HIGHSD <chr>, ...
```

FHL Analysis Transplanted Individual Detection

Merge Sample Metadata

```
sample_metadata<- read_excel(here("raw_data", "qPCR_raw_data", "ElectronicArchive_CruiseSampleMetadata_20180601.xlsx"))

results_data_cleaned_run3 %>%
  filter(., `Target Name`=="Pycno") %>%
  filter(., !str_detect(`Task`, "STANDARD")) %>%
  mutate(., Quantity=if_else(is.na(Quantity), 0, Quantity)) %>%
  mutate(., Concentration = (Quantity*10*(100/2)/ 1)) %>% # Quantity is the concentration generated by
  mutate(., Observation = if_else(Quantity ==0, "Not Detected", "Detected")) %>%
  mutate(., `Sample Name`= if_else(str_detect(`Sample Name`, "Negative Control"), "NTC", `Sample Name`)) %>%
  ungroup() %>%
  mutate(., Sample_Name= str_remove(`Sample Name`, ".FH25")) %>%
  left_join(sample_metadata, by=c("Sample_Name"="Sample_Name")) %>%
  mutate(., `Sample Name` = if_else(`Sample Name`=="Ext_blank_UWFH25", "Extraction Blank", `Sample Name`)) %>%
  mutate(., Negative_control = if_else(is.na(Negative_control), TRUE, Negative_control)) %>%
  group_by(`Sample Name`) %>%
  mutate(Technical_Replicate = row_number(`Sample Name`)) %>%
  mutate(., Station = if_else(is.na(Station) & Negative_control==TRUE | Task=="NTC", `Sample Name`, Station)) %>%
  mutate(., Name_plotting = str_c(Station, " Bottle ", Biological_Replicate, " PCR ", Technical_Replicate)) %>%
  mutate(., Name_plotting = if_else(is.na(Name_plotting), str_c(`Sample Name`, " PCR ", Technical_Replicate))) %>%
  mutate(., Name_plotting_b = str_c(Station, " Bottle ", Biological_Replicate)) %>%
```

```
mutate(., Name_plotting_b = if_else(is.na(Name_plotting_b), `Sample Name`, Name_plotting_b)) %>%
mutate(., Type = if_else(is.na(Station), "Extraction Blank", Station)) %>%
  mutate(., Type = if_else(str_detect(Type, ".NC"), "Field Blank", Type)) %>%
mutate(., Sample_Type = if_else(Negative_control==FALSE, "Sample", "Blank")) -> pycno_sample_data_4_plotting_run3
```

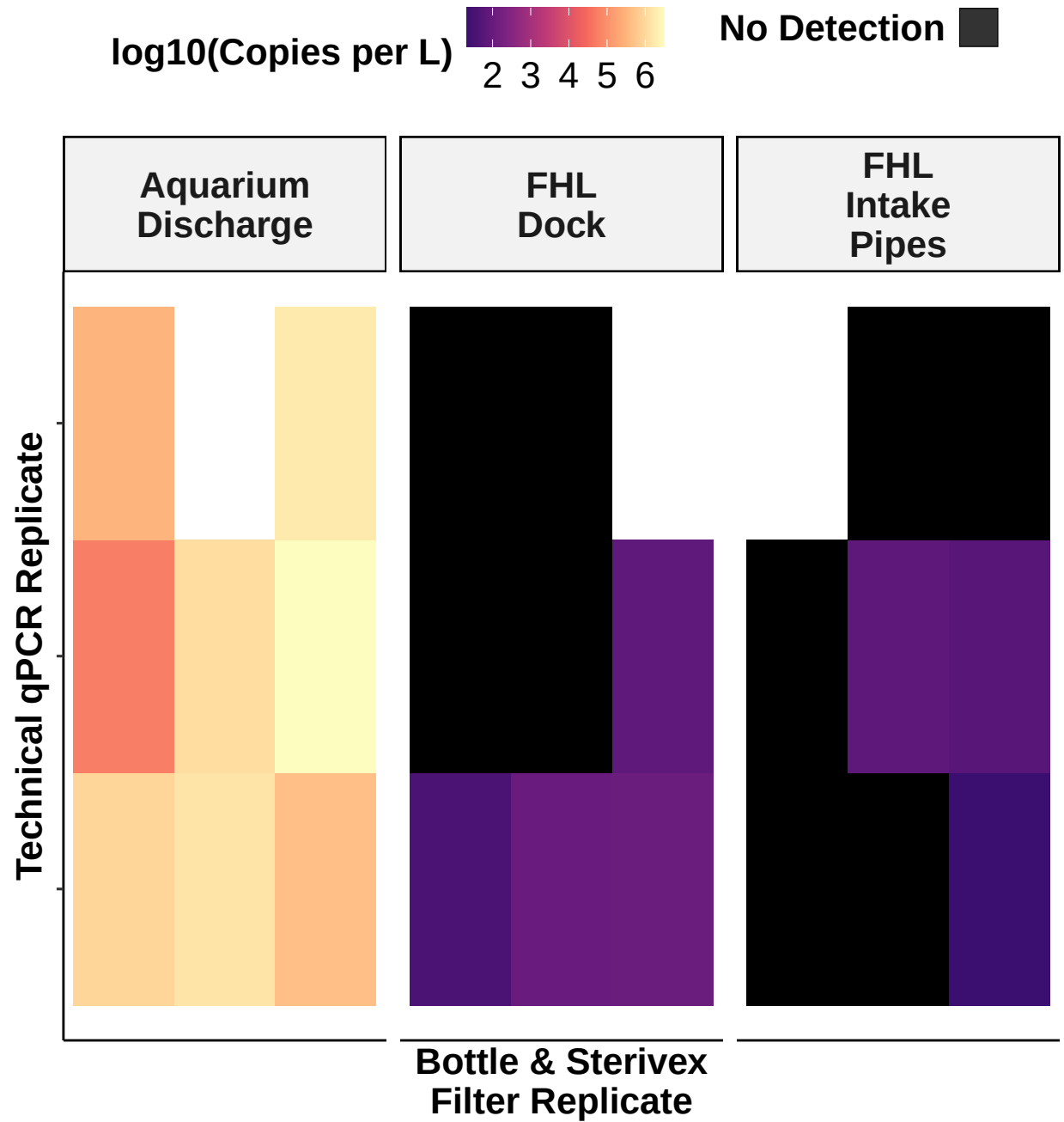
Figure sa5

```
pycno_sample_data_4_plotting_run3$Type2 <- pycno_sample_data_4_plotting_run3$Type
pycno_sample_data_4_plotting_run3$Type2 <- as.factor(pycno_sample_data_4_plotting_run3$Type2)
levels(pycno_sample_data_4_plotting_run3$Type2)[levels(pycno_sample_data_4_plotting_run3$Type2) == "Aquifer"] == "FHL"
levels(pycno_sample_data_4_plotting_run3$Type2)[levels(pycno_sample_data_4_plotting_run3$Type2) == "FHL"] == "FHL"
levels(pycno_sample_data_4_plotting_run3$Type2)[levels(pycno_sample_data_4_plotting_run3$Type2) == "FHL"] == "FHL"

pycno_sample_data_4_plotting_run3 %>%
  filter(., str_detect(Field_collection_method, "Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
    mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., Concentration = if_else(Concentration == 0, NA, Concentration)) %>%
  mutate(., Plate = "3") -> plate_3_data_analysis

plate_3_data_analysis %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=log10(Concentration), color="")) + geom_tile() + theme_minimal() +
  guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +
  labs(fill = "log10(Copies per L)") -> sa5

sa5
```



```
ggsave(filename = here("Figures", "FHL_mesocosm_field_trial_run3.png"),
        width=10,
        height = 4,
        dpi = 400,
        units = c("in"))
```

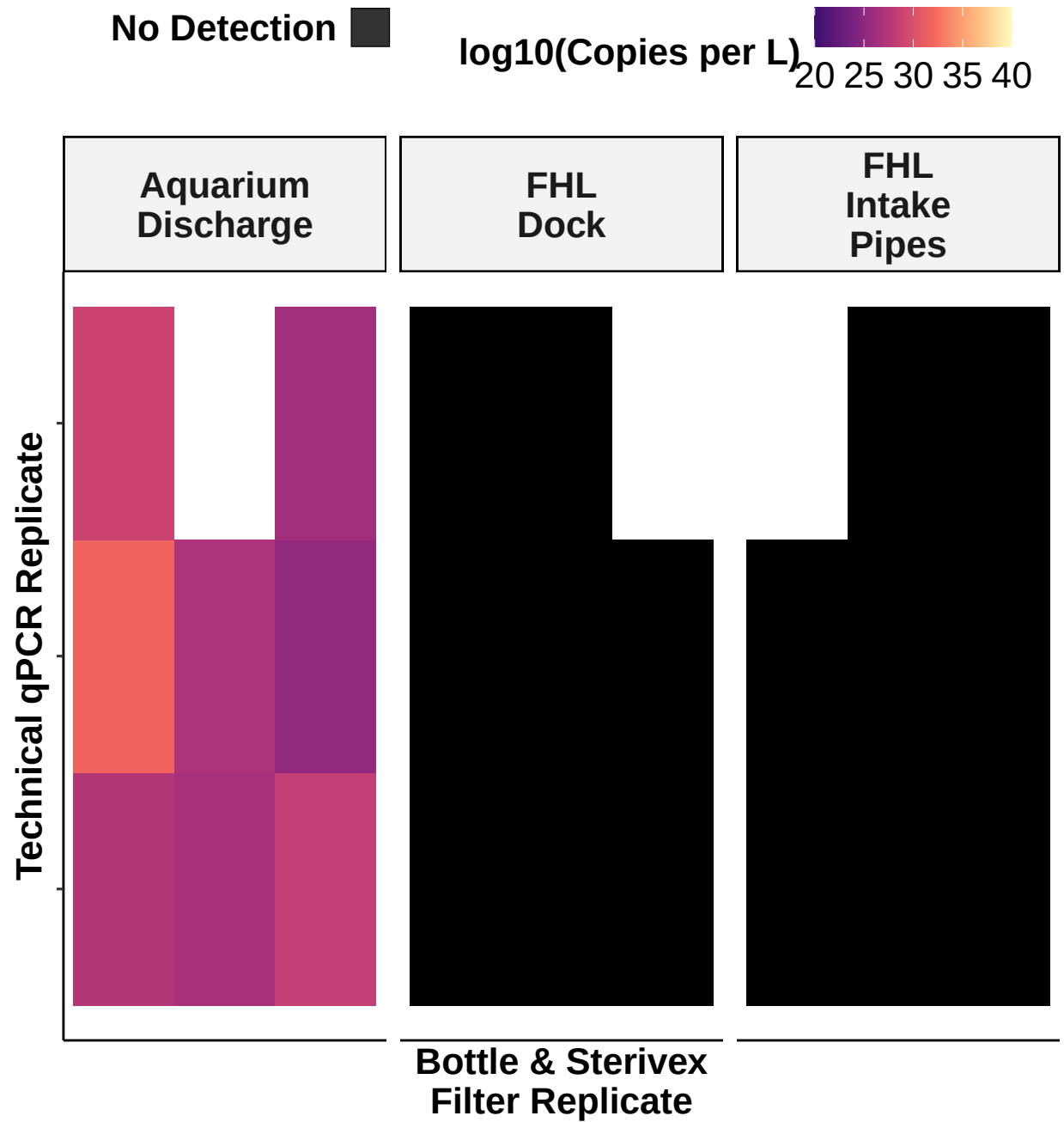
Figure sa6


```

pycno_sample_data_4_plotting_run3 %>%
  filter(., str_detect(Field_collection_method, "Sterivex")) %>%
  filter(., Sample_Type != "Blank") %>%
    mutate(across(Sample_Type, ~factor(., levels=c("Sample", "Blank")))) %>%
  group_by(Name_plotting_b) %>%
  mutate(., Replicate = row_number()) %>%
  ungroup() %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT = replace(CT, CT>40, NA)) %>%
  ggplot(., aes(x=Name_plotting_b, y=Replicate, fill=CT, color="")) + geom_tile() + theme_pubr() + theme(a
    guides(colour=guide_legend("No Detection", override.aes=list(colour="black"))) +
  labs(fill = "log10(Copies per L)") -> sa6

```

sa6

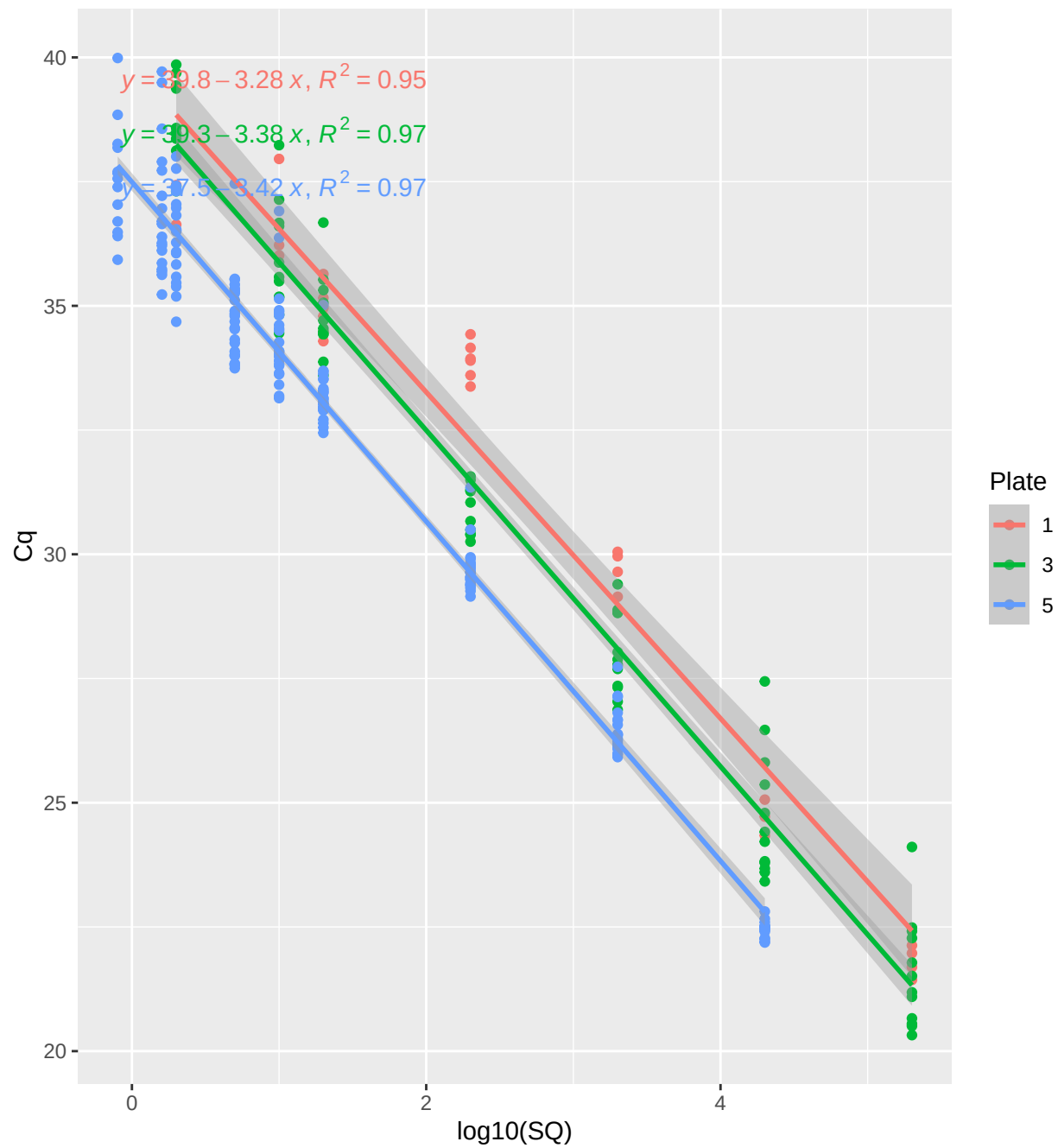


```
ggsave(filename = here("Figures", "FHL_mesocosm_field_trial_Ct_run3.png"),
        width=10,
        height = 4,
        dpi = 400,
        units = c("in"))
```

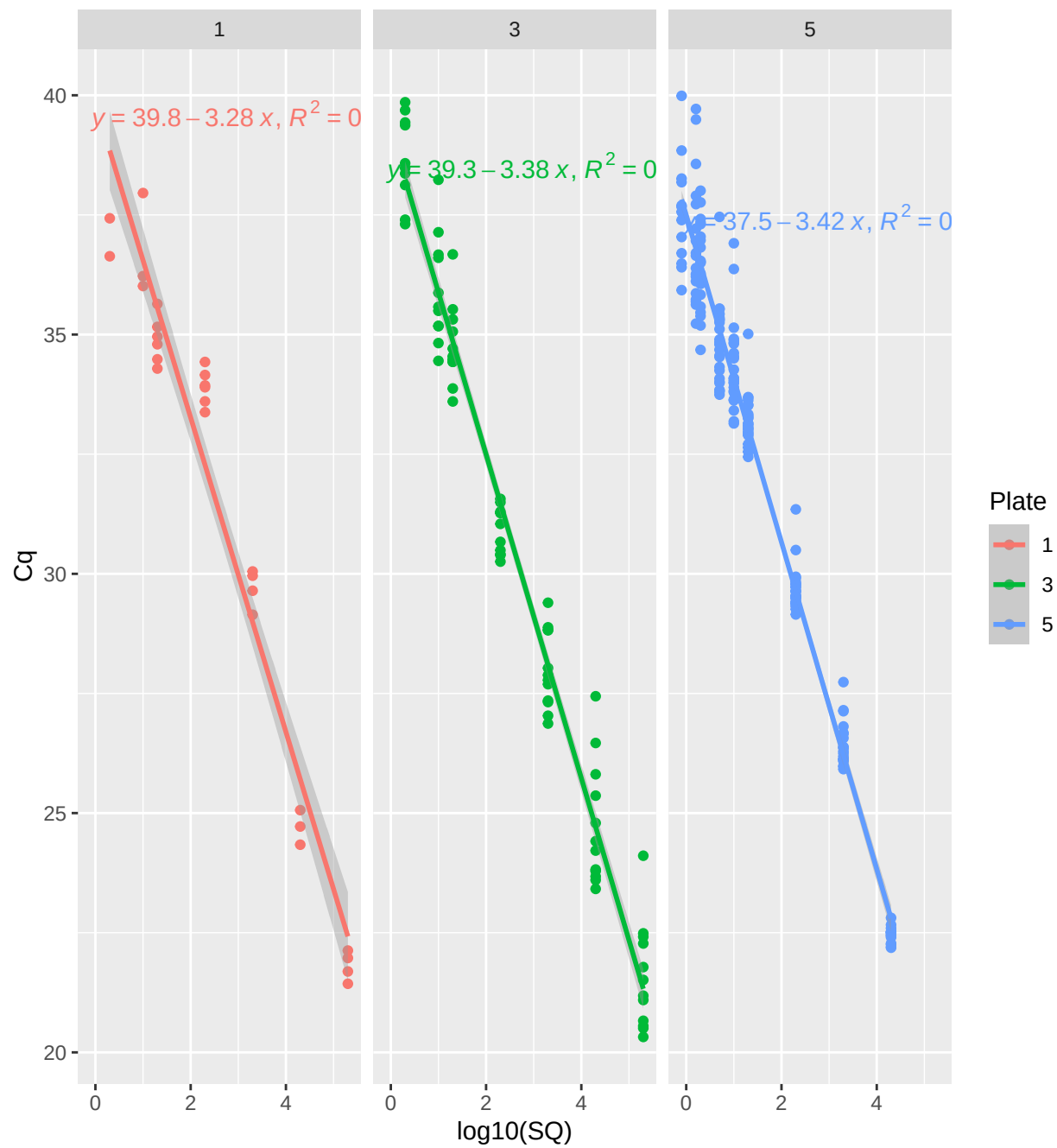
Combine Standard Plates

```
plate1_standards <- read.delim(file = here("LOD_LOQ/qPCR_Pycno_1_standards.csv"), header=T, sep=",") %>%
plate3_standards <- read.delim(file = here("LOD_LOQ/qPCR_Pycno_3_standards.csv"), header=T, sep=",") %>%
plate5_standards <- read.delim(file = here("LOD_LOQ/qPCR_Pycno_5_v3_standards.csv"), header=T, sep=",")
rbind(plate1_standards, plate3_standards, plate5_standards) -> plates_135_combined
write.csv(plates_135_combined, file = here("LOD_LOQ/plates_135_combined.csv"), row.names = F)
```

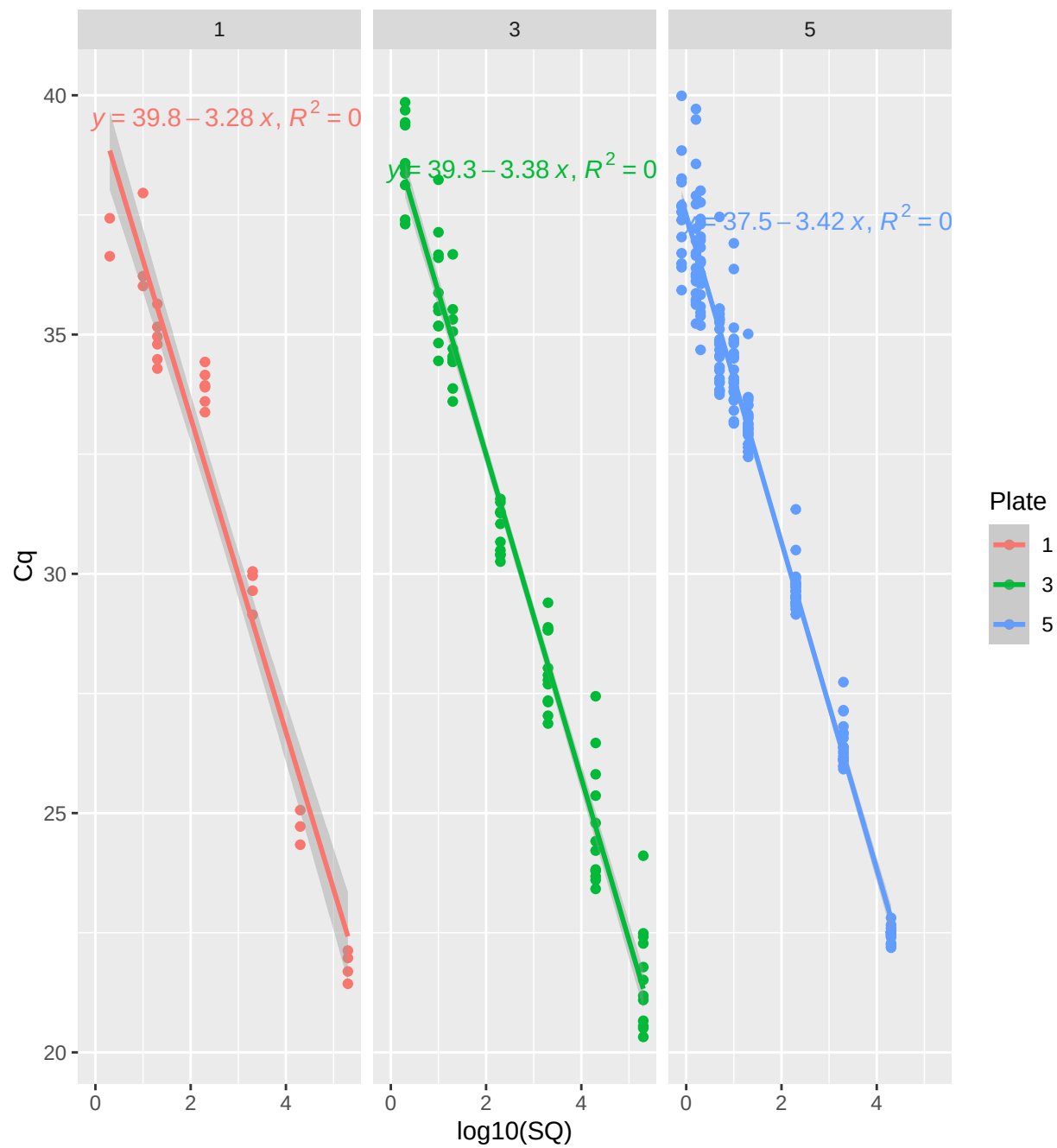
```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  ggplot(aes(x= log10(SQ), y=Cq, color= Plate)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")))
```



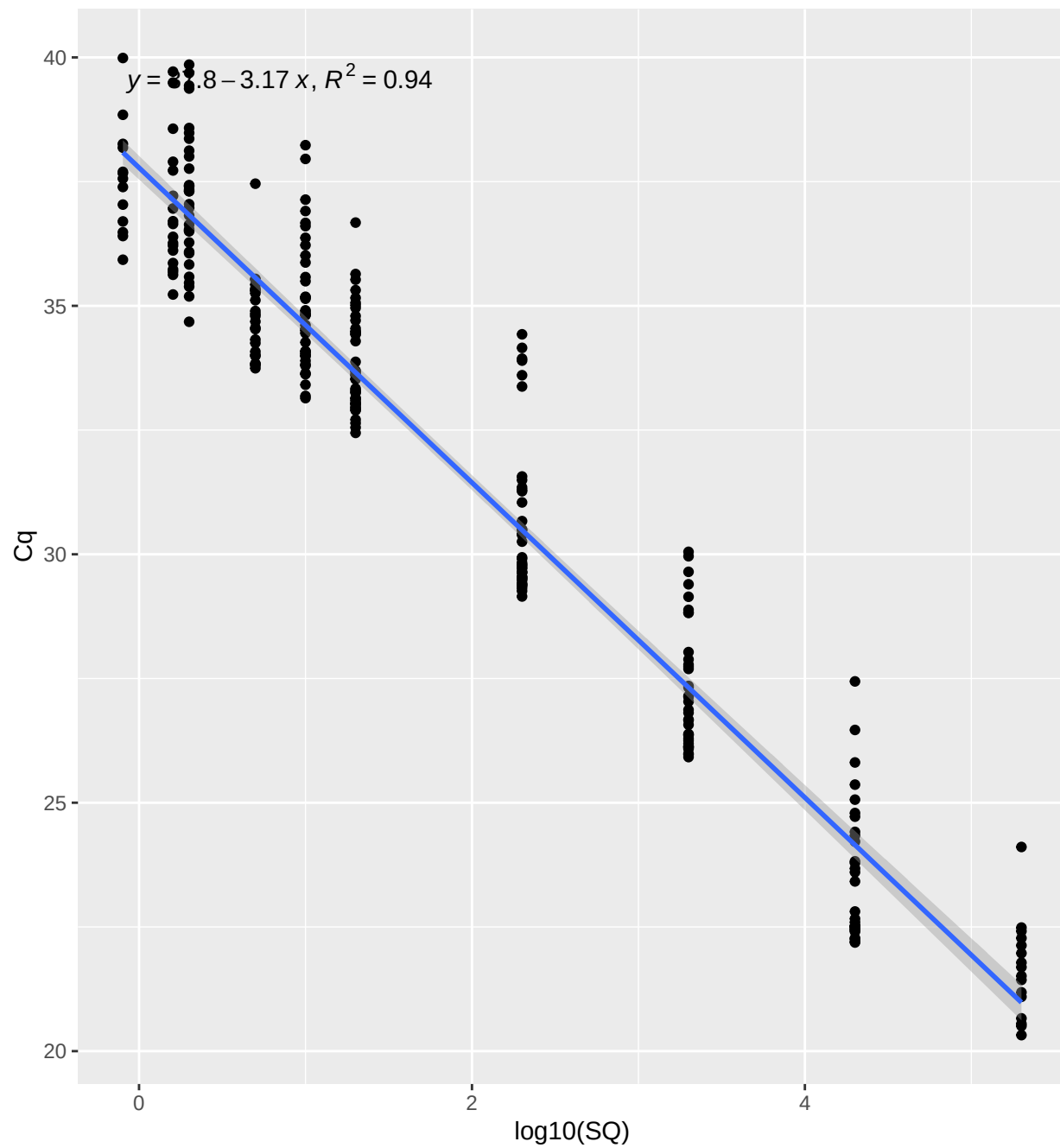
```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  ggplot(aes(x= log10(SQ), y=Cq, color= Plate)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2"))) +facet_wrap(~Plate)
```



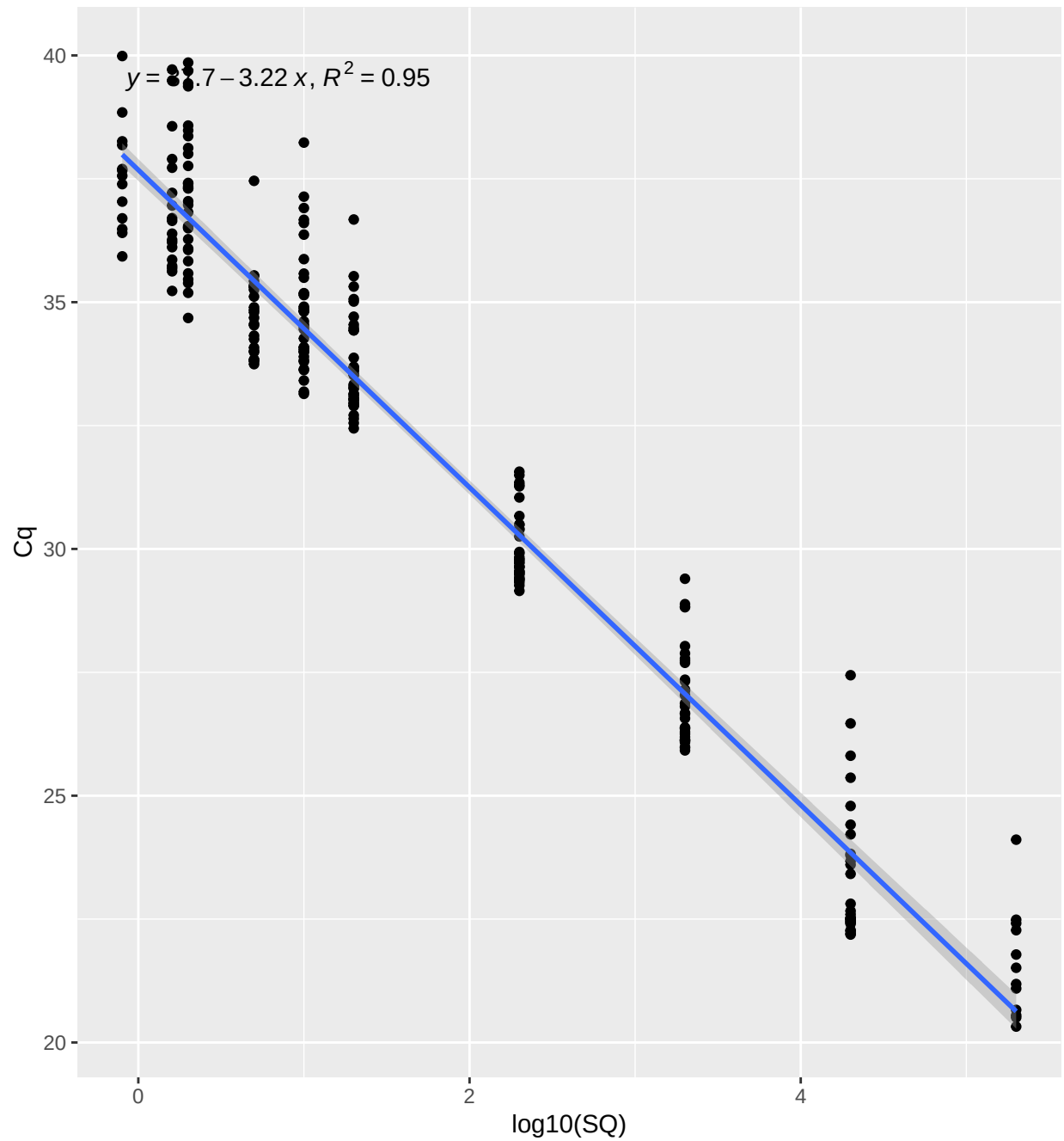
```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  ggplot(aes(x= log10(SQ), y=Cq, color= Plate)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")))+facet_wrap(~Plate)
```



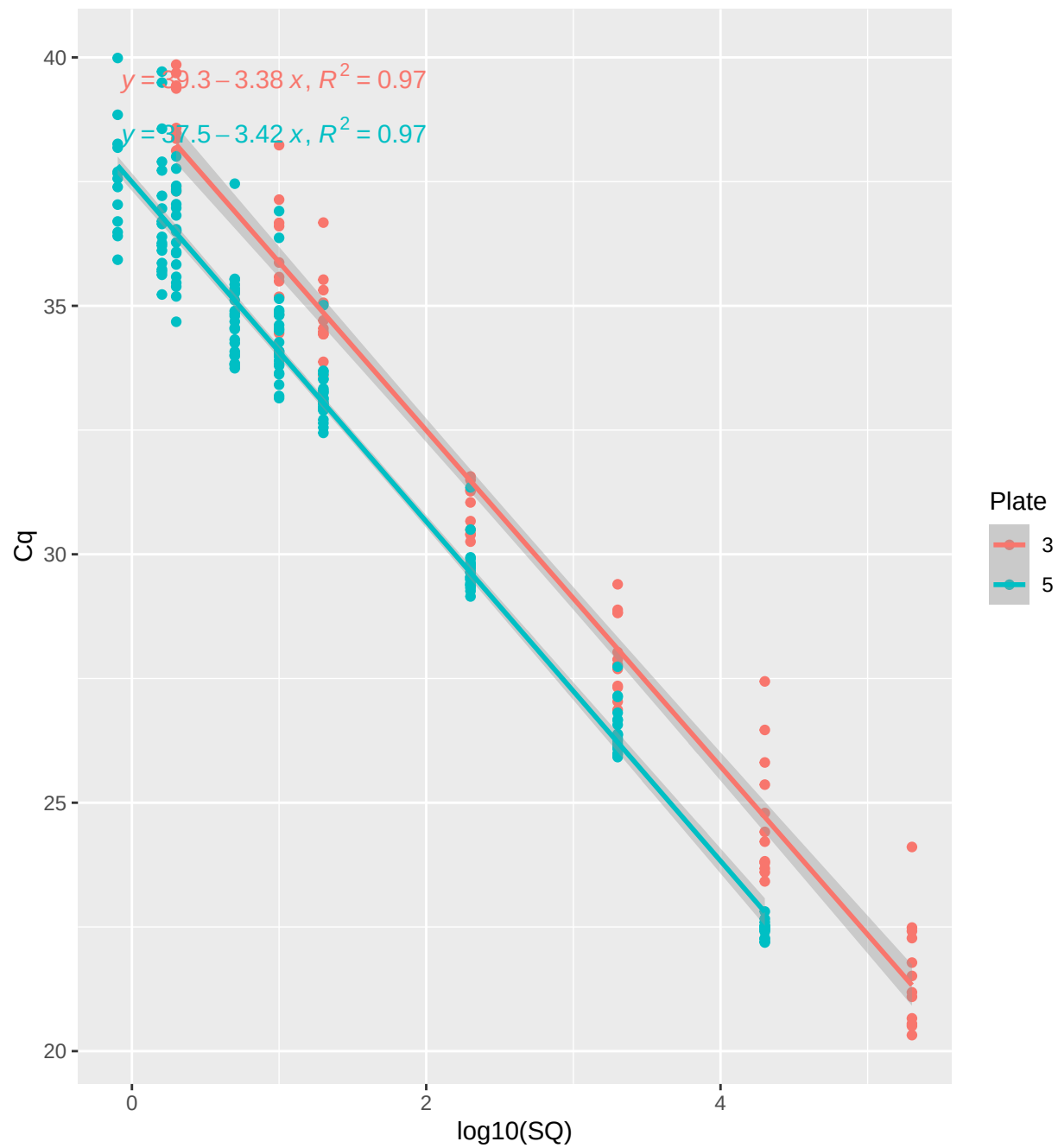
```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  ggplot(aes(x= log10(SQ), y=Cq)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")))
```



```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  filter(., Plate != "1") %>%
  ggplot(aes(x= log10(SQ), y=Cq)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")))
```

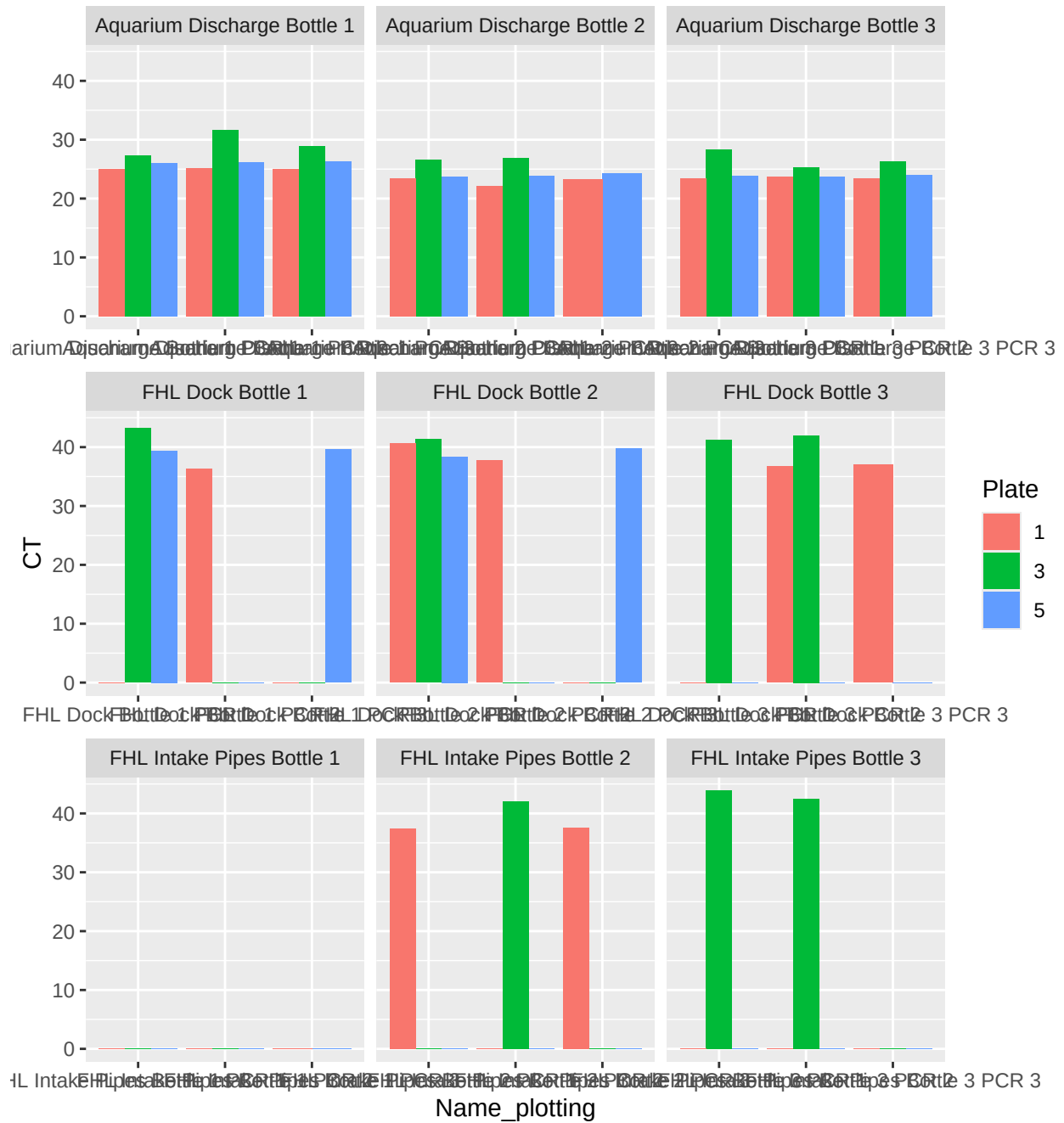


```
plates_135_combined %>%
  mutate(., Plate= as.character(Plate)) %>%
  mutate(., SQ=SQ*TemplateVolume) %>%
  filter(., Plate != "1") %>%
  ggplot(aes(x= log10(SQ), y=Cq, color=Plate)) + geom_point() + stat_poly_line() +
  stat_poly_eq(use_label(c("eq", "R2")))
```

```
bind_rows(plate_5_data_analysis, plate_3_data_analysis, plate_1_data_analysis) -> combo_plate_analysis

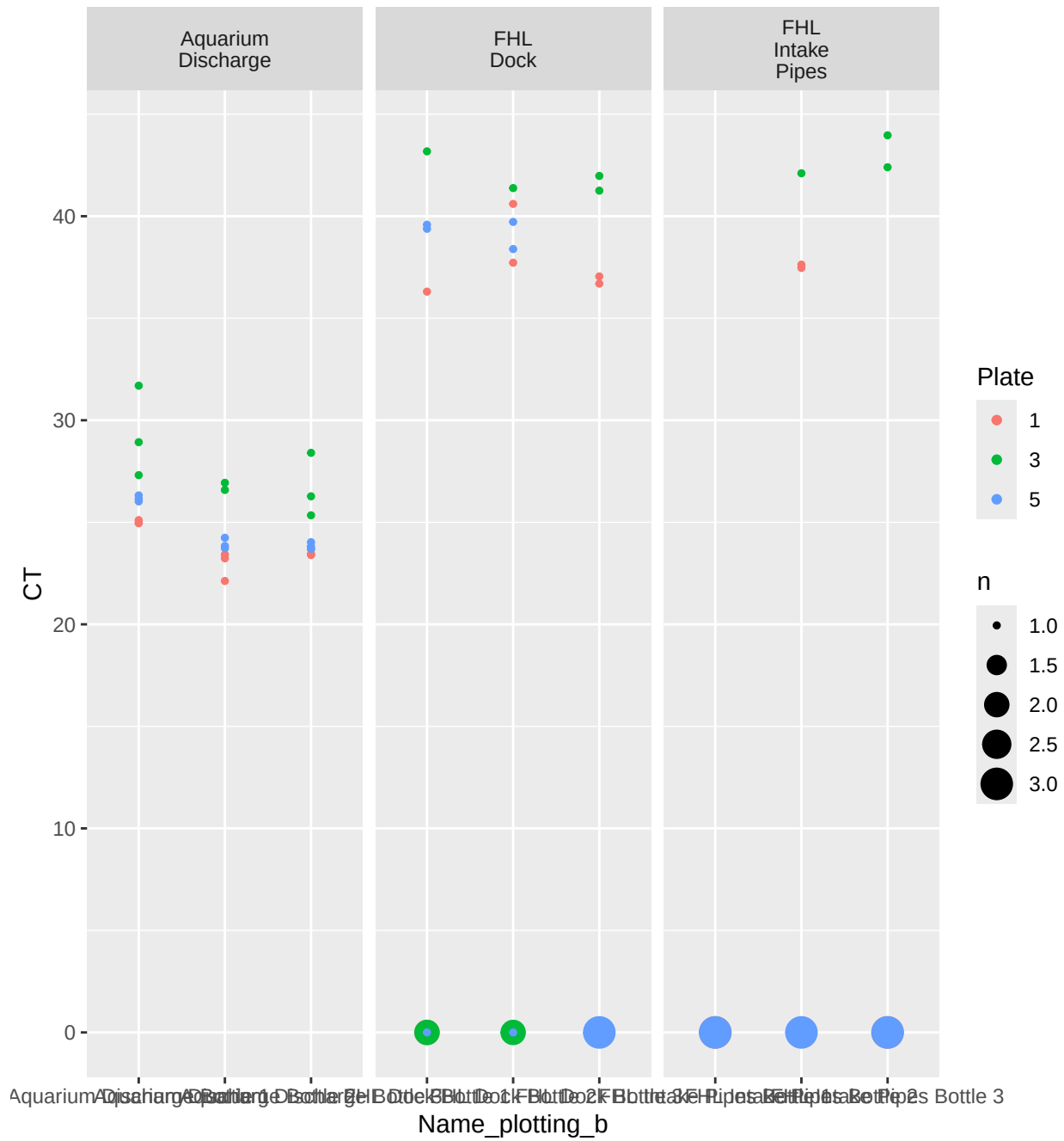
combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  ggplot(., aes(x=Name_plotting, y= CT, fill= Plate)) +geom_col(stat = "identity",position="dodge") +fa
```



```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  ggplot(., aes(x=Name_plotting_b, y= CT, color= Plate)) +geom_count() +facet_wrap(~Type2, scales="free

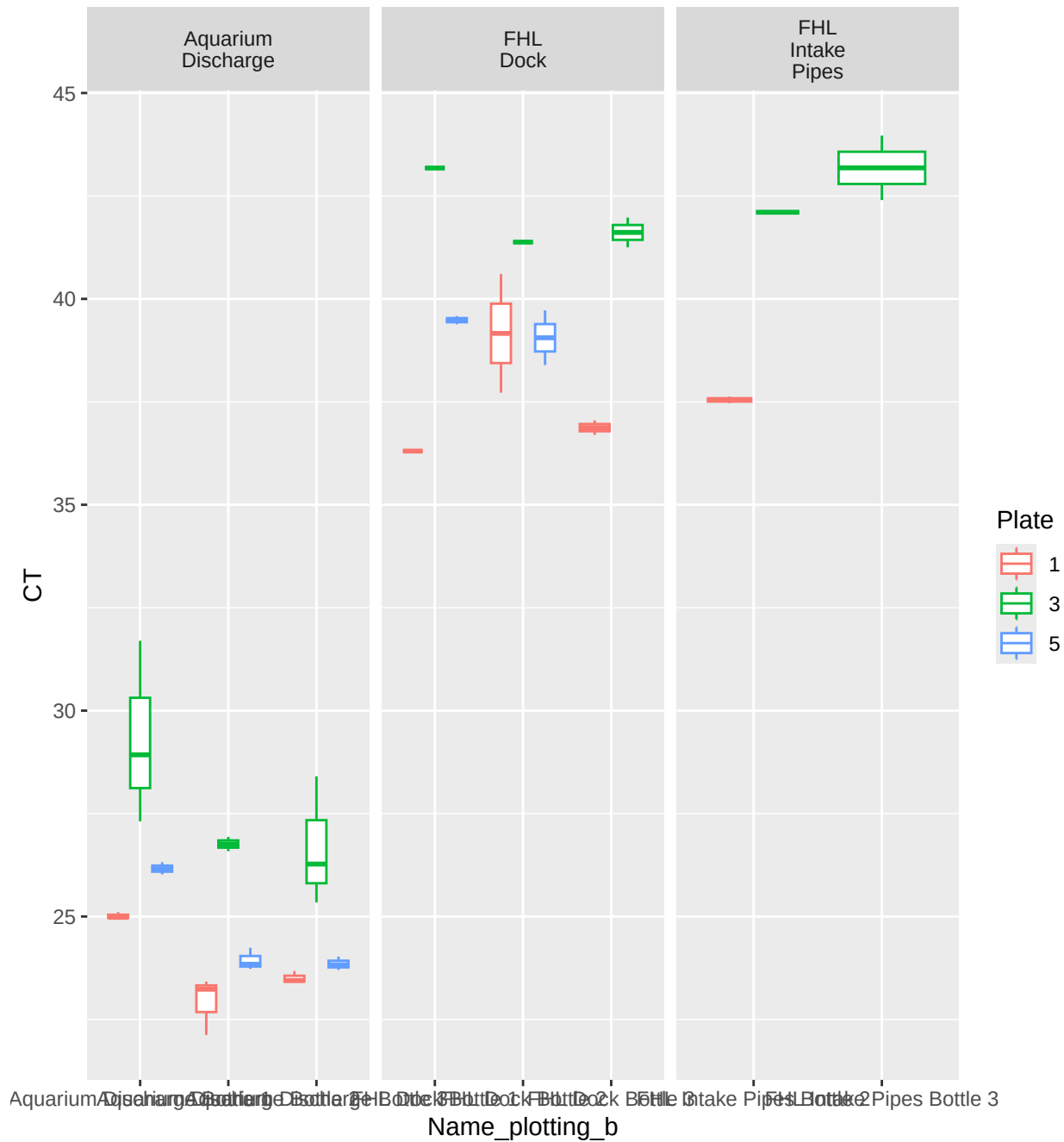
```



```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  filter(., CT >0) %>%
  ggplot(., aes(x=Name_plotting_b, y= CT, color= Plate)) +geom_boxplot() +facet_wrap(~Type2, scales="fr

```



```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  dplyr::group_by(Plate, Type, Name_plotting_b) %>%
  filter(., CT > 0) %>%
  dplyr::summarise(mean_CT= mean(CT), sd_CT=sd(CT), n=n())

```

```

## # A tibble: 20 x 6
## # Groups:   Plate, Type [8]
##   Plate Type      Name_plotting_b      mean_CT  sd_CT    n

```

##	<chr>	<chr>	<chr>	<dbl>	<dbl>	<int>
##	1	1	Aquarium Discharge Aquarium Discharge Bottle 1	25.0	0.0820	3
##	2	1	Aquarium Discharge Aquarium Discharge Bottle 2	22.9	0.700	3
##	3	1	Aquarium Discharge Aquarium Discharge Bottle 3	23.5	0.149	3
##	4	1	FHL Dock FHL Dock Bottle 1	36.3	NA	1
##	5	1	FHL Dock FHL Dock Bottle 2	39.2	2.04	2
##	6	1	FHL Dock FHL Dock Bottle 3	36.9	0.250	2
##	7	1	FHL Intake Pipes FHL Intake Pipes Bottle 2	37.5	0.114	2
##	8	3	Aquarium Discharge Aquarium Discharge Bottle 1	29.3	2.22	3
##	9	3	Aquarium Discharge Aquarium Discharge Bottle 2	26.8	0.247	2
##	10	3	Aquarium Discharge Aquarium Discharge Bottle 3	26.7	1.57	3
##	11	3	FHL Dock FHL Dock Bottle 1	43.2	NA	1
##	12	3	FHL Dock FHL Dock Bottle 2	41.4	NA	1
##	13	3	FHL Dock FHL Dock Bottle 3	41.6	0.510	2
##	14	3	FHL Intake Pipes FHL Intake Pipes Bottle 2	42.1	NA	1
##	15	3	FHL Intake Pipes FHL Intake Pipes Bottle 3	43.2	1.11	2
##	16	5	Aquarium Discharge Aquarium Discharge Bottle 1	26.2	0.148	3
##	17	5	Aquarium Discharge Aquarium Discharge Bottle 2	23.9	0.269	3
##	18	5	Aquarium Discharge Aquarium Discharge Bottle 3	23.9	0.163	3
##	19	5	FHL Dock FHL Dock Bottle 1	39.5	0.142	2
##	20	5	FHL Dock FHL Dock Bottle 2	39.1	0.940	2

```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  dplyr::group_by(Plate, Type, Name_plotting_b) %>%
  filter(., CT > 0) %>%
  dplyr::summarise(mean_CT= mean(CT), sd_CT=sd(CT), n=n())

```

```

## # A tibble: 20 x 6
## # Groups:   Plate, Type [8]
##   Plate Type      Name_plotting_b      mean_CT      sd_CT      n
##   <chr> <chr>      <chr>      <dbl>      <dbl> <int>
## 1 1 1 Aquarium Discharge Aquarium Discharge Bottle 1 25.0 0.0820 3
## 2 1 1 Aquarium Discharge Aquarium Discharge Bottle 2 22.9 0.700 3
## 3 1 1 Aquarium Discharge Aquarium Discharge Bottle 3 23.5 0.149 3
## 4 1 1 FHL Dock FHL Dock Bottle 1 36.3 NA 1
## 5 1 1 FHL Dock FHL Dock Bottle 2 39.2 2.04 2
## 6 1 1 FHL Dock FHL Dock Bottle 3 36.9 0.250 2
## 7 1 1 FHL Intake Pipes FHL Intake Pipes Bottle 2 37.5 0.114 2
## 8 3 3 Aquarium Discharge Aquarium Discharge Bottle 1 29.3 2.22 3
## 9 3 3 Aquarium Discharge Aquarium Discharge Bottle 2 26.8 0.247 2
## 10 3 3 Aquarium Discharge Aquarium Discharge Bottle 3 26.7 1.57 3
## 11 3 3 FHL Dock FHL Dock Bottle 1 43.2 NA 1
## 12 3 3 FHL Dock FHL Dock Bottle 2 41.4 NA 1
## 13 3 3 FHL Dock FHL Dock Bottle 3 41.6 0.510 2
## 14 3 3 FHL Intake Pipes FHL Intake Pipes Bottle 2 42.1 NA 1
## 15 3 3 FHL Intake Pipes FHL Intake Pipes Bottle 3 43.2 1.11 2
## 16 5 5 Aquarium Discharge Aquarium Discharge Bottle 1 26.2 0.148 3
## 17 5 5 Aquarium Discharge Aquarium Discharge Bottle 2 23.9 0.269 3
## 18 5 5 Aquarium Discharge Aquarium Discharge Bottle 3 23.9 0.163 3
## 19 5 5 FHL Dock FHL Dock Bottle 1 39.5 0.142 2
## 20 5 5 FHL Dock FHL Dock Bottle 2 39.1 0.940 2

```

```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  dplyr::group_by(Plate, Type) %>%
  filter(., CT > 0) %>%
  dplyr::summarise(mean_CT= mean(CT), sd_CT=sd(CT), n=n())

```

```

## # A tibble: 8 x 5
## # Groups:   Plate [3]
##   Plate Type      mean_CT sd_CT      n
##   <chr> <chr>      <dbl> <dbl> <int>
## 1 1     Aquarium Discharge  23.8 1.00      9
## 2 1     FHL Dock           37.7 1.72      5
## 3 1     FHL Intake Pipes    37.5 0.114     2
## 4 3     Aquarium Discharge  27.7 1.98      8
## 5 3     FHL Dock           41.9 0.880     4
## 6 3     FHL Intake Pipes    42.8 1.00      3
## 7 5     Aquarium Discharge  24.7 1.15      9
## 8 5     FHL Dock           39.3 0.602     4

```

```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  dplyr::group_by(Plate, Type) %>%
  filter(., CT < 1) %>%
  dplyr::summarise(n=n())

```

```

## # A tibble: 6 x 3
## # Groups:   Plate [3]
##   Plate Type      n
##   <chr> <chr>    <int>
## 1 1     FHL Dock      4
## 2 1     FHL Intake Pipes  7
## 3 3     FHL Dock      4
## 4 3     FHL Intake Pipes  5
## 5 5     FHL Dock      5
## 6 5     FHL Intake Pipes  9

```

```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  filter(., CT > 0) -> fhl_pos_CT

```

```

model_aov_fhl_pos_CT <- aov(CT ~ Plate + Type + Plate*Type, data=fhl_pos_CT)

```

```

summary(model_aov_fhl_pos_CT)

```

```

##           Df Sum Sq Mean Sq F value    Pr(>F)
## Plate      2  248.1   124.0  70.509 3.54e-13 ***
## Type       2 2143.0  1071.5  609.049 < 2e-16 ***
## Plate:Type  3    2.5     0.8   0.469   0.705
## Residuals 36   63.3     1.8

```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
tukey_hsd(model_aov_fhl_pos_CT)
```

```
## # A tibble: 42 x 9
##   term          group1      group2 null.value estimate conf.low conf.high  p.adj
## * <chr>         <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Plate         1          3          0      4.65      3.49      5.82 3.51e-11
## 2 Plate         1          5          0     -0.713    -1.92      0.497 3.32e- 1
## 3 Plate         3          5          0     -5.37     -6.59     -4.14 2.89e-12
## 4 Type          Aquarium D~ FHL D~      0      14.2      13.1      15.3 0
## 5 Type          Aquarium D~ FHL I~      0      13.8      12.2      15.4 0
## 6 Type          FHL Dock   FHL I~      0     -0.347    -2.05      1.36 8.73e- 1
## 7 Plate:Type 1:Aquarium~ 3:Aqu~      0      3.87      1.74      5.99 2.25e- 5
## 8 Plate:Type 1:Aquarium~ 5:Aqu~      0      0.837    -1.22      2.90 9.13e- 1
## 9 Plate:Type 1:Aquarium~ 1:FHL~      0      13.9      11.4      16.3 0
## 10 Plate:Type 1:Aquarium~ 3:FHL~      0      18.1      15.5      20.8 0
## # i 32 more rows
## # i 1 more variable: p.adj.signif <chr>
```

```
model_aov_fhl_pos_CT_bottle <- aov(CT ~ Plate + Name_plotting_b + Plate*Name_plotting_b, data=fhl_pos_C
```

```
summary(model_aov_fhl_pos_CT_bottle)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Plate          2  248.1  124.05  131.02 1.22e-13 ***
## Name_plotting_b  7 2176.9  310.98  328.48 < 2e-16 ***
## Plate:Name_plotting_b 10   9.2   0.92   0.97  0.493
## Residuals      24   22.7   0.95
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
tukey_hsd(model_aov_fhl_pos_CT_bottle)
```

```
## # A tibble: 307 x 9
##   term          group1 group2 null.value estimate conf.low conf.high  p.adj
## * <chr>         <chr> <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Plate         1      3          0      4.65      3.78      5.53 4.25e-12
## 2 Plate         1      5          0     -0.713    -1.62      0.194 1.43e- 1
## 3 Plate         3      5          0     -5.37     -6.29     -4.45 6.36e-13
## 4 Name_plotting_b Aquar~ Aquar~      0     -2.15     -3.71     -0.583 2.84e- 3
## 5 Name_plotting_b Aquar~ Aquar~      0     -2.15     -3.67     -0.634 1.97e- 3
## 6 Name_plotting_b Aquar~ FHL D~      0      13.3      11.4      15.2 2.13e-14
## 7 Name_plotting_b Aquar~ FHL D~      0      13.4      11.6      15.2 2.12e-14
## 8 Name_plotting_b Aquar~ FHL D~      0      11.4      9.46      13.3 2.93e-14
## 9 Name_plotting_b Aquar~ FHL I~      0      12.0      9.85      14.1 5.04e-14
## 10 Name_plotting_b Aquar~ FHL I~      0      13.0      10.5      15.5 1.9 e-13
## # i 297 more rows
## # i 1 more variable: p.adj.signif <chr>
```

```

combo_plate_analysis %>%
  mutate(., CT=as.numeric(CT)) %>%
  mutate(., CT= replace_na(CT, 0)) %>%
  filter(., CT > 0) %>%
  filter(., Type == "FHL Dock" ) -> fhl_dock_CT

model_aov_fhl_dock <- aov(CT ~ Plate, data=fhl_dock_CT)

summary(model_aov_fhl_dock)

```

```

##              Df Sum Sq Mean Sq F value    Pr(>F)
## Plate          2  40.80   20.399    13.41 0.00148 **
## Residuals     10   15.21    1.521
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

tukey_hsd(model_aov_fhl_dock)

```

```

## # A tibble: 3 x 9
##   term group1 group2 null.value estimate conf.low conf.high  p.adj
## * <chr> <chr> <chr>      <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
## 1 Plate 1      3      0      4.27     2.00     6.54 0.00112
## 2 Plate 1      5      0      1.59    -0.674    3.86 0.181
## 3 Plate 3      5      0     -2.68    -5.07    -0.286 0.0292
## # i 1 more variable: p.adj.signif <chr>

```

Figure 5

```

pycno_sample_data_4_plotting %>%
  mutate(., Plate="Pycno Run 5") %>%
  dplyr::select(Plate, `Sample Name`, CT, Quantity, `Quantity Mean`, `Quantity SD`, Concentration= Conc

pycno_sample_data_4_plotting_run3 %>%
  mutate(., Plate="Pycno Run 3") %>%
  dplyr::select(Plate, `Sample Name`, CT, Quantity, `Quantity Mean`, `Quantity SD`, Concentration, Biol

pycno_sample_data_4_plotting_run1 %>%
  mutate(., Plate="Pycno Run 1") %>%
  dplyr::select(Plate, `Sample Name`, CT, Quantity, `Quantity Mean`, `Quantity SD`, Concentration, Biol

colnames(LoD_plate3_analysis)

```

```

## [1] "Plate"           "Sample Name"      "CT"
## [4] "Quantity"        "Quantity Mean"    "Quantity SD"
## [7] "Concentration"   "Biological_Replicate" "Technical_Replicate"
## [10] "Name_plotting"   "Name_plotting_b"  "Station"

```



```
colnames(LoD_plate2_analysis)
```

```
## [1] "Plate"           "Sample Name"      "CT"
## [4] "Quantity"        "Quantity Mean"    "Quantity SD"
## [7] "Concentration"    "Biological_Replicate" "Technical_Replicate"
## [10] "Name_plotting"    "Name_plotting_b"  "Station"
```

```
colnames(LoD_plate1_analysis)
```

```
## [1] "Plate"           "Sample Name"      "CT"
## [4] "Quantity"        "Quantity Mean"    "Quantity SD"
## [7] "Concentration"    "Biological_Replicate" "Technical_Replicate"
## [10] "Name_plotting"    "Name_plotting_b"  "Station"
```

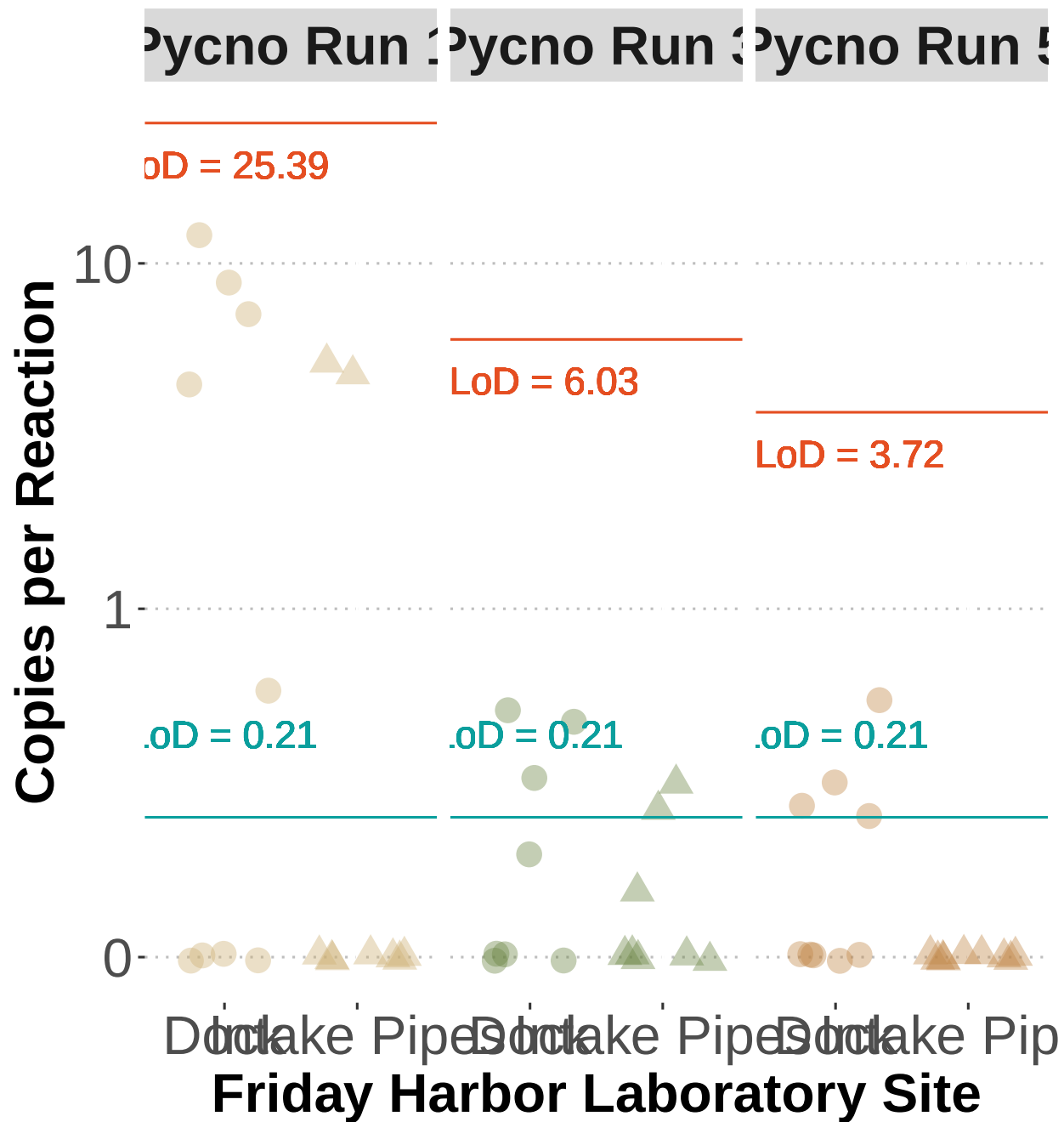
```
bind_rows(LoD_plate3_analysis, LoD_plate2_analysis, LoD_plate1_analysis) -> LoD_plate_analysis_combo
```

```
loblodloq_df <- read.delim(, file=here("LOD_LOQ","final_lob_loq_table.csv"), sep=",") %>%
  filter(., Type=="qPCR") %>% pivot_wider(names_from = LoD_method, values_from = c(LOB,LOD,LOQ)) %>%
  mutate(., LOB_eLowQuant = if_else(is.na(LOB_eLowQuant),0, LOB_eLowQuant),
          LOD_eLowQuant = if_else(is.na(LOD_eLowQuant),0.2134723, LOD_eLowQuant),
          LOQ_eLowQuant = if_else(is.na(LOQ_eLowQuant),0.8078273, LOQ_eLowQuant))
```

```
wes_palette("AsteroidCity1") -> AC
```

```
nbreaks=5
breaks <- c(0,10^0,10^1,10^2,1000)
```

```
LoD_plate_analysis_combo %>%
  left_join(loblodloq_df) %>%
  mutate(., Station = recode(Station,"FHL Dock"="Dock",
                             "FHL Intake Pipes"="Intake Pipes")) %>%
  filter(., Station %in% c("Dock","Intake Pipes")) %>%
  mutate(., Quantity=Quantity*2) %>%
  ggplot(aes(y=Quantity, x= Station, color=Plate, shape =Station)) + geom_jitter(size=5, alpha=0.4) + t
  scale_color_manual(values =c(AC[2],AC[4],AC[5])) +
  geom_hline(aes(yintercept=LOD_eLowQuant),color=AC[1]) +
  geom_text(aes(y = LOD_eLowQuant, x = Inf, label = paste0("eLowQuant LoD = ", round(LOD_eLowQuant, 2)),
               vjust = -2.5, hjust=1.3, color = AC[1], size=6))+ geom_hline(aes(yintercept=LOD_Klymus), co
  geom_text(aes(y = LOD_Klymus, x = Inf, label = paste0("Klymus et al. 2020 LoD = ", round(LOD_Klymus, 1
  scale_y_continuous(trans = pseudo_log_trans(sigma = 0.1, base = 10), breaks=breaks)
```



```
ggsave(filename = here("Figures", "Figure_5_field_trial_LoD.png"),
        width=14,
        height = 8,
        dpi = 400,
        units = c("in"))
```